

Are High Densities Biasing Observed Galaxy Properties?

Investigating Nebular Properties in Extreme Conditions Across Cosmic Time

Zorayda Martinez

2nd Year Graduate Student

UT Austin

Galaxies Seminar March 31st, 2025

**Observable
galaxy properties
enhance our
understanding of
galaxy evolution**

**galaxy
sizes**

**emission
lines**

**SFR
densities**

**chemical
enrichment**

**BH
masses**

**electron
densities**

galaxy sizes

SFR
densities

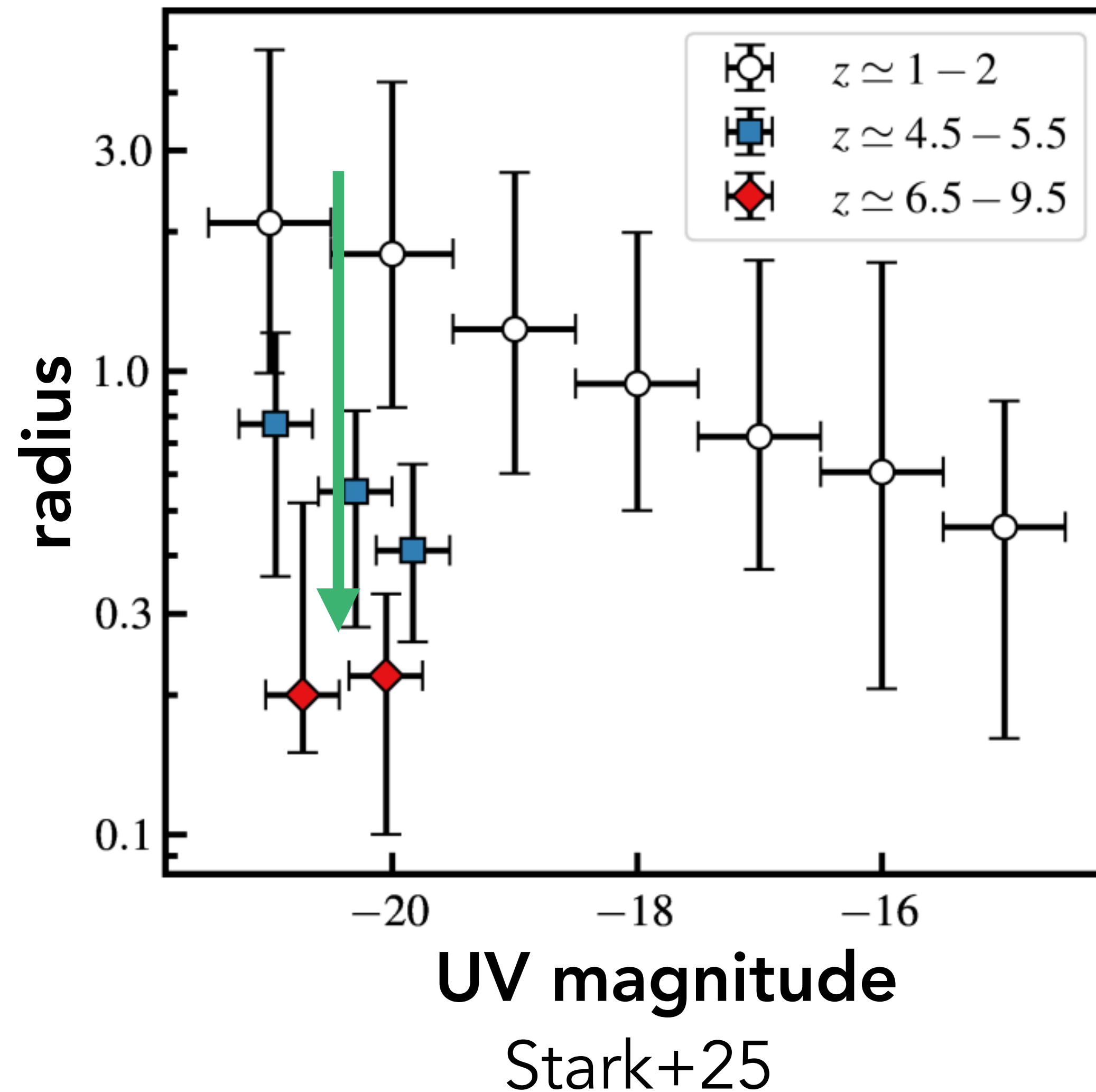
BH
masses

emission
lines

chemical
enrichment

electron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



high- z
galaxies are
compact

each increasing
redshift bin points
occupies a position
corresponding to
smaller galaxy size

galaxy sizes

**SFR
densities**

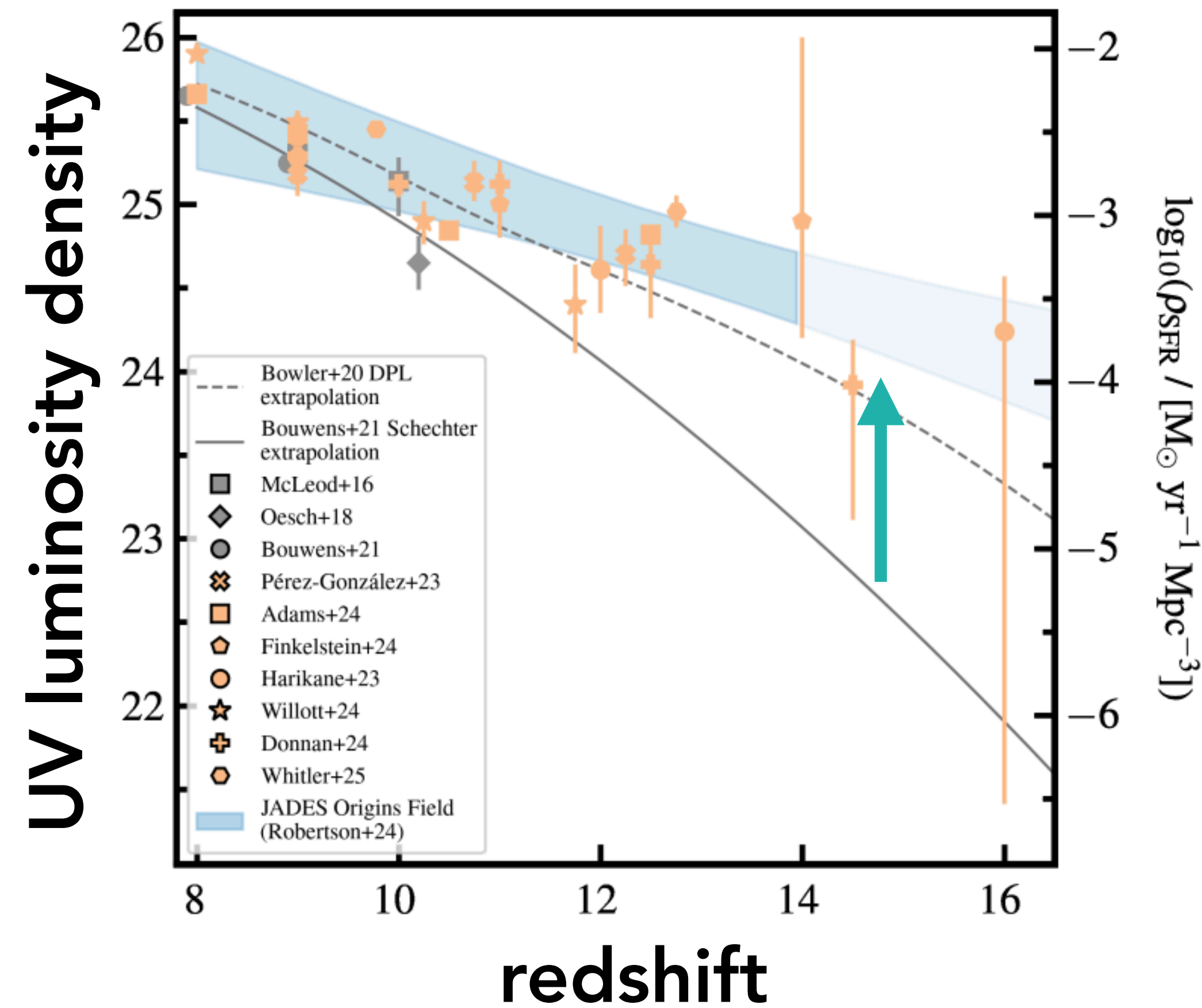
BH
masses

emission
lines

chemical
enrichment

electron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



high- z
galaxies
have high
SFR density

observed ρ_{SFR} is
higher than pre-
JWST predictions

galaxy sizes

SFR
densities

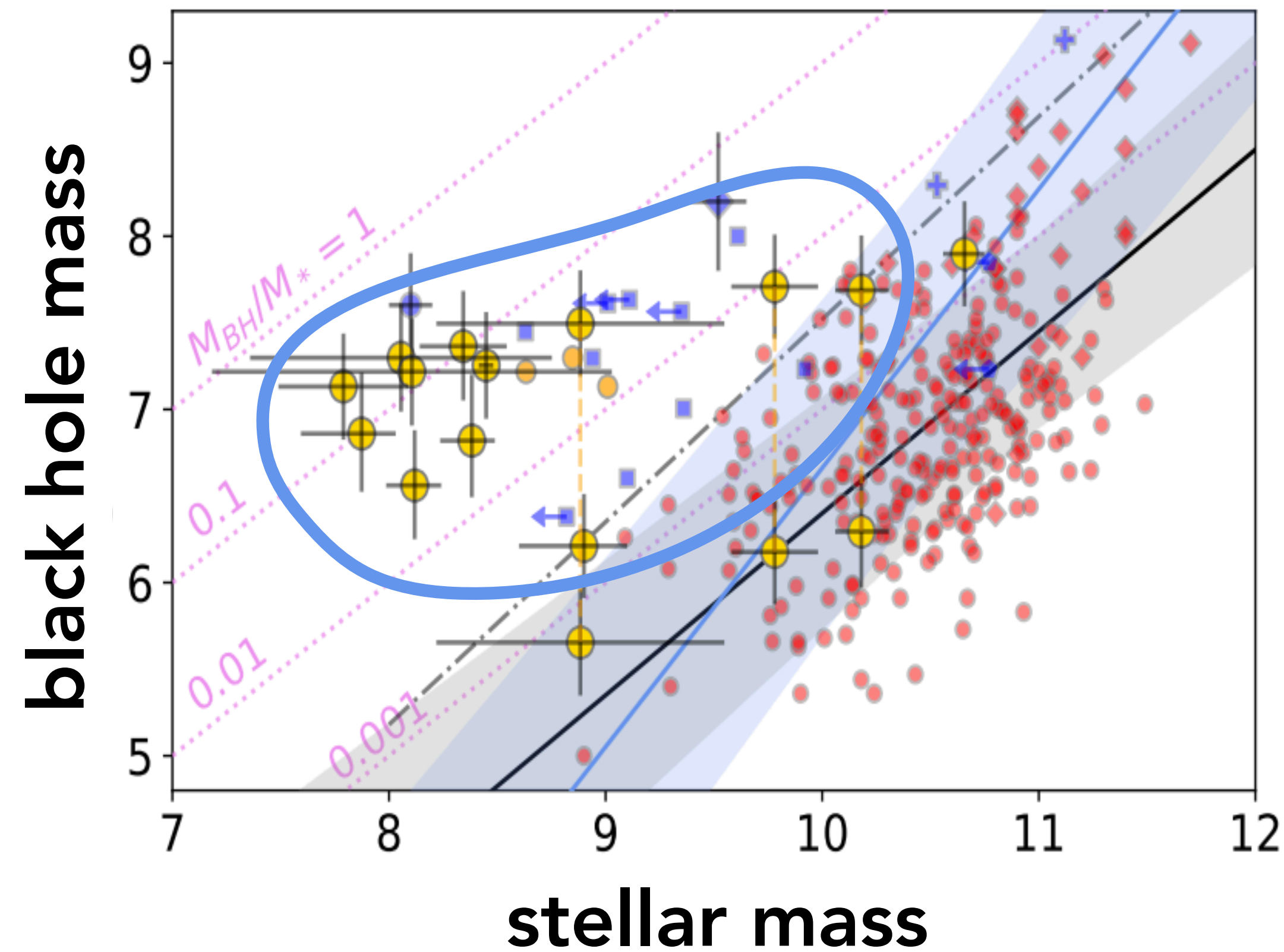
BH
masses

emission
lines

chemical
enrichment

electron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



Maiolino+25

high- z
galaxies are
large BH
masses

$M_{\star} - M_{\text{BH}}$ offset
to higher M_{BH}
from local
relation

galaxy sizes

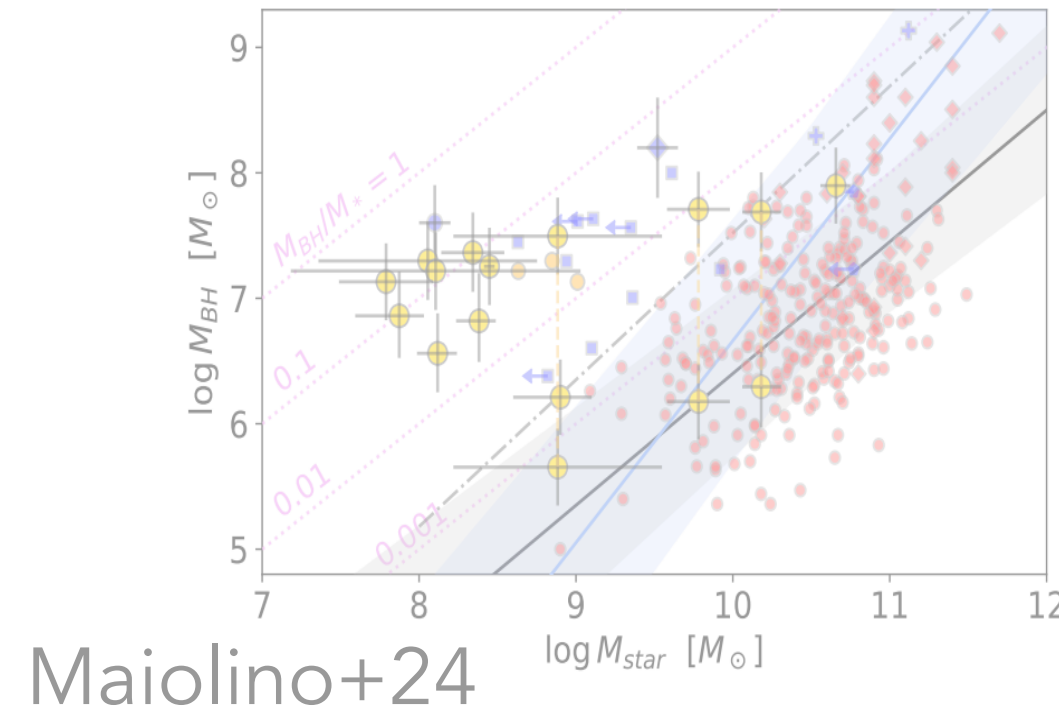
SFR
densities

BH
masses

emission
lines

chemical
enrichment

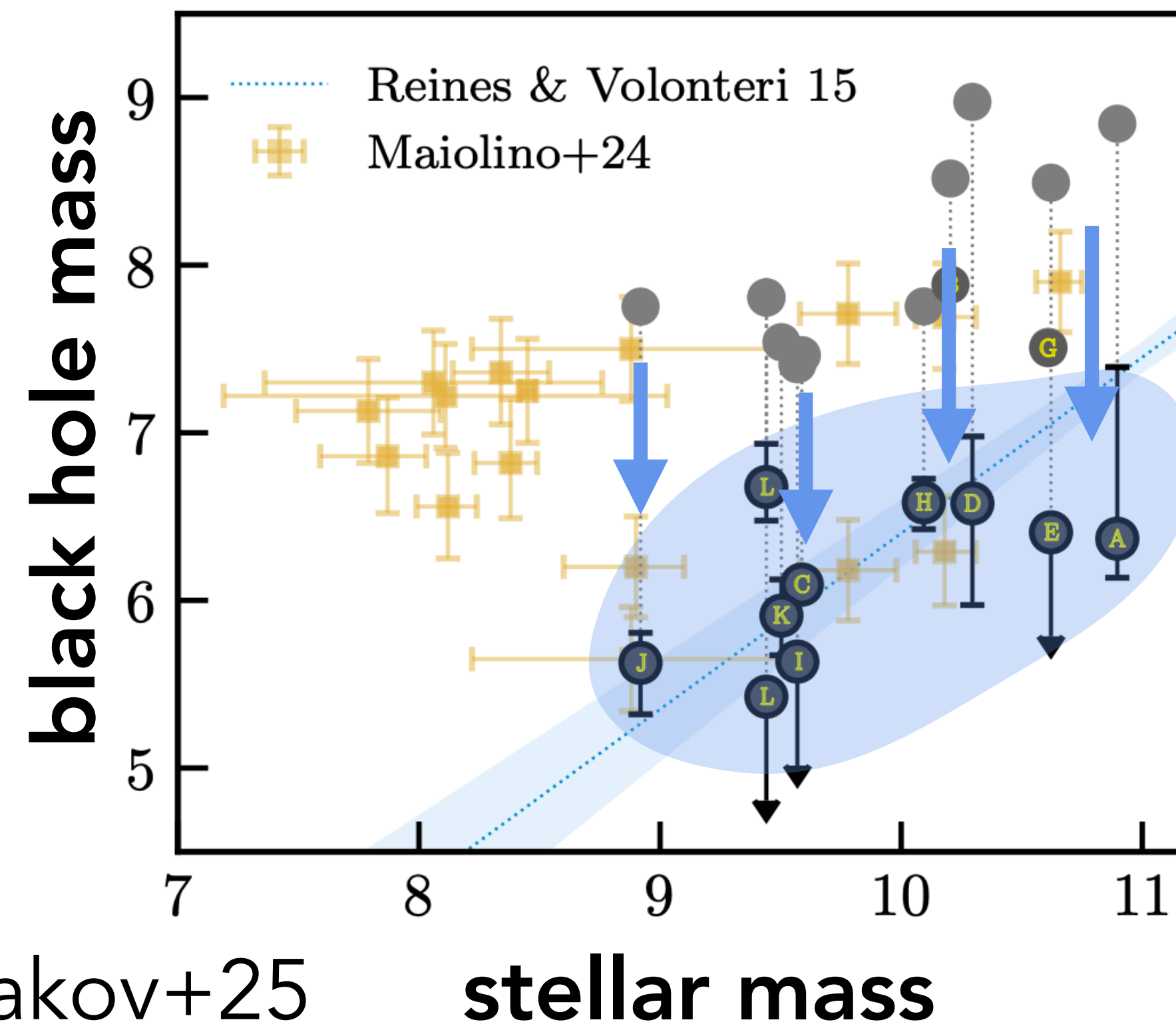
electron
densities



high- z
galaxies are
large BH
masses

$M_\star - M_{BH}$ offset
to higher M_{BH}
from local
relation

High- z galaxies
are unlike
anything we
see in the local
universe

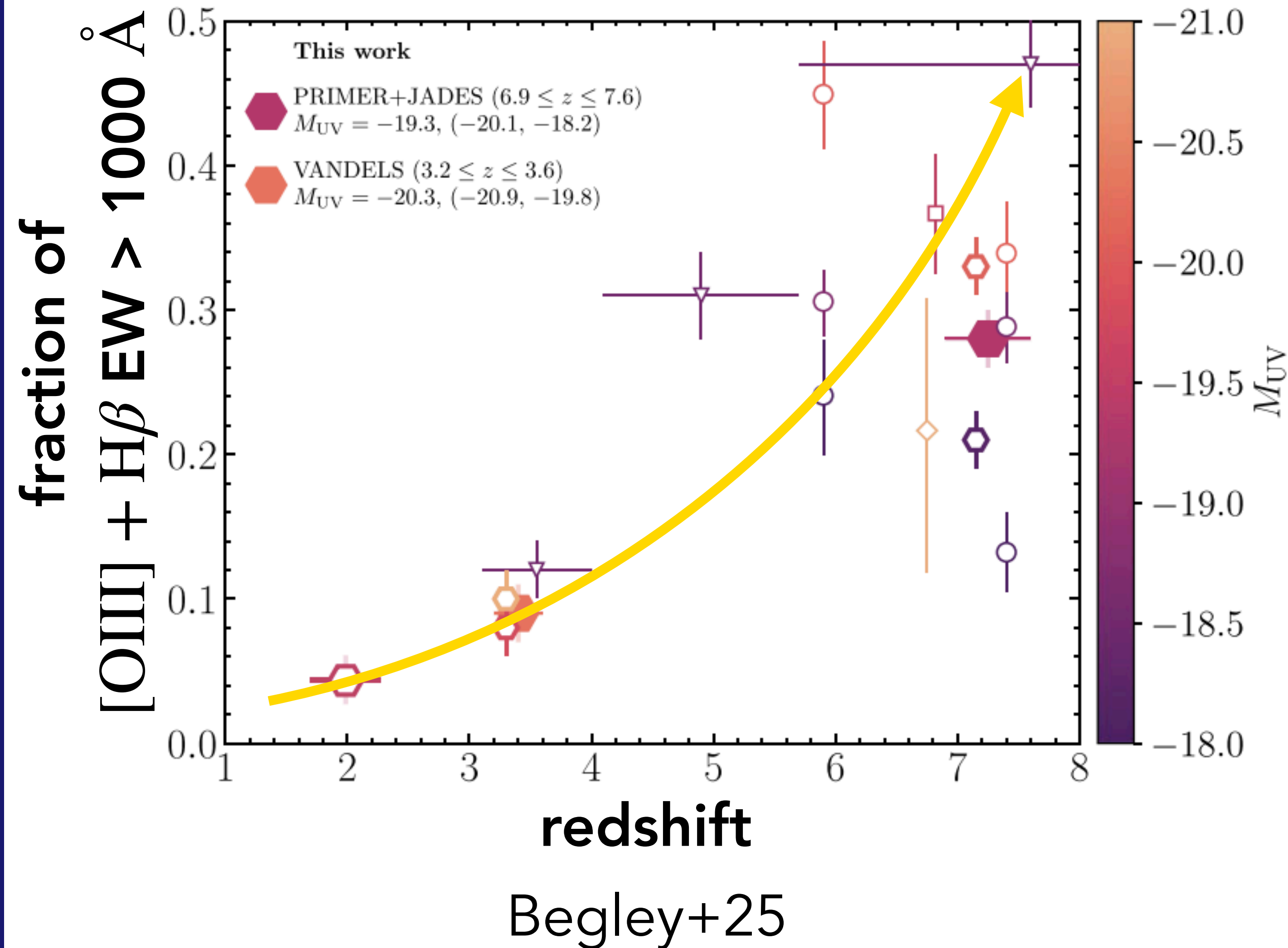


modeling electron scattering
in dense ionized gas allows
the authors derive M_{BH} that
correspond to the local
 $M_\star - M_{BH}$ relation

galaxy sizes

SFR
densitiesBH
massesemission
lineschemical
enrichmentelectron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



high- z
galaxies have
extreme
emission lines

more objects
with very large
 $[\text{OIII}] + \text{H}\beta \text{ EW}$
at high- z

galaxy sizes

SFR
densities

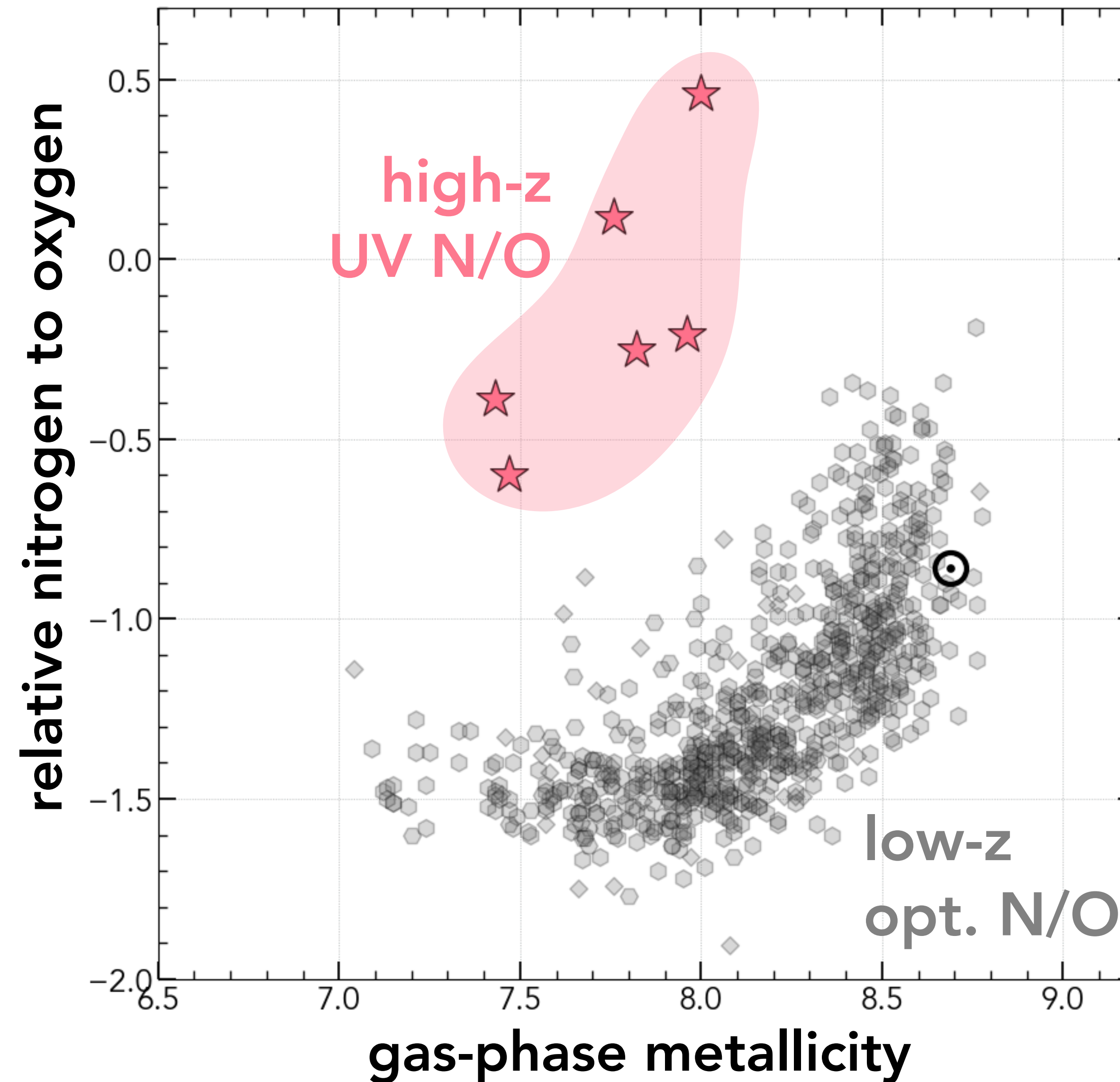
BH
masses

emission
lines

chemical
enrichment

electron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



high- z
galaxies have
unusual
chemical
enhancement

relative abundances
offset from local
trends at high- z

galaxy sizes

SFR
densities

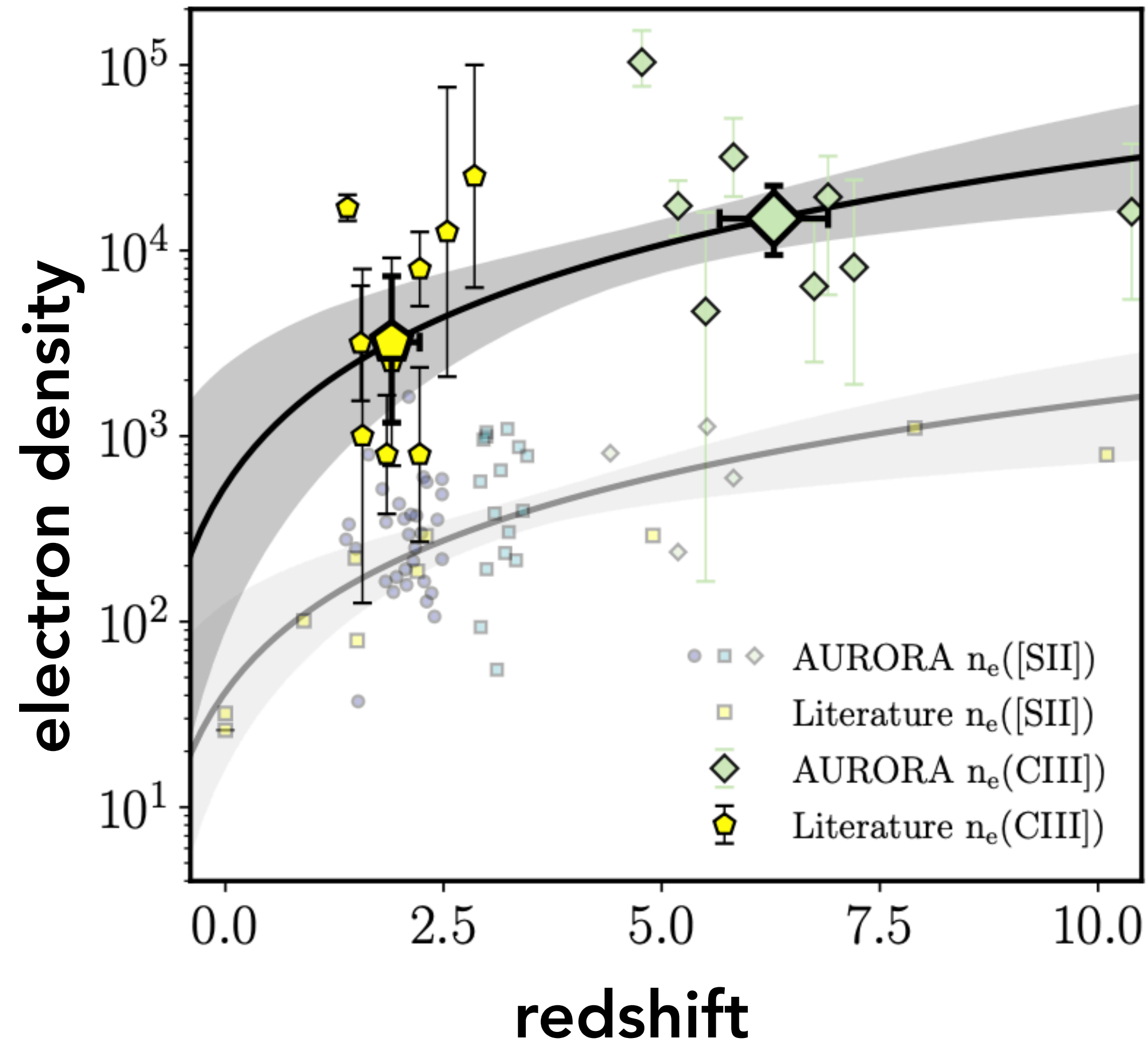
BH
masses

emission
lines

chemical
enrichment

electron
densities

High- z galaxies
are unlike
anything we
see in the local
universe



Topping+25

high- z
galaxies are
dense

electron density
increases with
redshift

Photoionization Modeling

Modeling + Observational Study

**Are high densities
biasing observed
galaxy properties?**

**How much does the
interpretation of those
properties change at
high densities?**

Photoionization Modeling

Are high densities
biasing observed
galaxy properties?

Modeling + Observational Study

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models

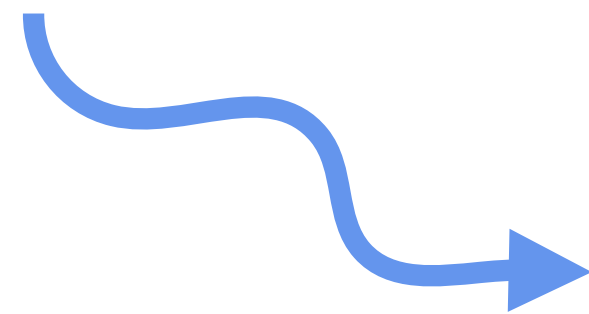
Modeling + Observational Study

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties

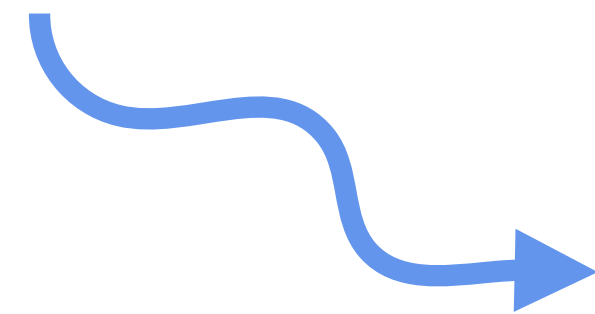
Modeling + Observational Study

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



create tools
to interpret
high-z
sources

Modeling + Observational Study

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



create tools
to interpret
high-z
sources

Modeling + Observational Study

UV N-emitting sample
from $0 < z < 11$

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



create tools
to interpret
high-z
sources

Modeling + Observational Study

UV N-emitting sample
from $0 < z < 11$



uniform abundance
determination

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



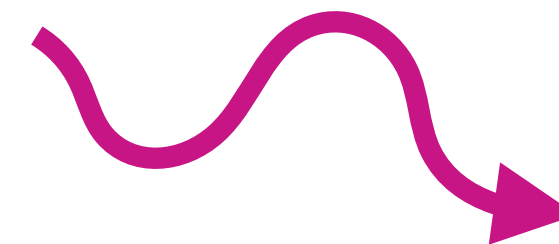
create tools
to interpret
high-z
sources

Modeling + Observational Study

UV N-emitting sample
from $0 < z < 11$



uniform abundance
determination



test varied
approaches
to the
density
structure

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

Are high densities
biasing observed
galaxy properties?

Aside about abundance calculation...

Step 1: *Diagnose the physical conditions.*

Emission line ratios are used to determine n_e , T_e , and $\log U$.

Step 2: *Determine gas-phase metallicity.*

Using n_e , T_e , and emission line intensities can be used to derive the total oxygen abundance, $12 + \log \text{O}/\text{H}$.

Step 3: *Calculate ionization correction factors.*

Photoionization modeling uses $\log U$ and $12 + \log \text{O}/\text{H}$ to predict contribution from unseen ionic species.

Step 4: *Derive relative N/O abundance.*

The relative ionic abundance is determined with n_e , T_e , and emission line intensities. An ionization correction factor is applied to acquire $\log \text{N}/\text{O}$.

How much does
interpretation
of those
properties change
at high densities?

extreme galaxy
extreme

UV N-emitting
from $0 < z$

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



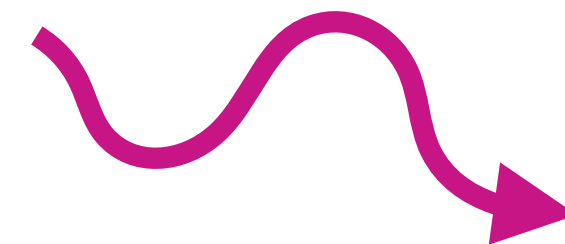
create tools
to interpret
high-z
sources

Modeling + Observational Study

UV N-emitting sample
from $0 < z < 11$



uniform abundance
determination



test varied
approaches
to the
density
structure

Are high densities
biasing observed
galaxy properties?

How much does
the interpretation
of those
properties change
at high densities?

New custom grid of Cloudy photoionization models

parameter	unit	min	max	steps
Z	$Z_{\odot}$	0.05	0.40	varies
$\log t$	yr	6.0	7.6	varies
$\log U$	dex	-3.5	-1.0	0.25
$\log n_e$	cm^{-3}	2.0	9.0	1.0
$\log \text{N/O}$	dex	-1.50	1.00	0.5
$\log \text{C/O}$	dex	-1.75	0.25	0.5

Cloudy 23.01

- Spectral synthesis code to simulate the conditions of interstellar matter
- Models the ionization, temperature, and chemical state of a cloud given ionizing source (e.g. stellar population synthesis models)

New

c

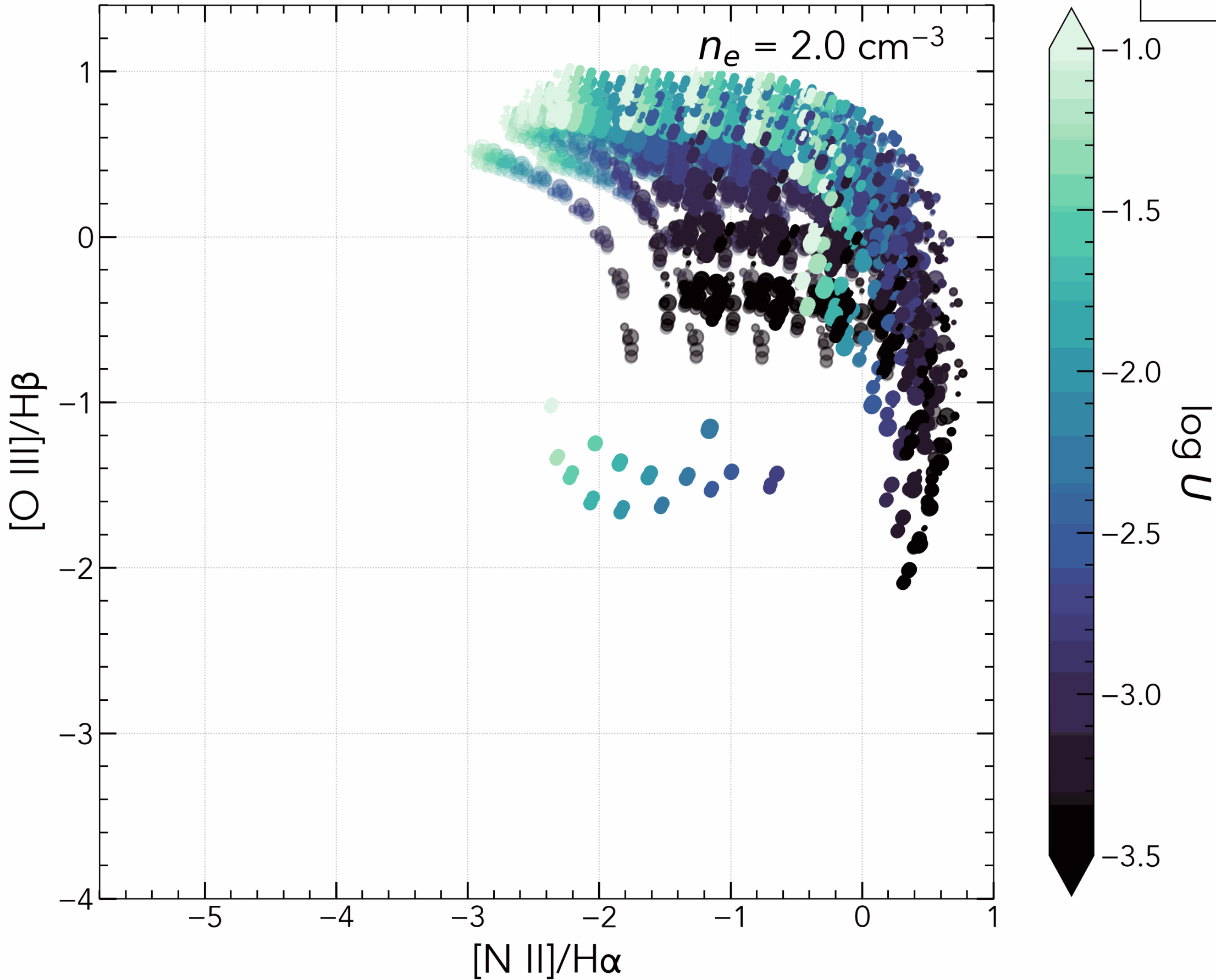
photoionization
models

				max	steps
				40	varies
			.6		varies
			-1.0		0.25
	$\log n_e$	cm^{-3}	2.0	9.0	1.0
	$\log \text{N/O}$	dex	-1.50	1.00	0.5
	$\log \text{C/O}$	dex	-1.75	0.25	0.5

New custom grid of Cloudy photoionization models

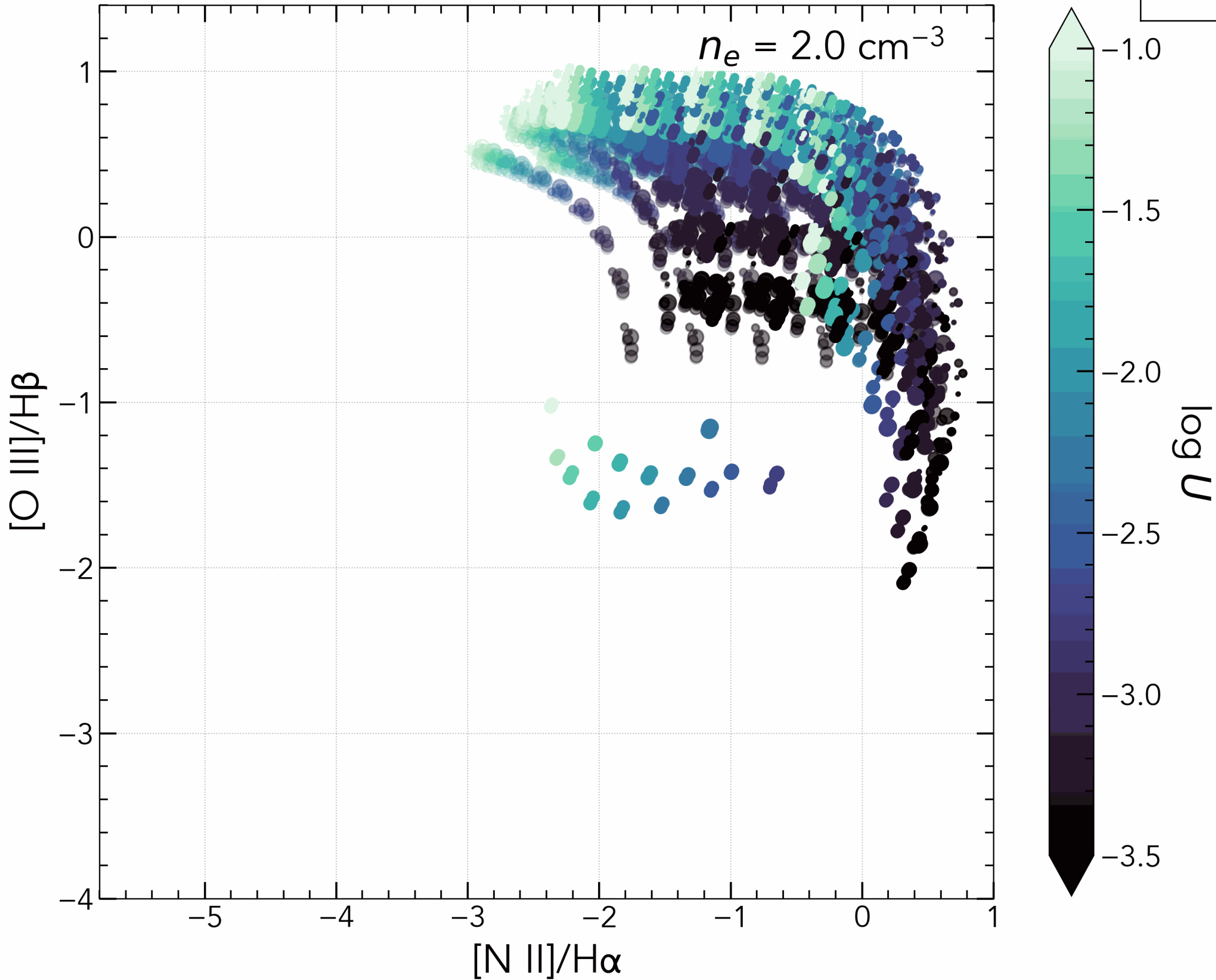
parameter	unit	min	max	steps
Z	$Z_{\odot}$	0.05	0.40	varies
$\log t$	yr	6.0	7.6	varies
$\log U$	dex	-3.5	-1.0	0.25
$\log n_e$	cm^{-3}	2.0	9.0	1.0
$\log \text{N/O}$	dex	-1.50	1.00	0.5
$\log \text{C/O}$	dex	-1.75	0.25	0.5

New custom grid of Cloudy photoionization models



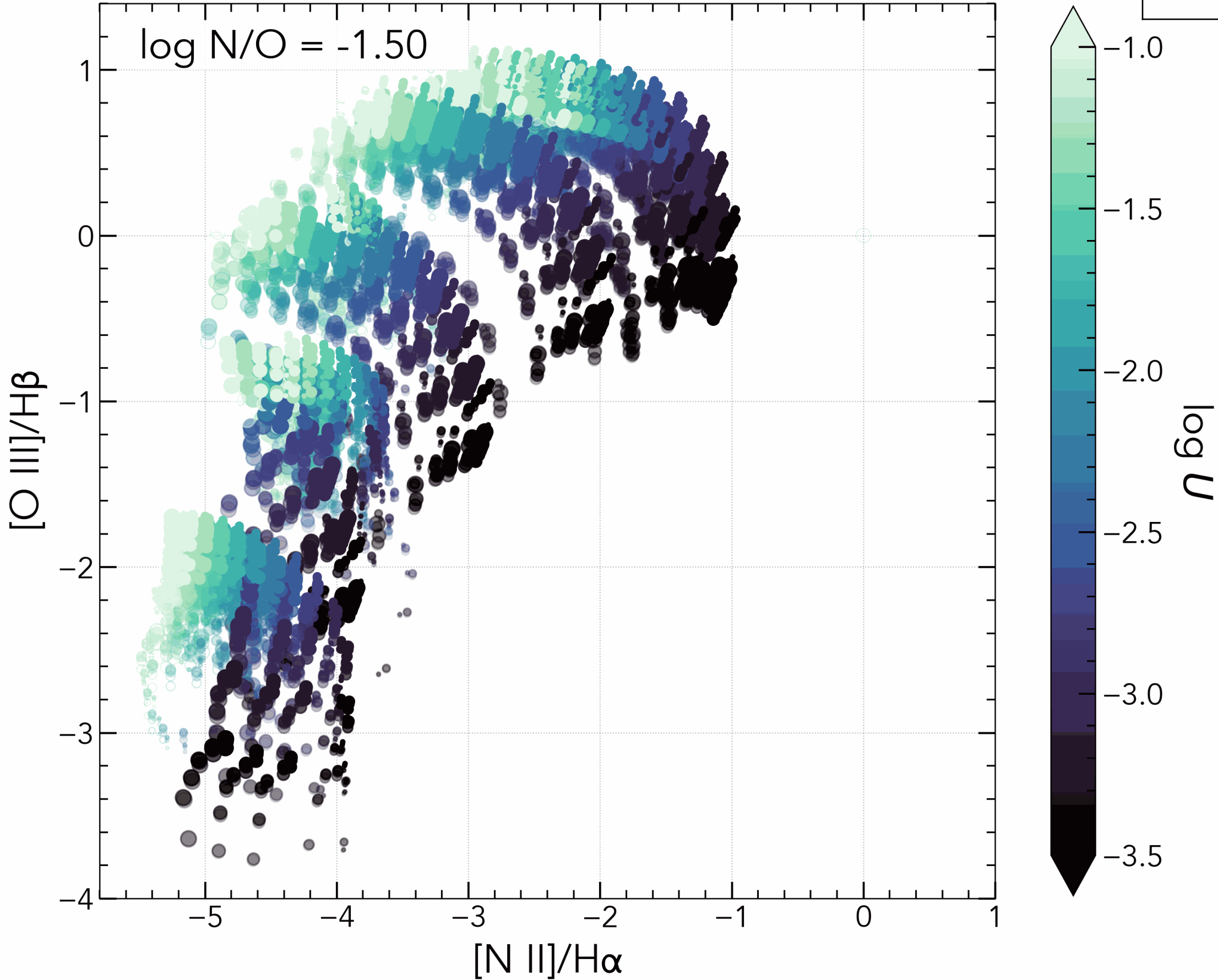
	parameter	units	min	max	steps
	Z	$Z_{\odot}$	0.05	0.40	varies
	$\log t$	yr	6.0	7.6	varies
	$\log U$	dex	-3.5	-1.0	0.25
	$\log n_e$	cm^{-3}	2.0	9.0	1.0
	$\log \text{N/O}$	dex	-1.50	1.00	0.5
	$\log \text{C/O}$	dex	-1.75	0.25	0.25

New custom grid of Cloudy photoionization models



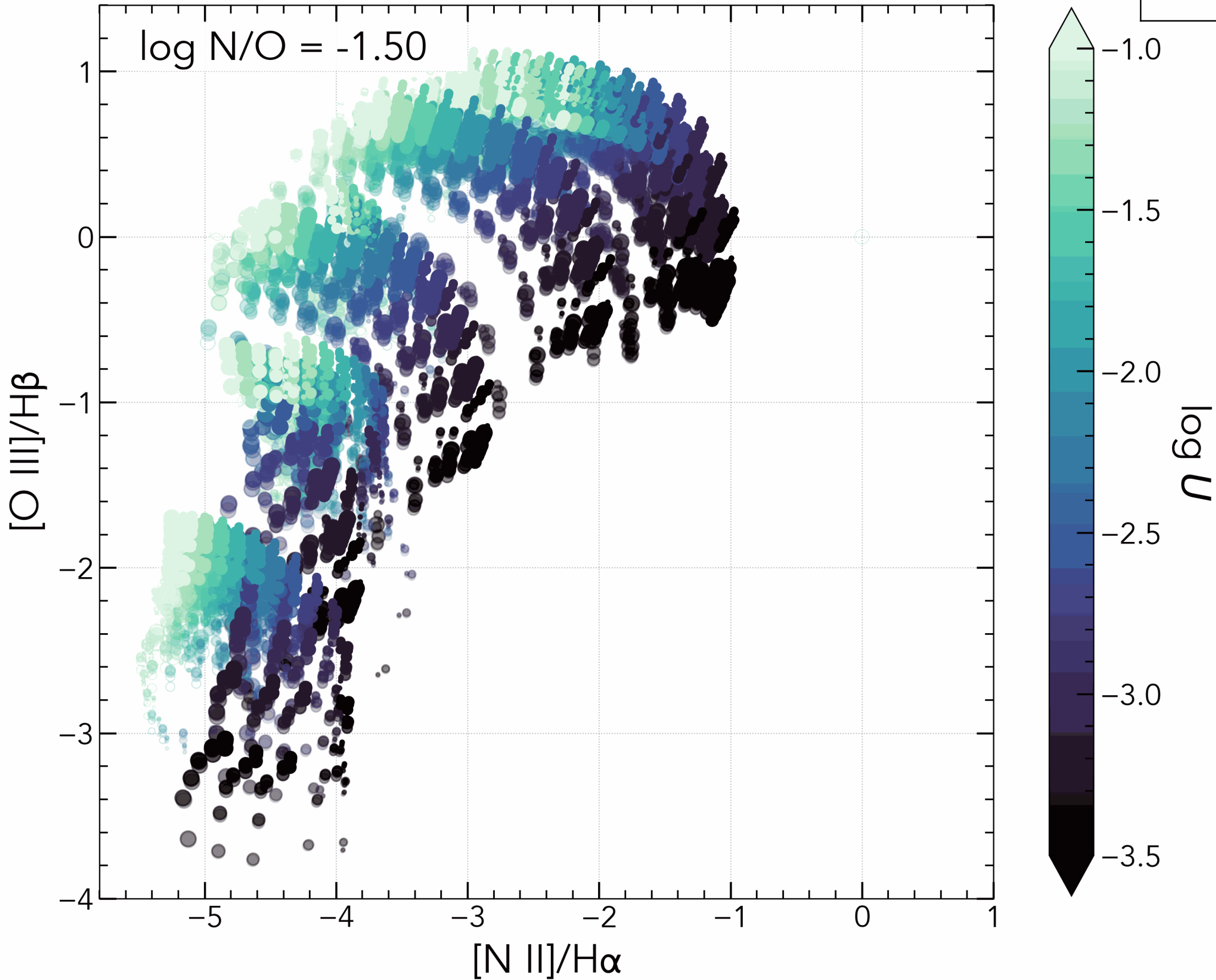
	parameter	units	min	max	steps
	Z	$Z_{\odot}$	0.05	0.40	varies
	$\log t$	yr	6.0	7.6	varies
	$\log U$	dex	-3.5	-1.0	0.25
	$\log n_e$	cm^{-3}	2.0	9.0	1.0
	$\log \text{N/O}$	dex	-1.50	1.00	0.5
	$\log \text{C/O}$	dex	-1.75	0.25	0.25

New custom grid of Cloudy photoionization models



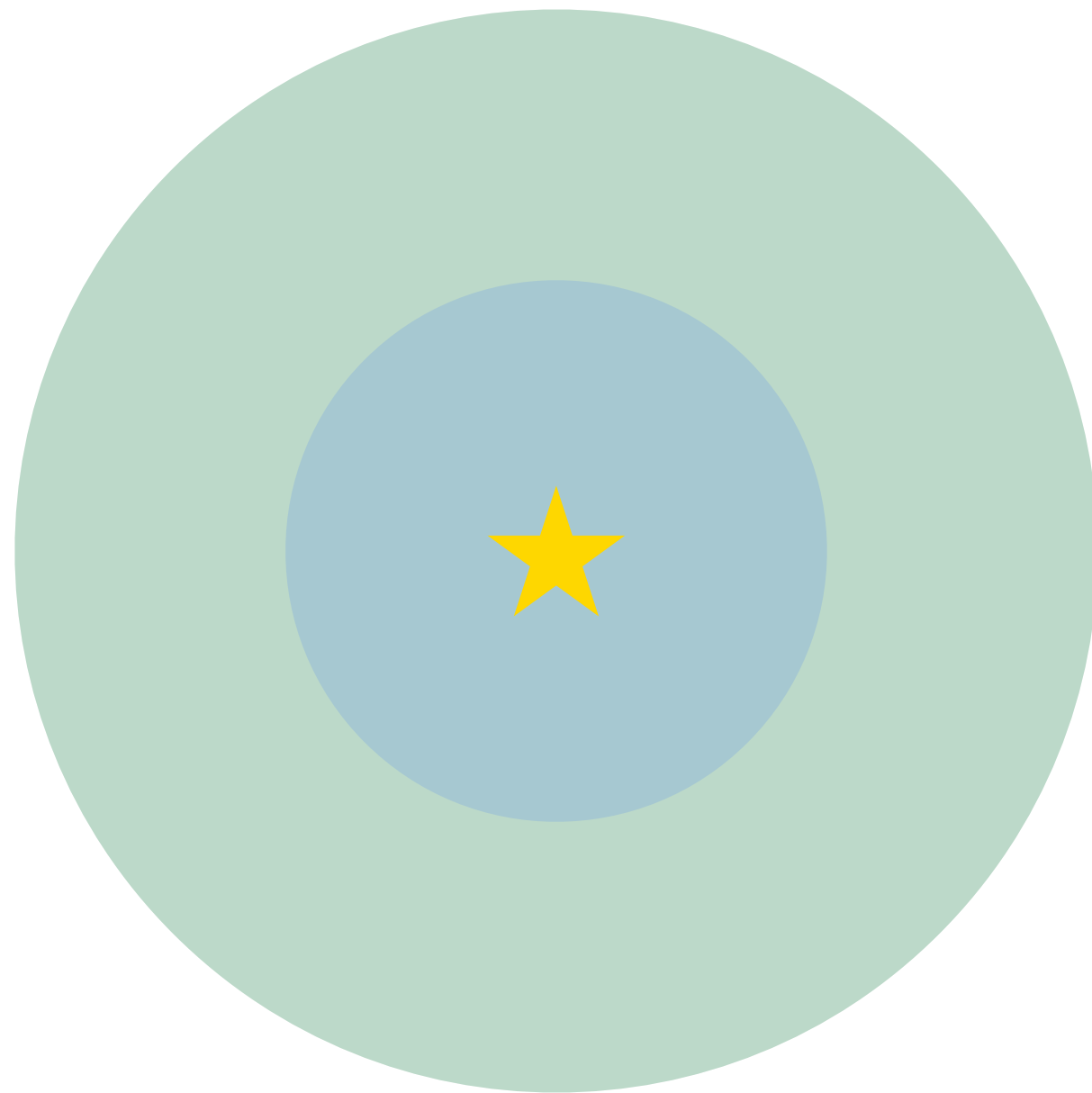
	parameter	units	min	max	steps
	Z	$Z_{\odot}$	0.05	0.40	varies
	log t	yr	6.0	7.6	varies
	log U	dex	-3.5	-1.0	0.25
	log n_e	cm^{-3}	2.0	9.0	1.0
	log N/O	dex	-1.50	1.00	0.5
	log C/O	dex	-1.75	0.25	0.5

New custom grid of Cloudy photoionization models

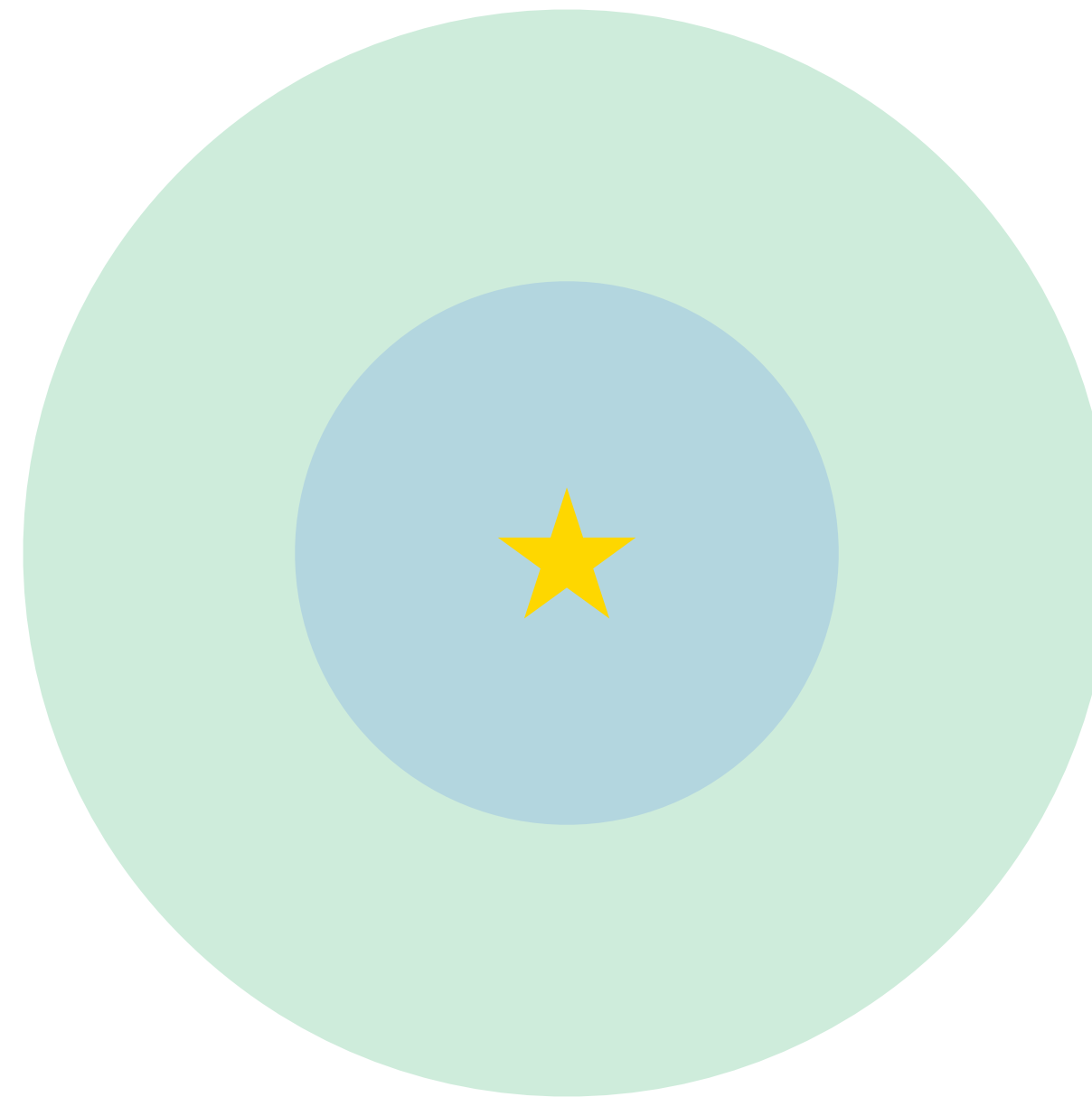


	parameter	units	min	max	steps
	Z	$Z_{\odot}$	0.05	0.40	varies
	log t	yr	6.0	7.6	varies
	log U	dex	-3.5	-1.0	0.25
	log n_e	cm^{-3}	2.0	9.0	1.0
	log N/O	dex	-1.50	1.00	0.5
	log C/O	dex	-1.75	0.25	0.5

uniform-density
nebula

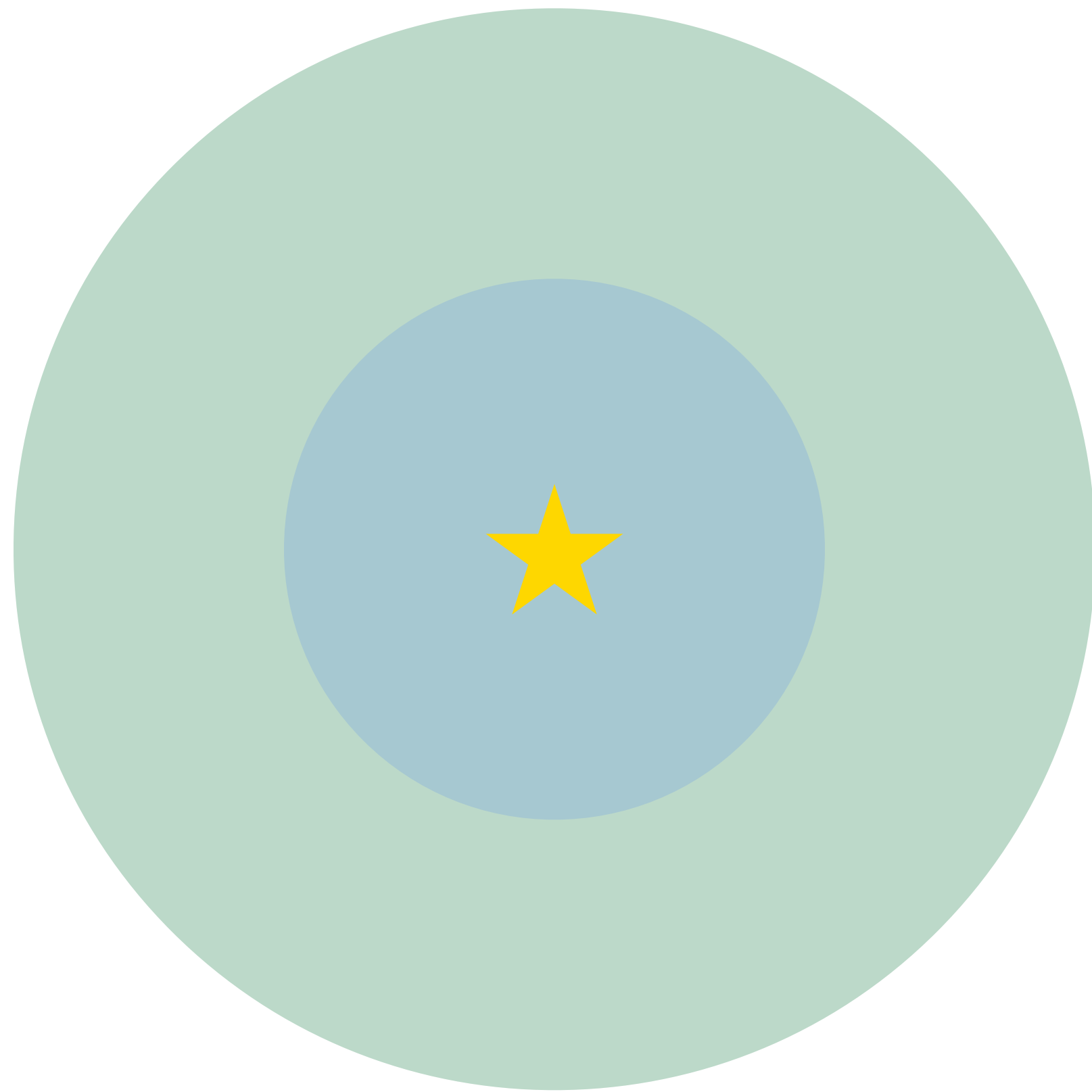


stratified-density
nebula

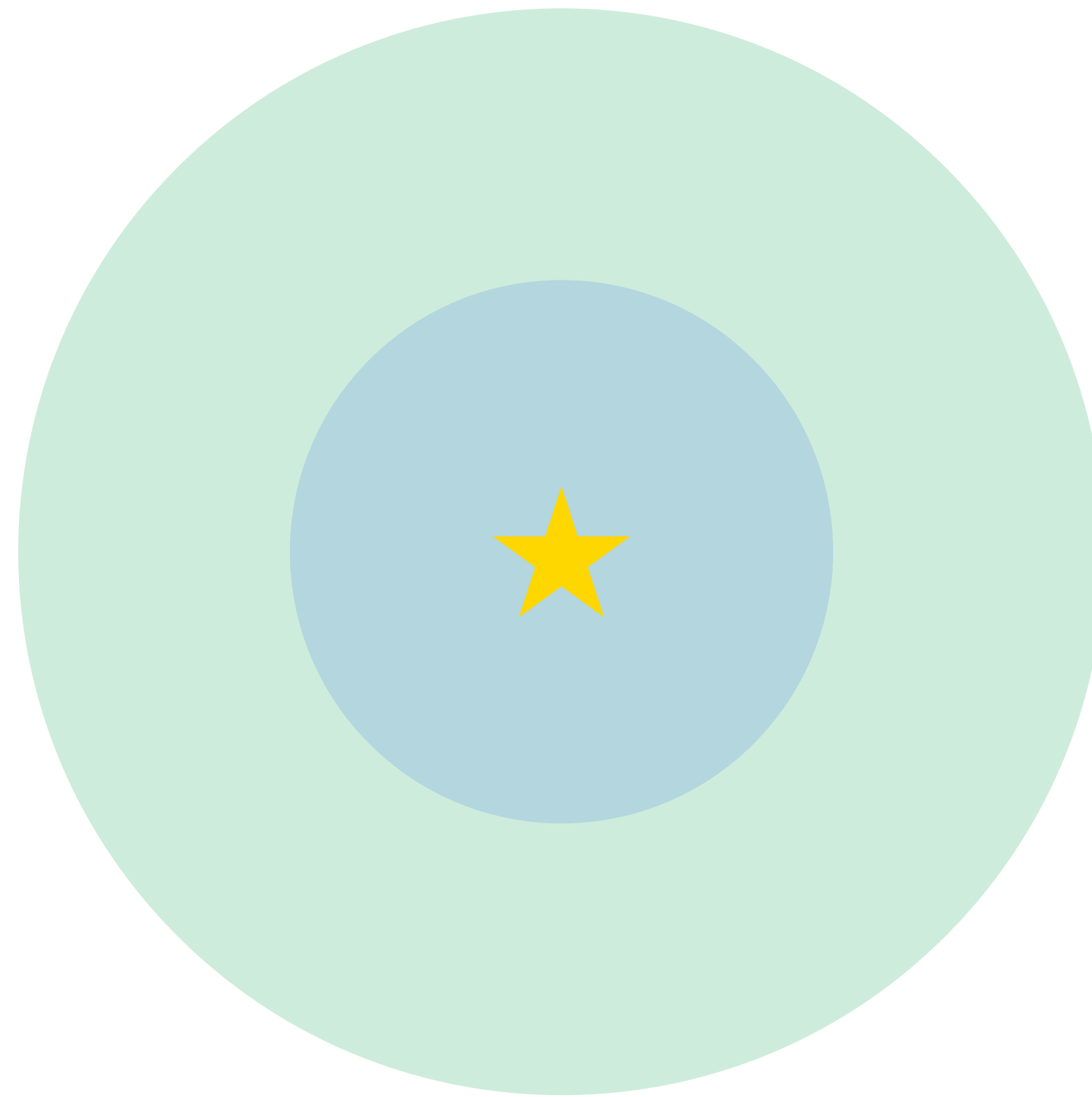


**We examine our
models in two
different contexts**

uniform-density
nebula

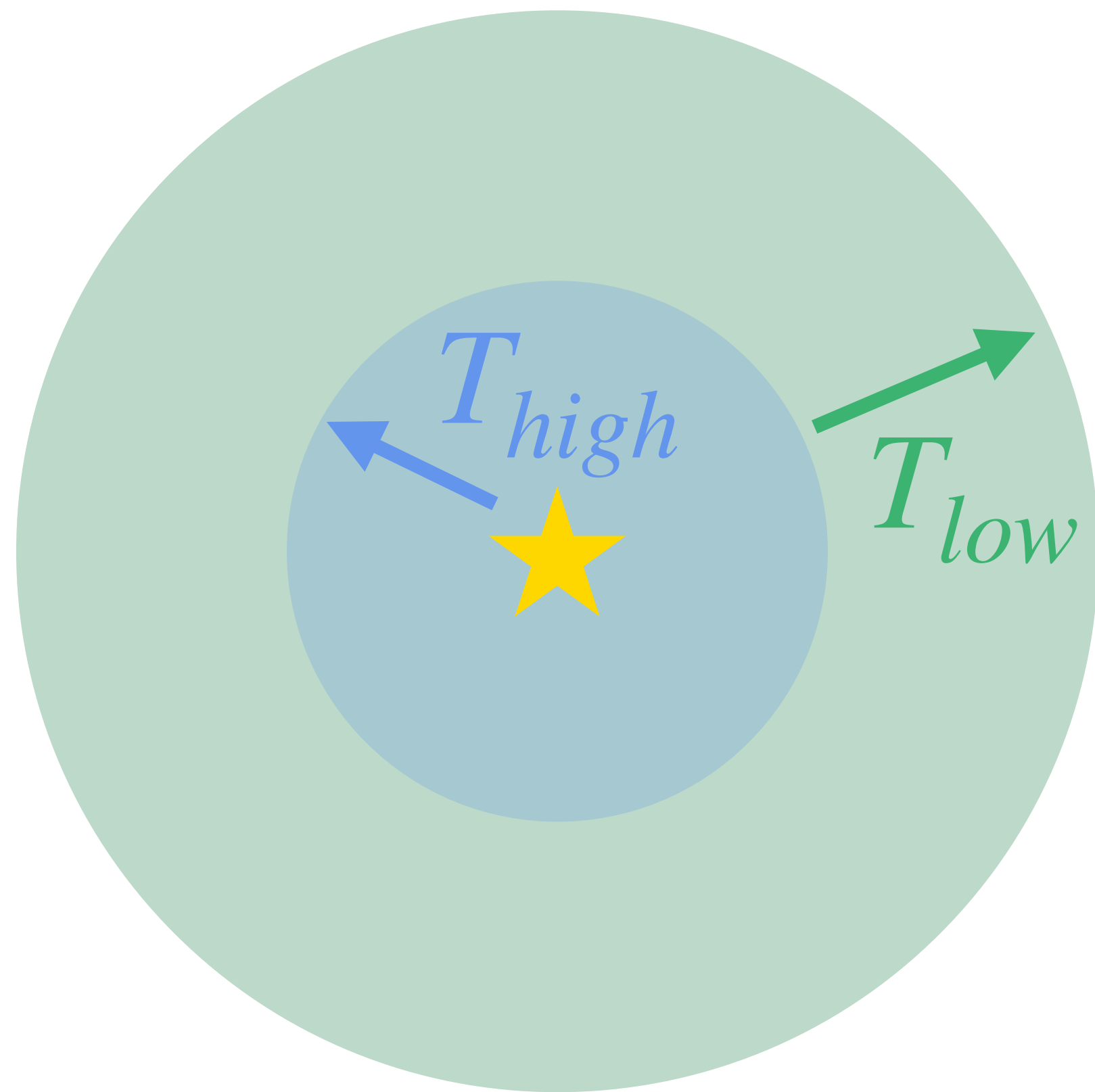


stratified-density
nebula

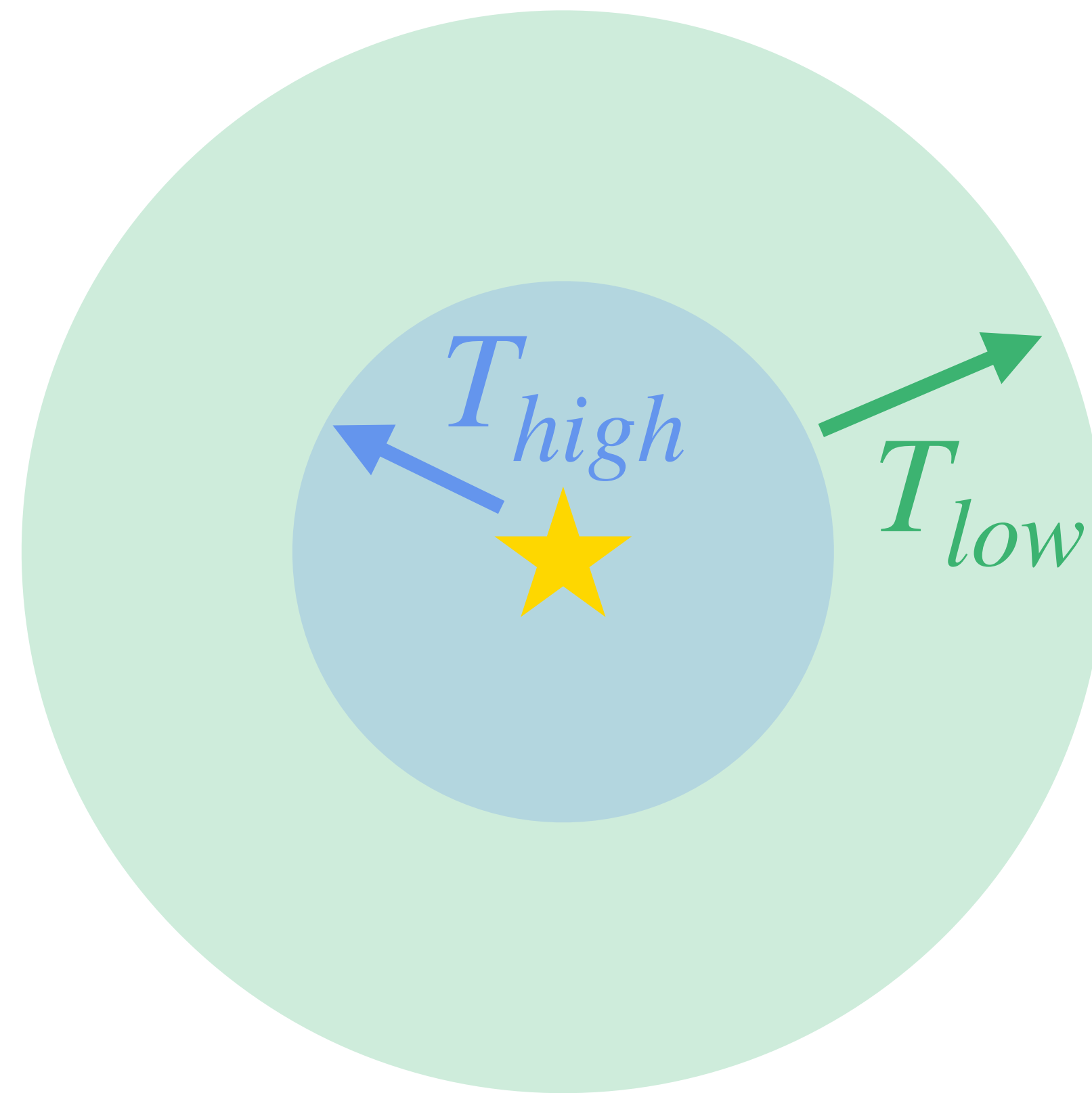


**We examine
our models
in two
different
contexts**

uniform-density
nebula

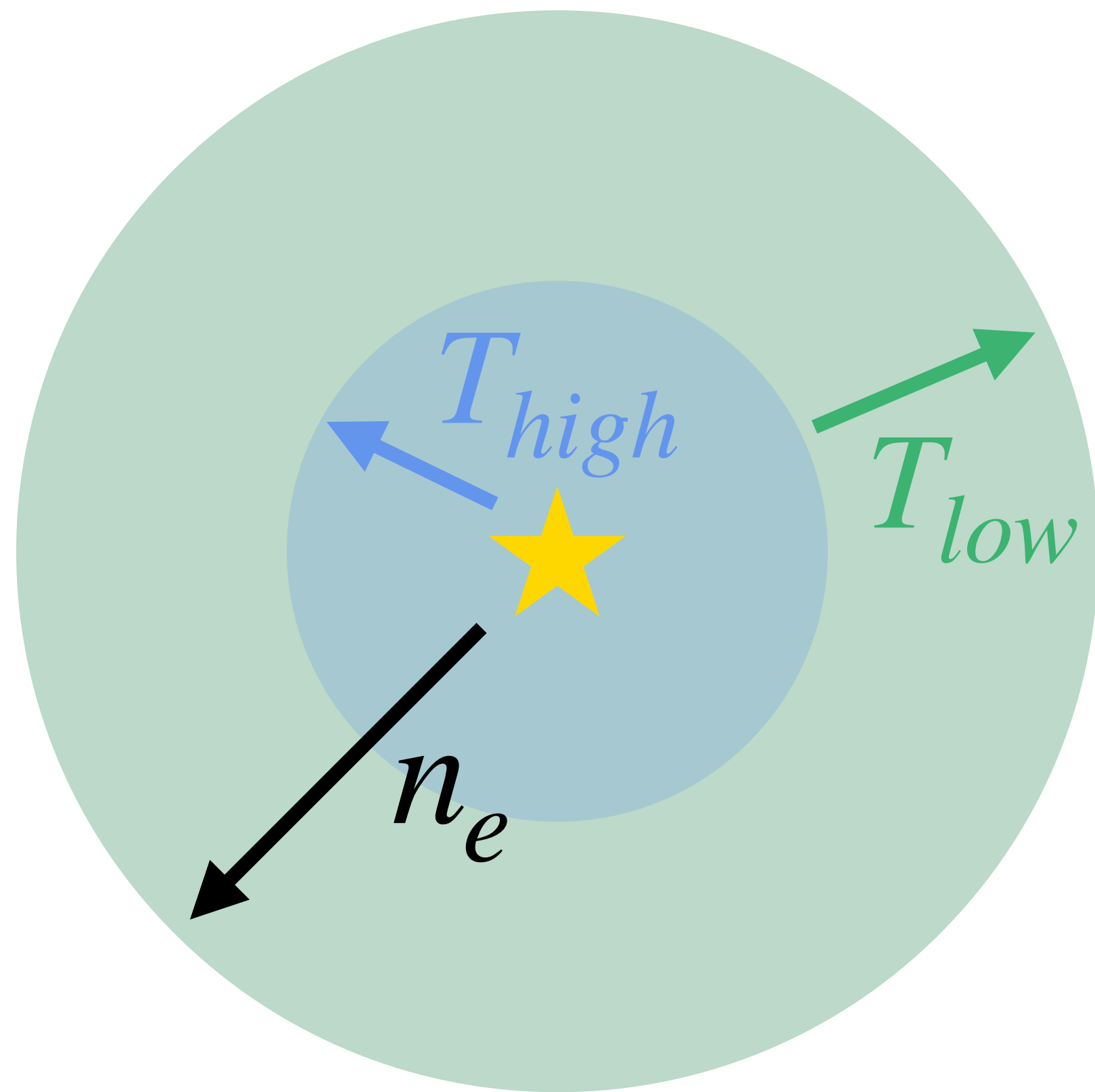


stratified-density
nebula

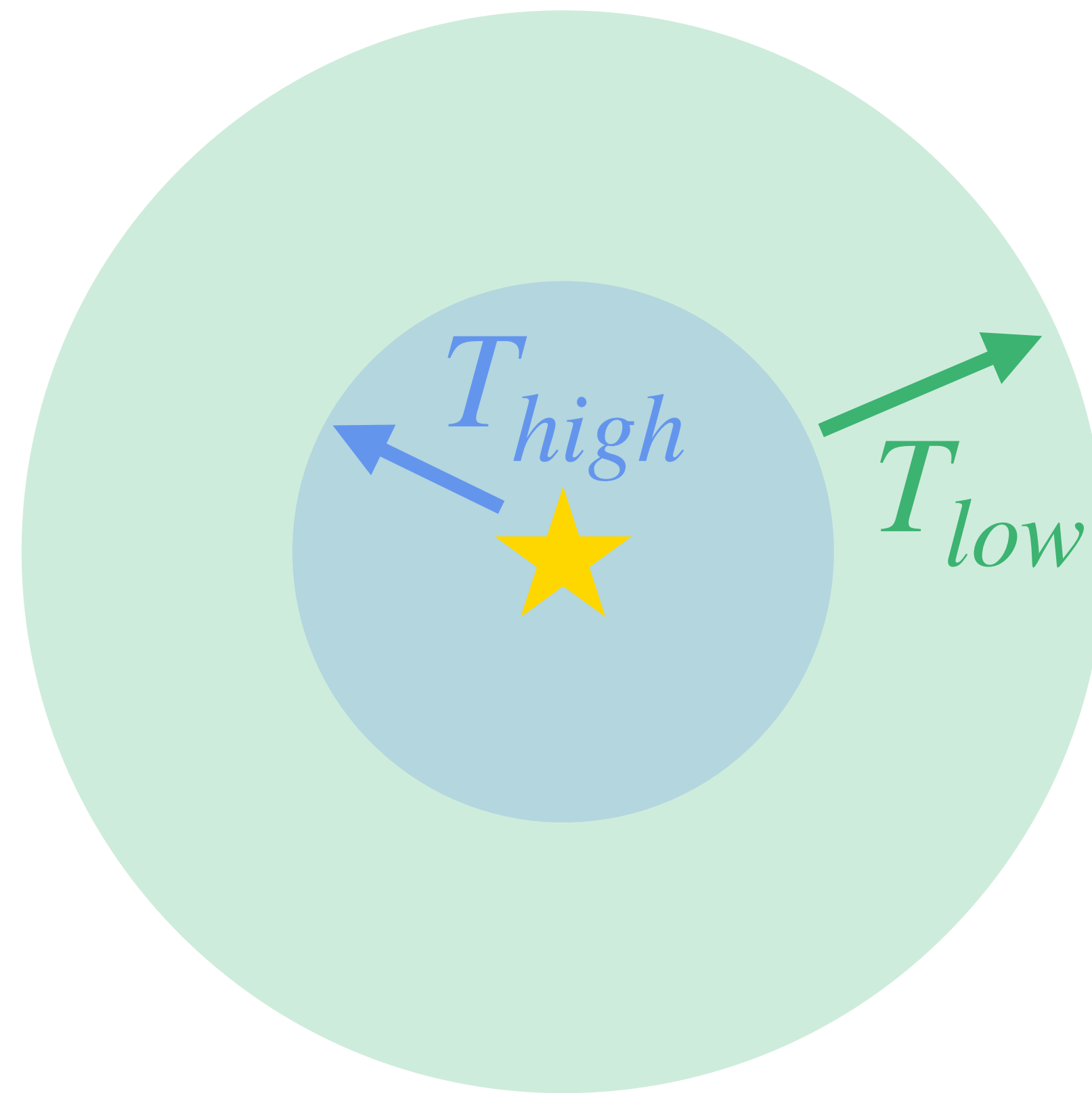


We examine
our models
in two
different
contexts

uniform-density
nebula

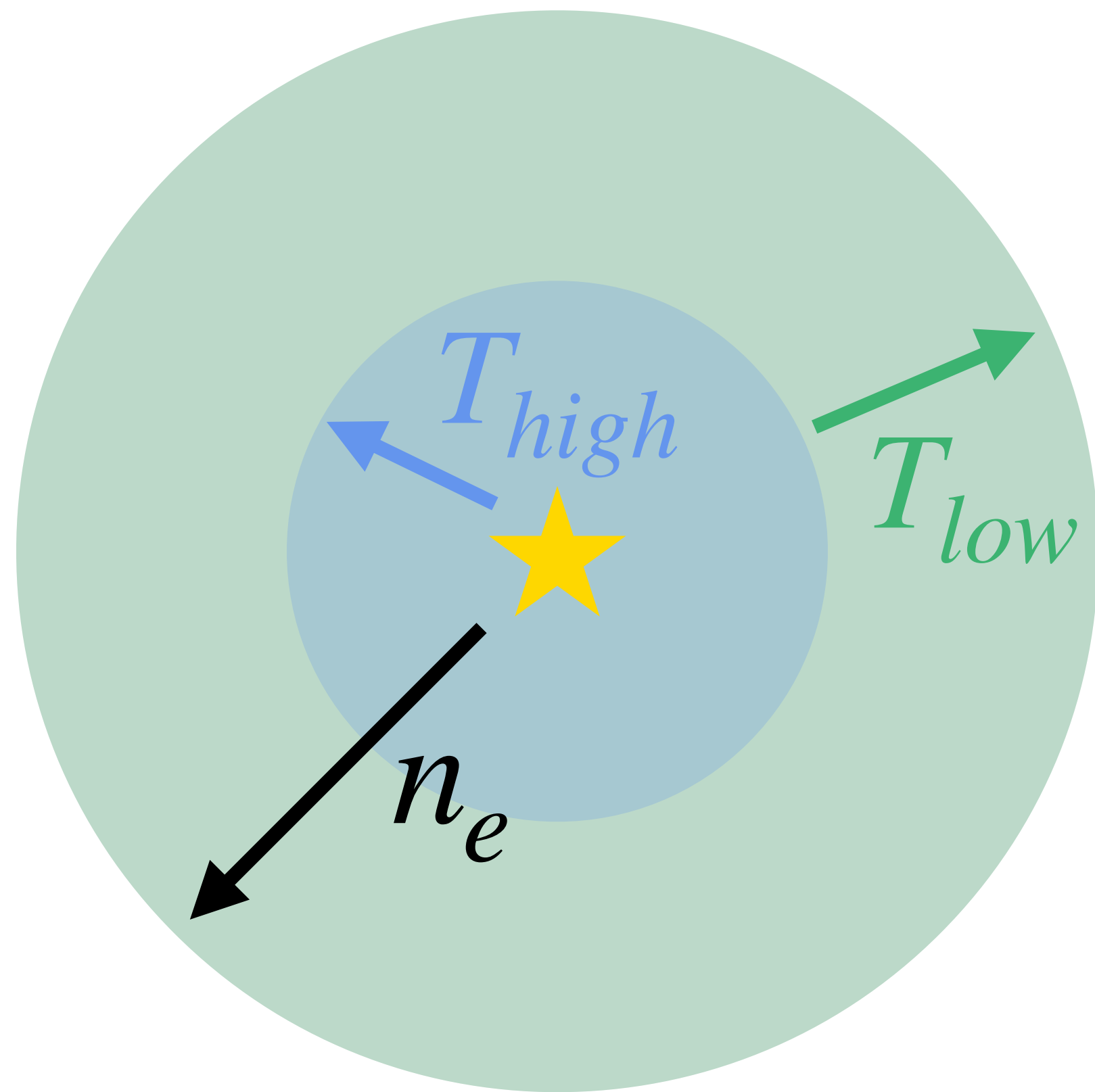


stratified-density
nebula

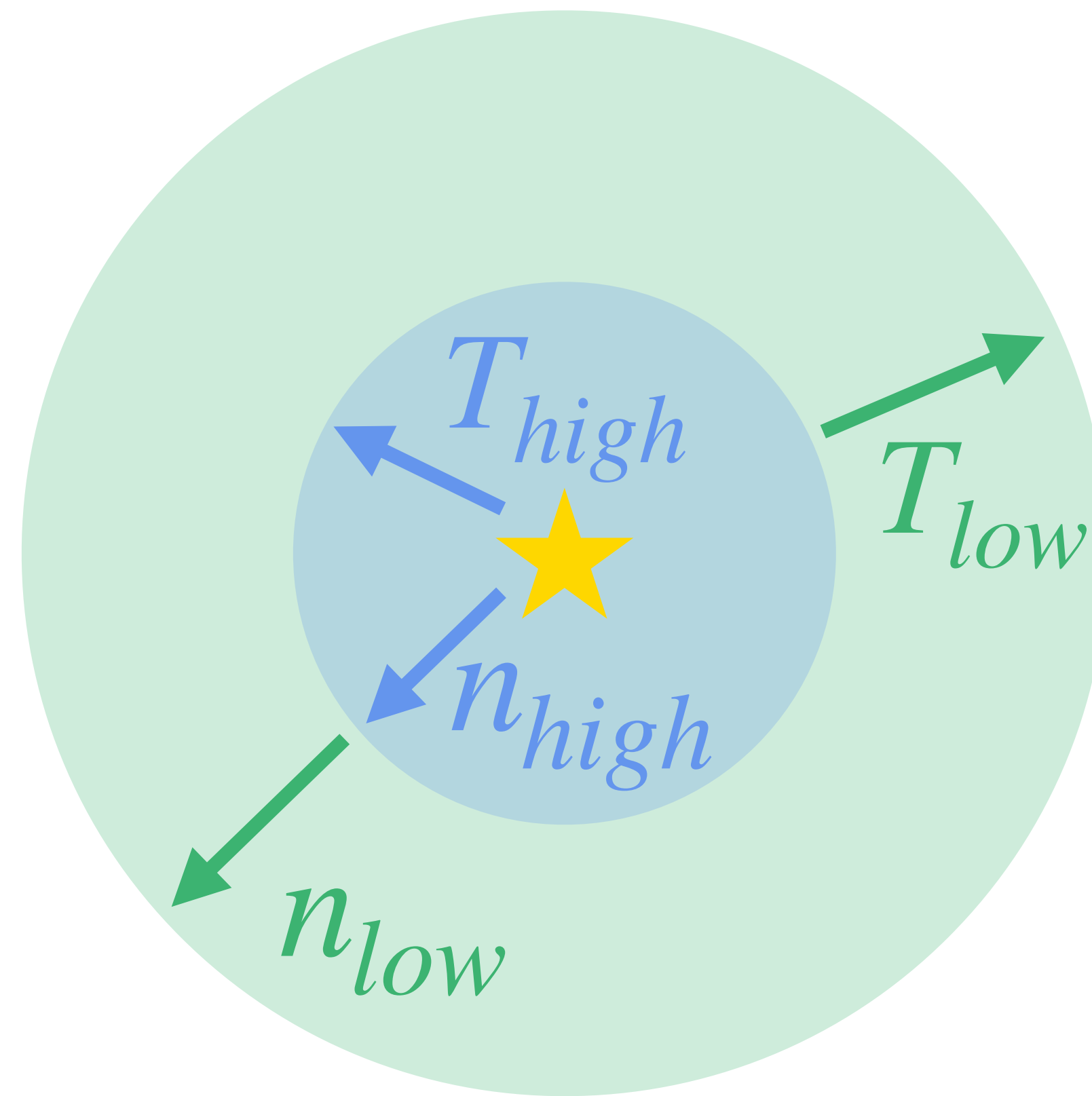


We examine
our models
in two
different
contexts

uniform-density
nebula

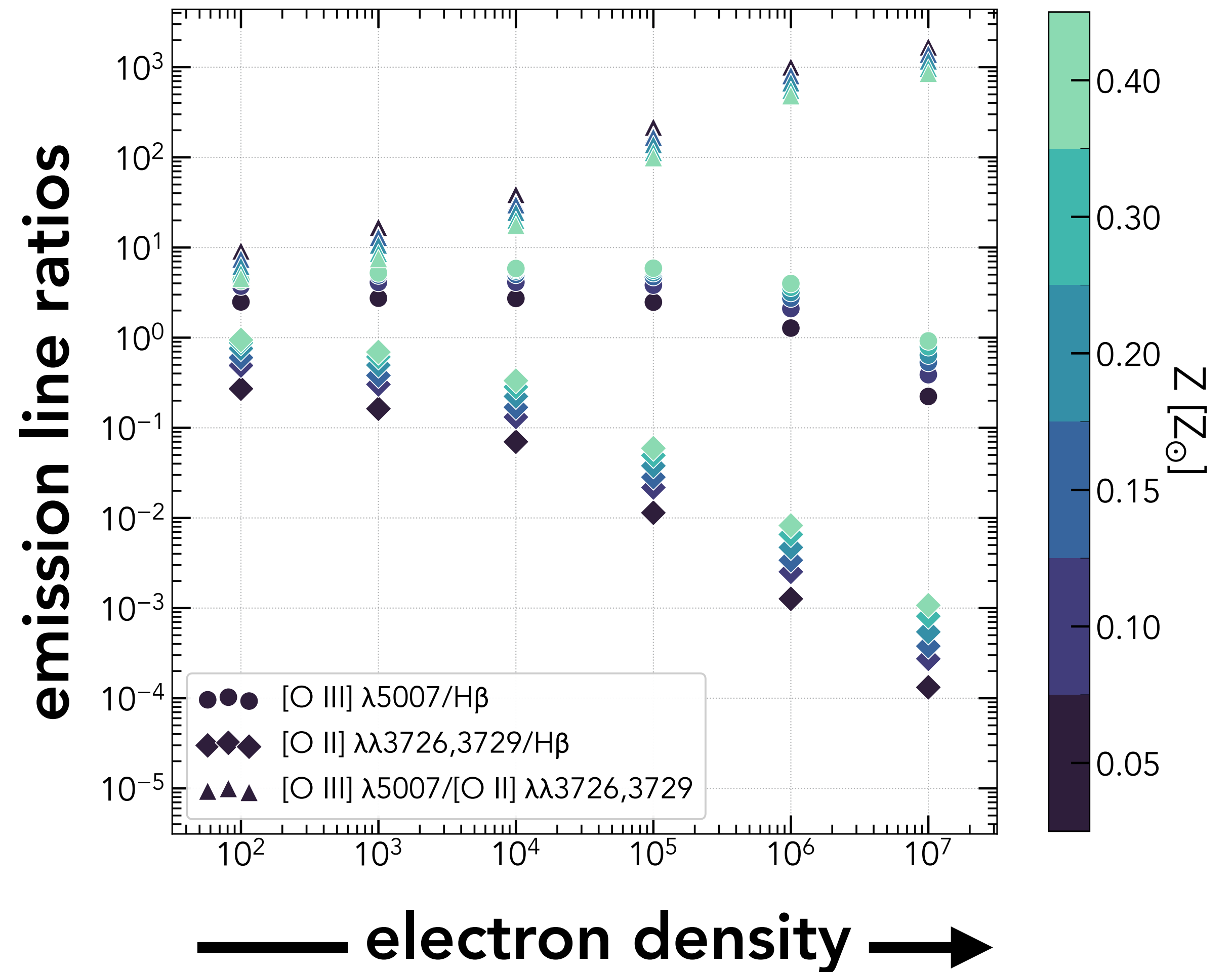


stratified-density
nebula

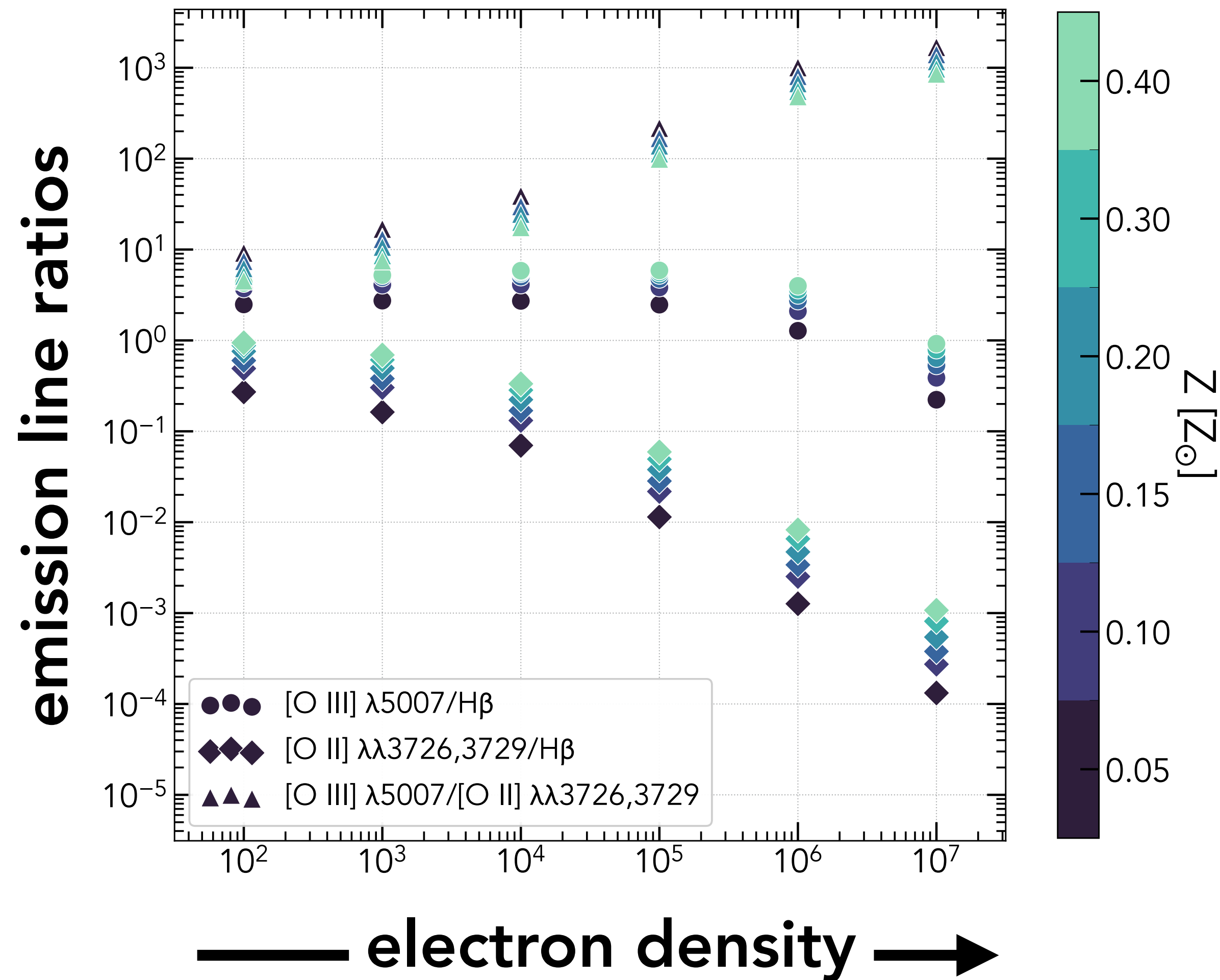


We examine
our models
in two
different
contexts

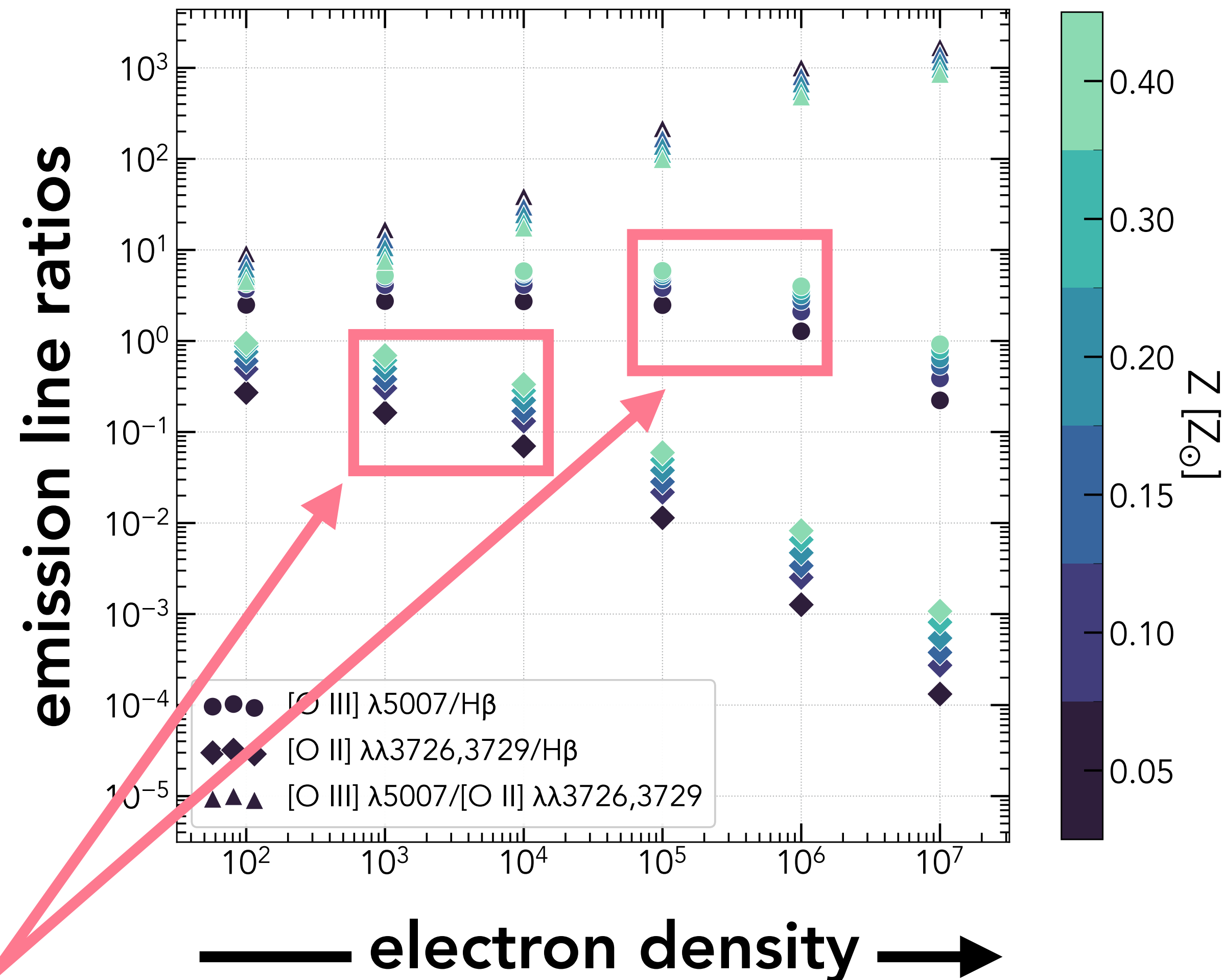
Crucial emission
line ratios are
extremely
sensitive to
density



Crucial
emission line
ratios are
extremely
sensitive to
density

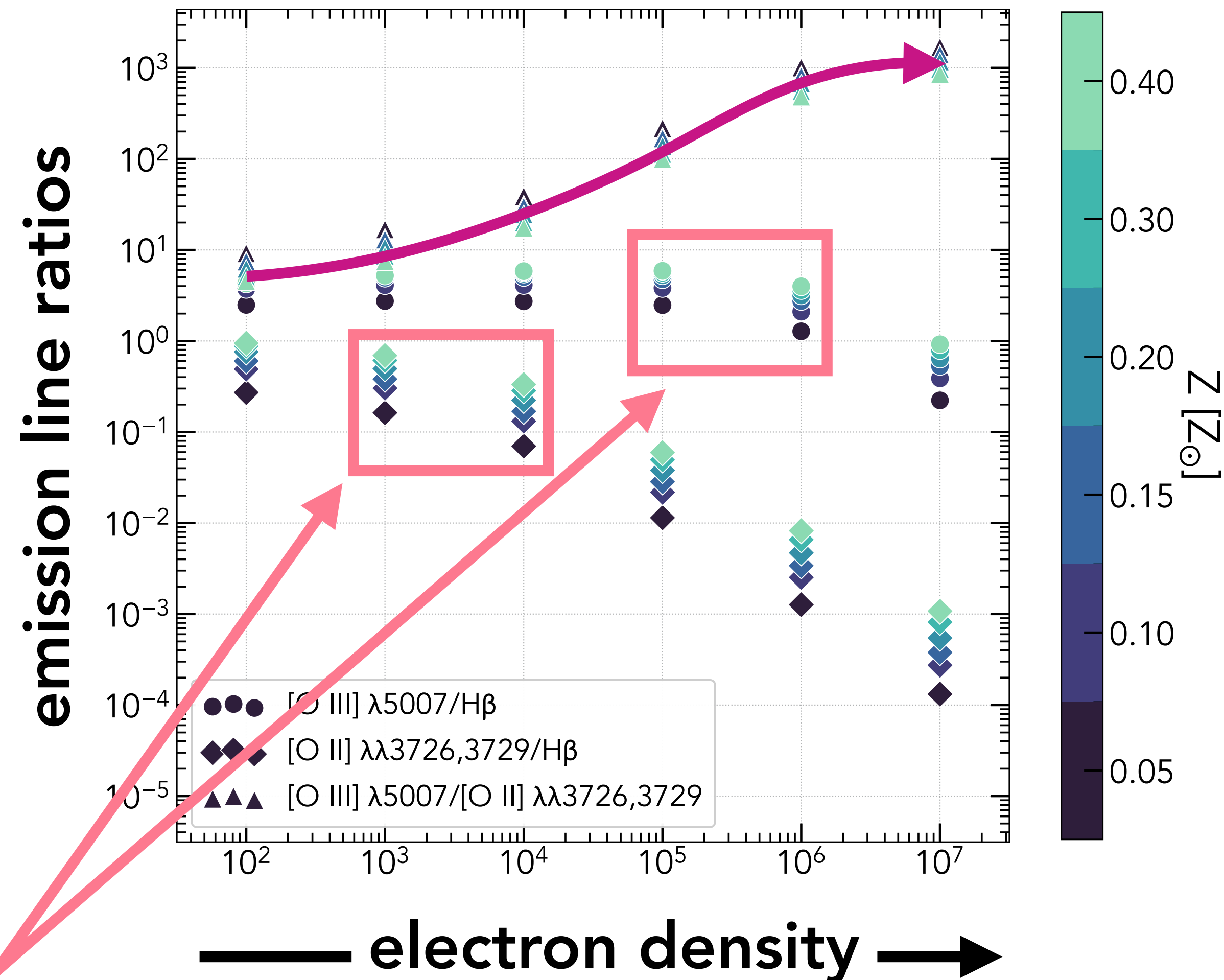


Crucial
emission line
ratios are
extremely
sensitive to
density



the emission line ratios
decrease as collisional de-
excitation begins to occur

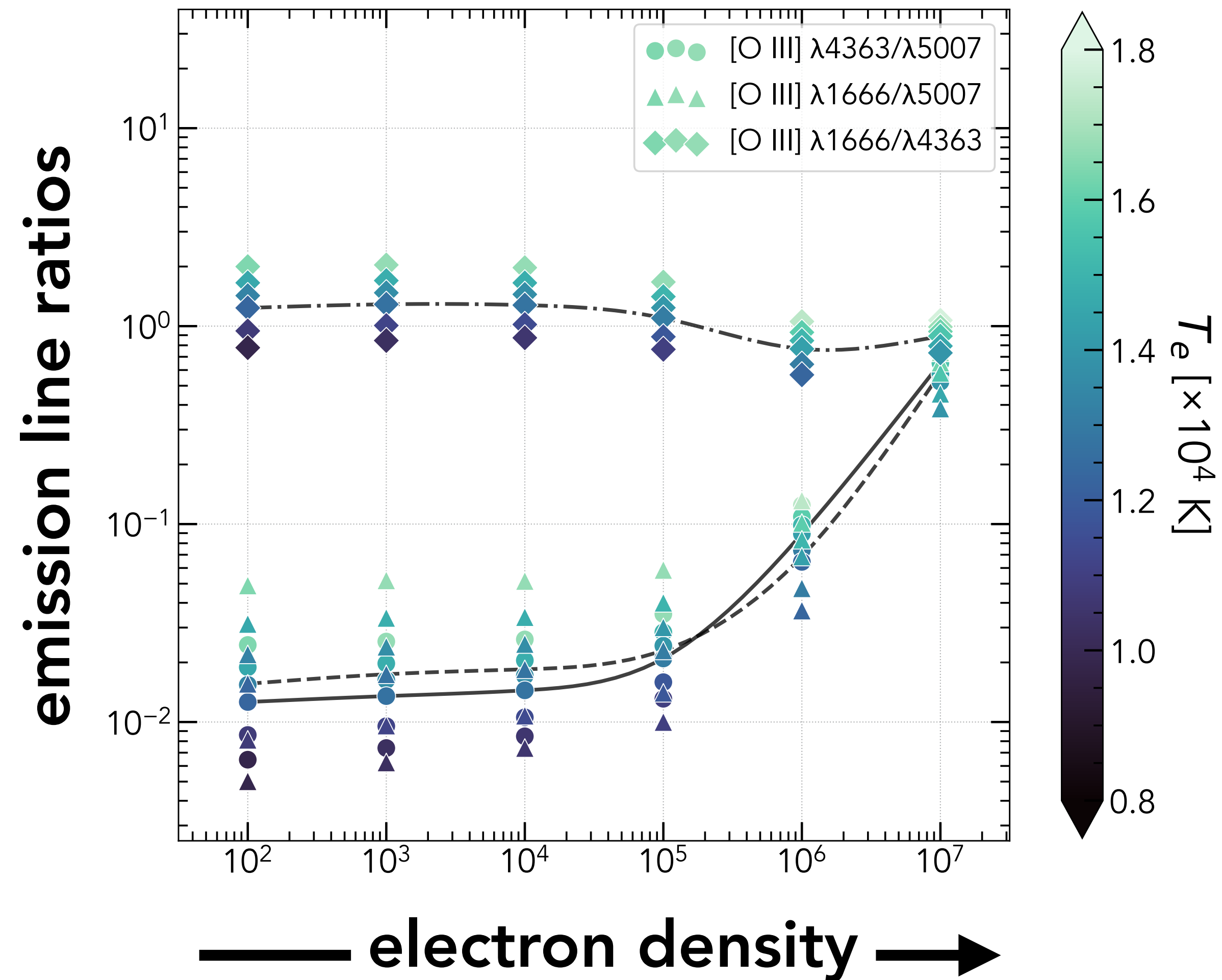
Crucial
emission line
ratios are
extremely
sensitive to
density



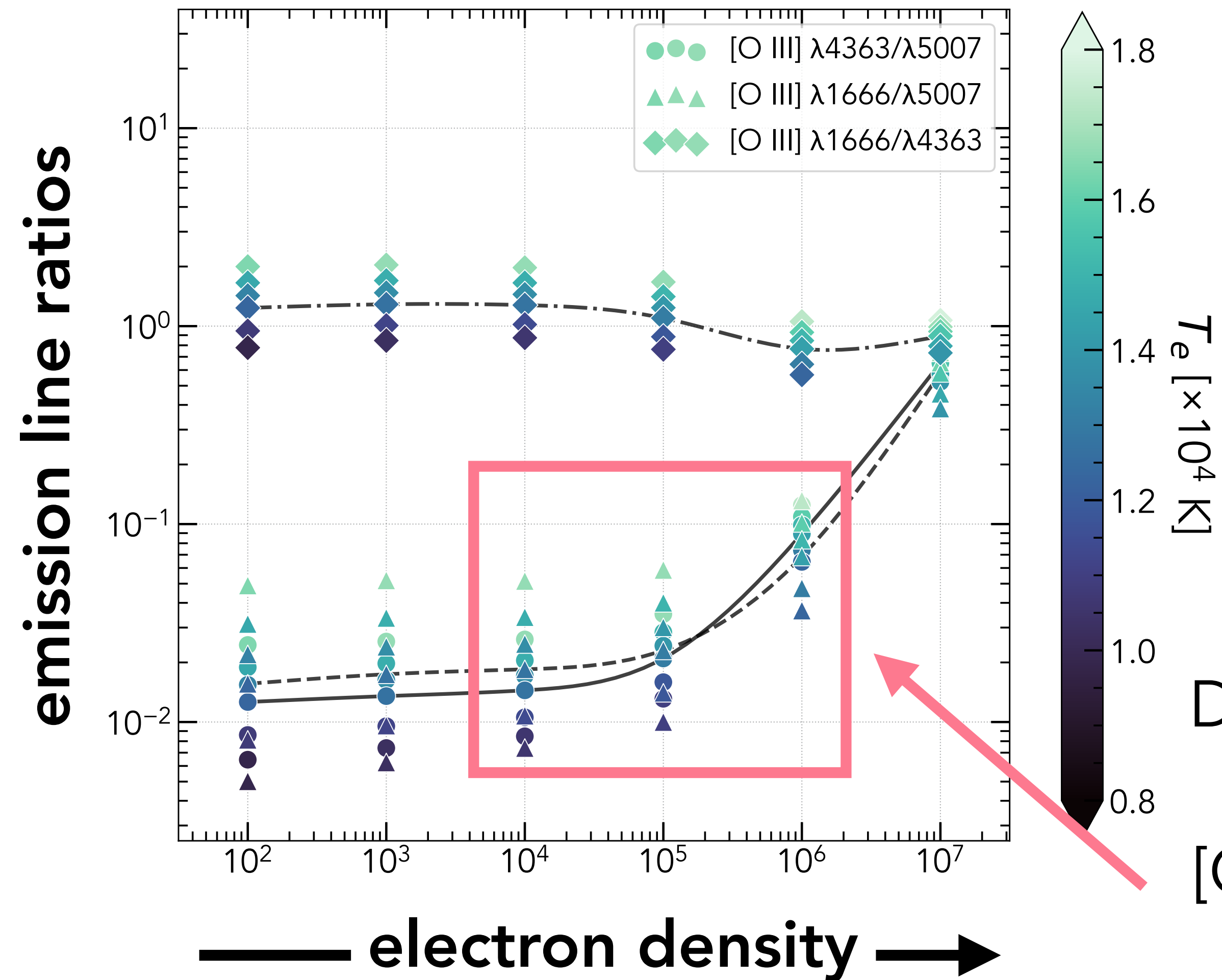
O_{32} ratios
increases
with
increasing
density

the emission line ratios
decrease as collisional de-
excitation begins to occur

Crucial
emission line
ratios are
extremely
sensitive to
density



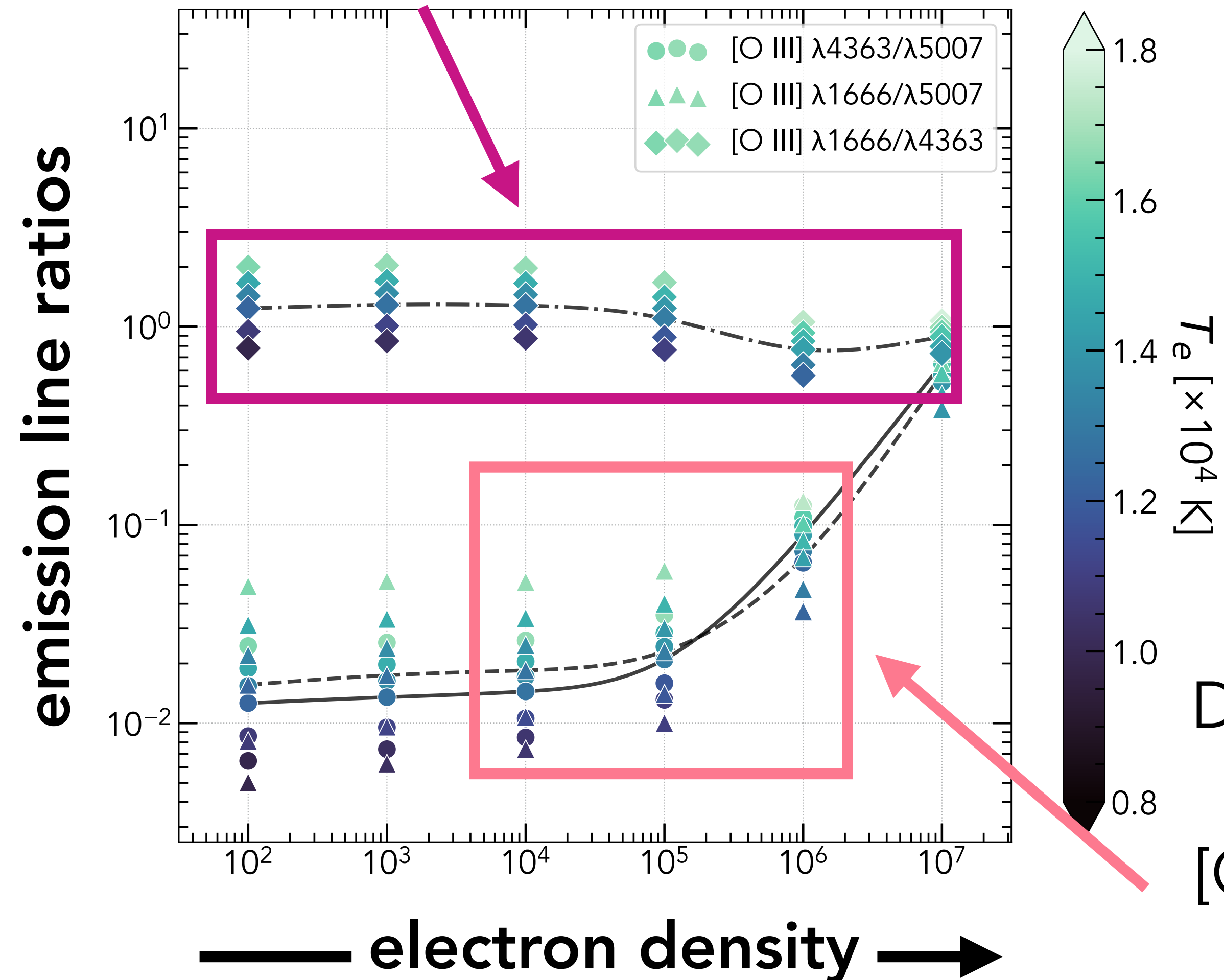
Crucial
emission line
ratios are
extremely
sensitive to
density



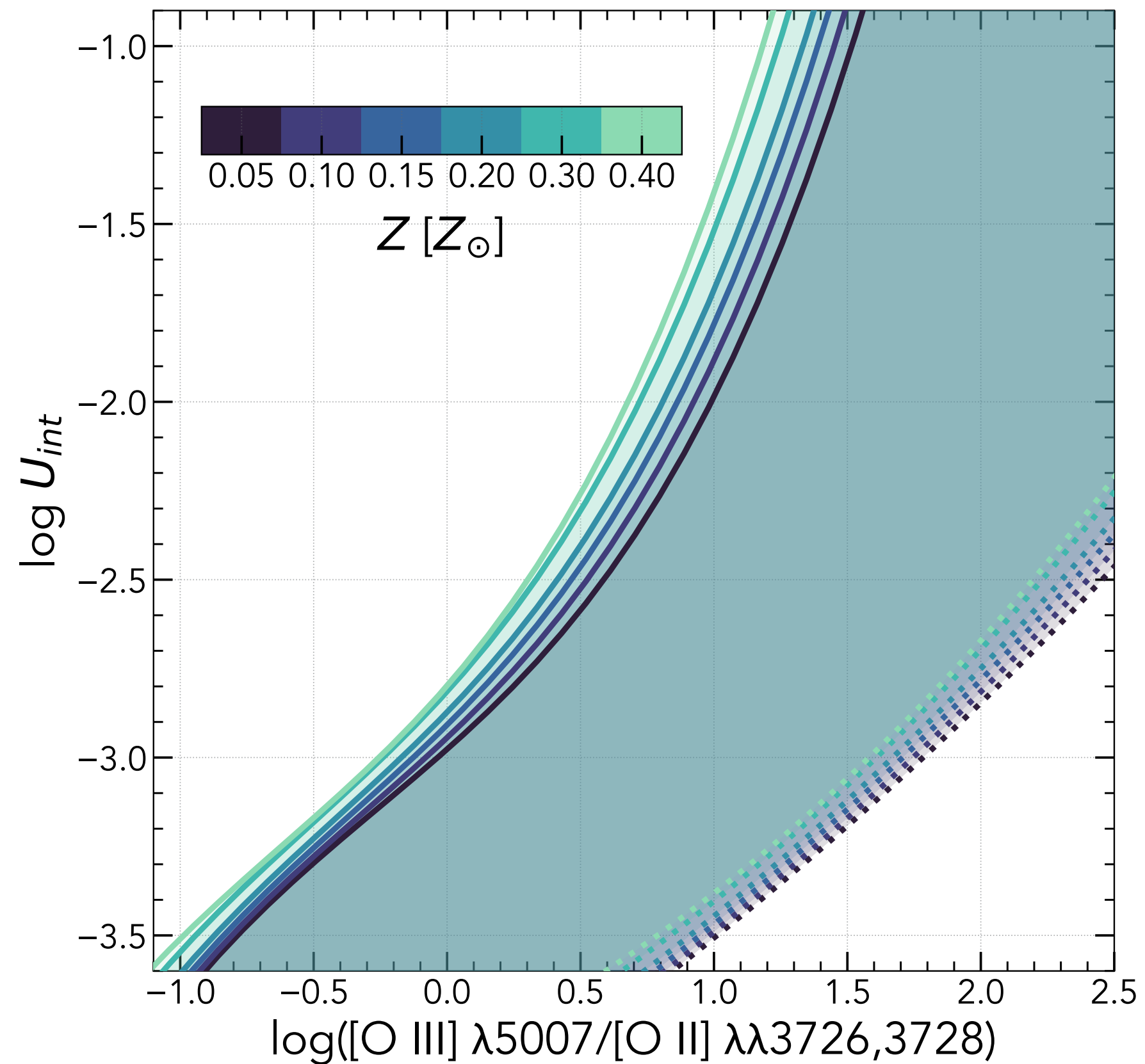
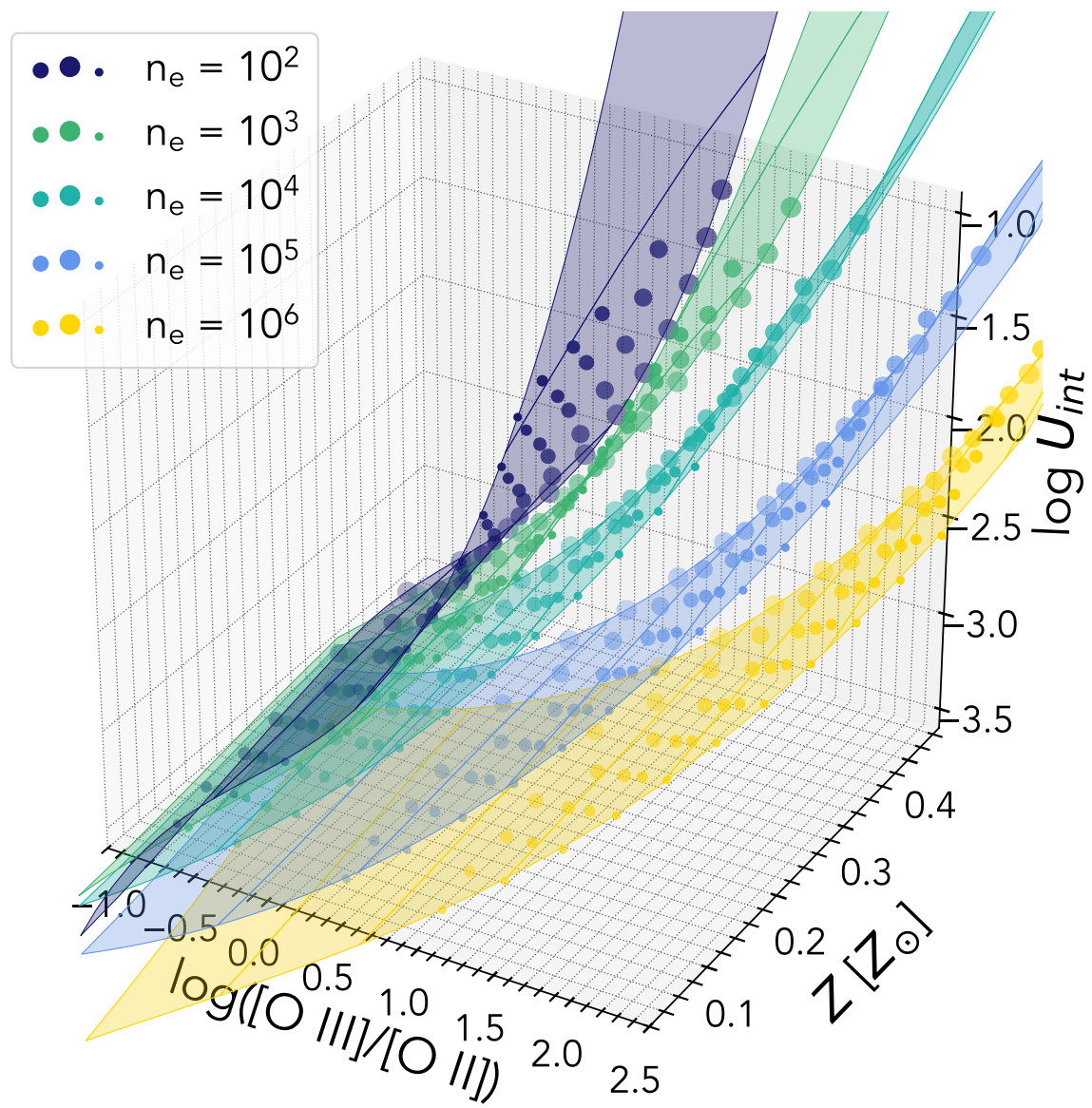
Diagnostics that
include
 $[O\ III]\ \lambda 5007$ are
begin to **vary**
with density

Crucial
emission line
ratios are
extremely
sensitive to
density

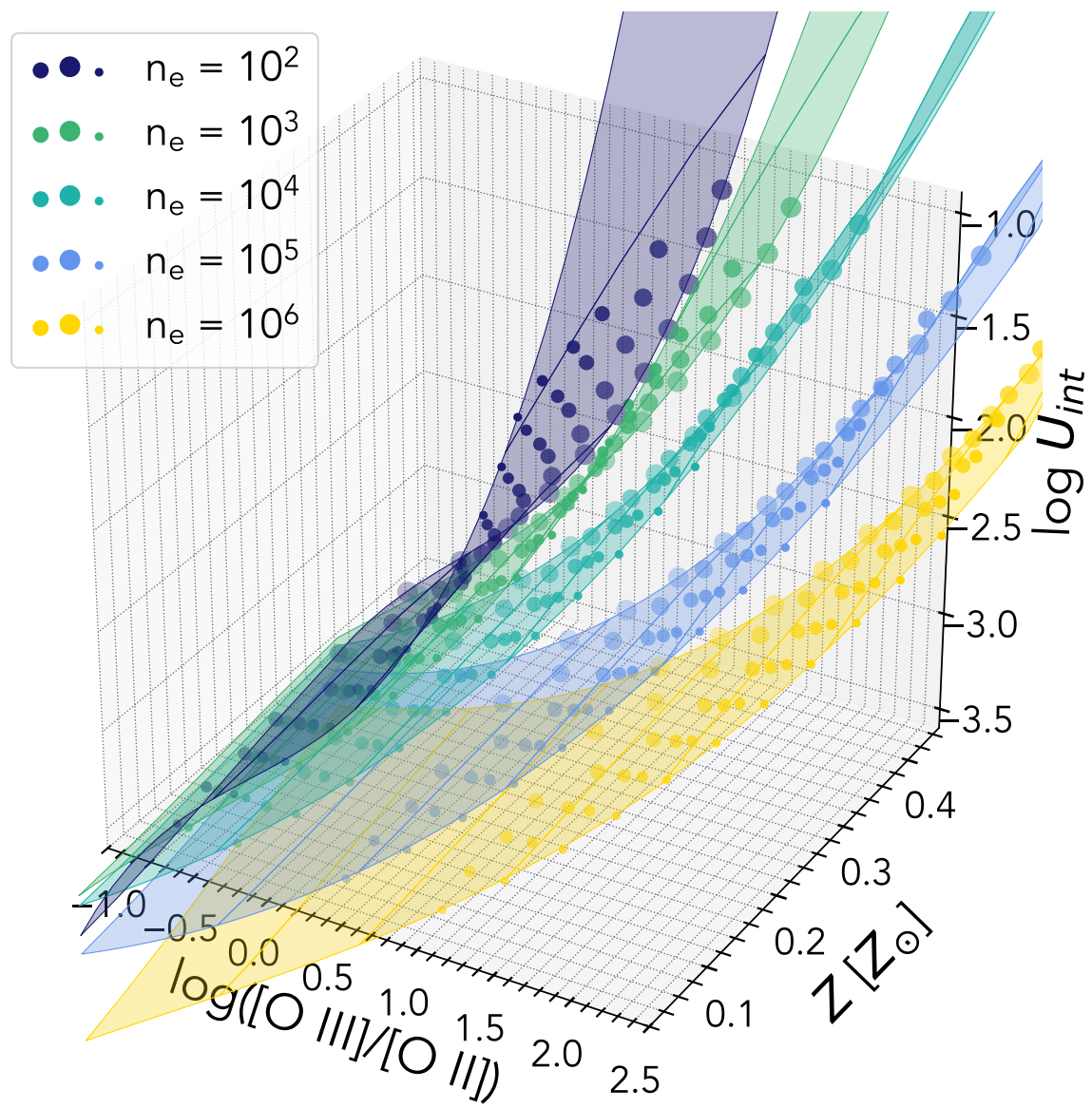
[O III] $\lambda 4363$ and [O III] $\lambda 1666$ have much higher critical
densities making them **more invariant to density**



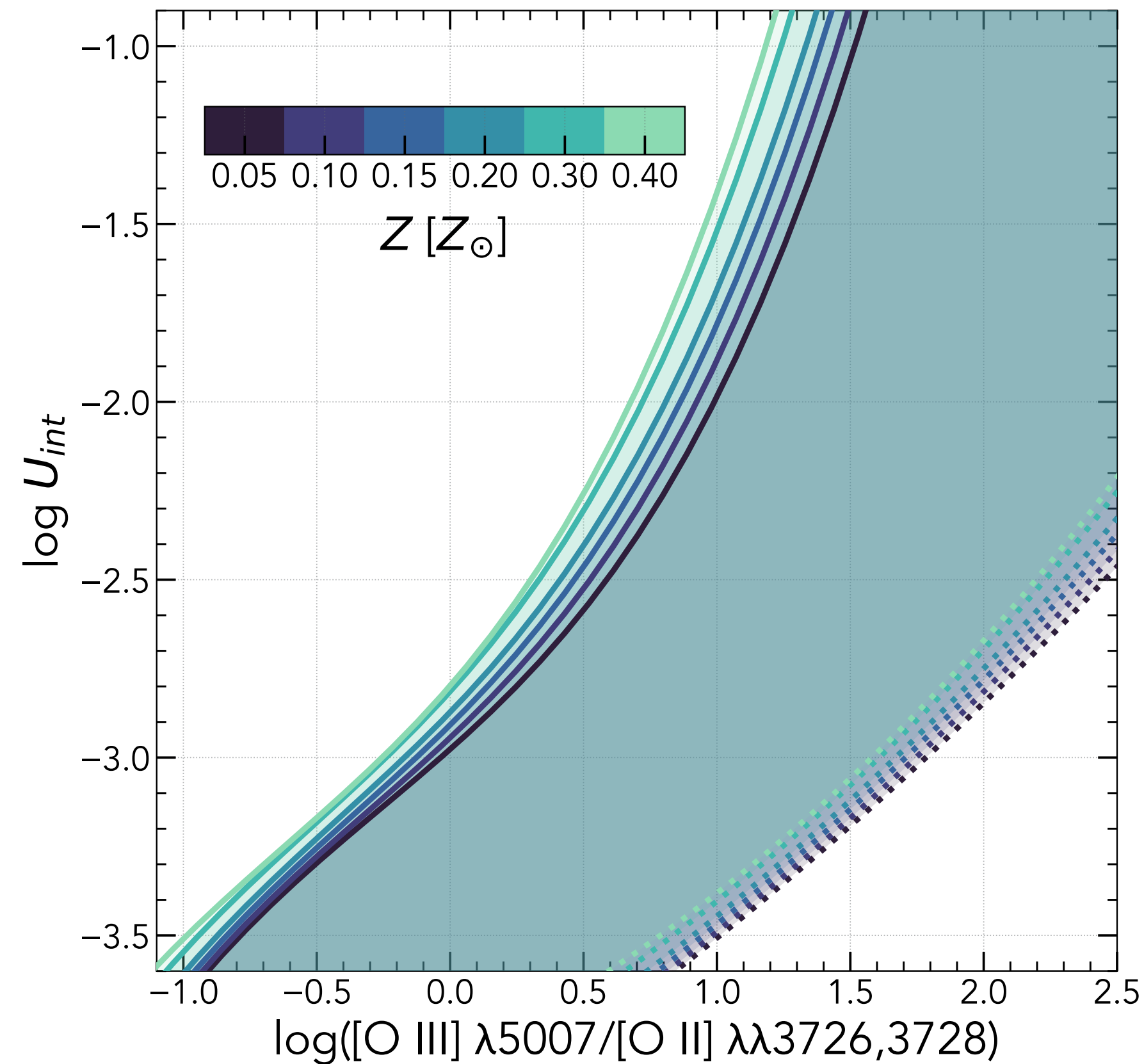
Diagnostics that
include
[O III] $\lambda 5007$ are
begin to **vary**
with density



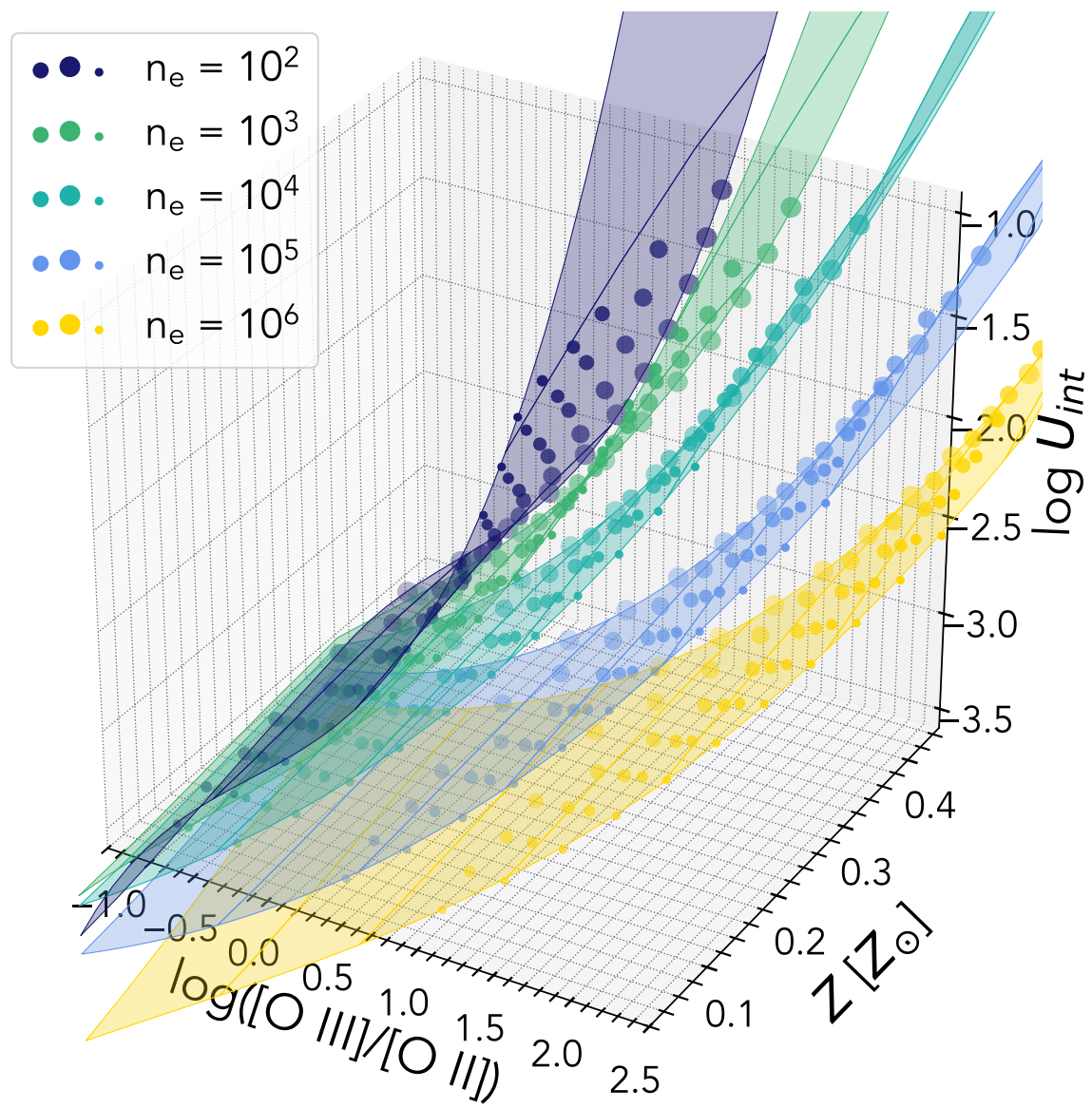
Classical ionization
parameter
diagnostics
drastically over
predict $\log U_{int}$
with increasing
density



Bicubic surface fits that take O_{32} and Z to return $\log U_{int}$

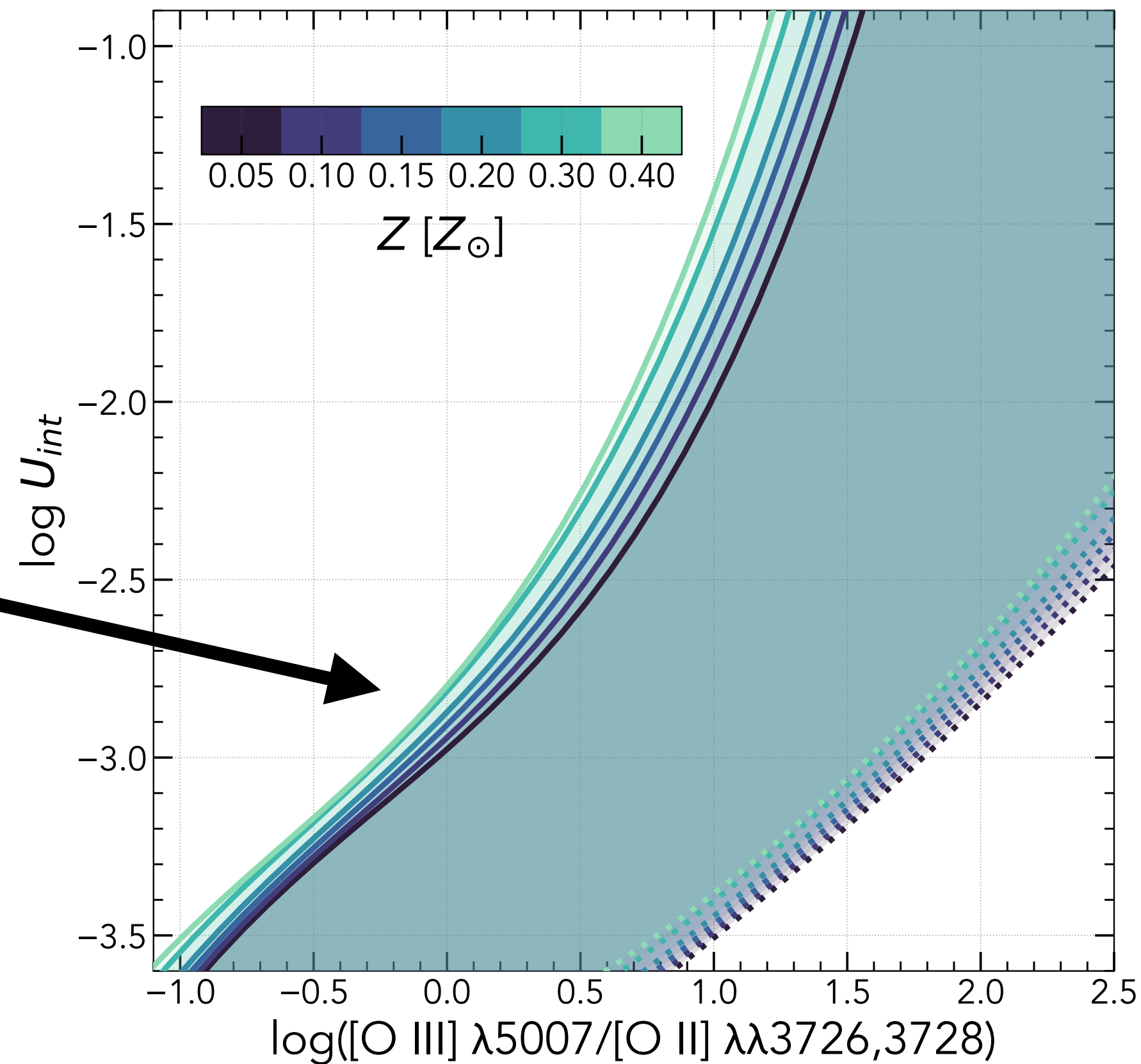


Classical
ionization
parameter
diagnostics
drastically
over predict
 $\log U_{int}$ with
increasing
density

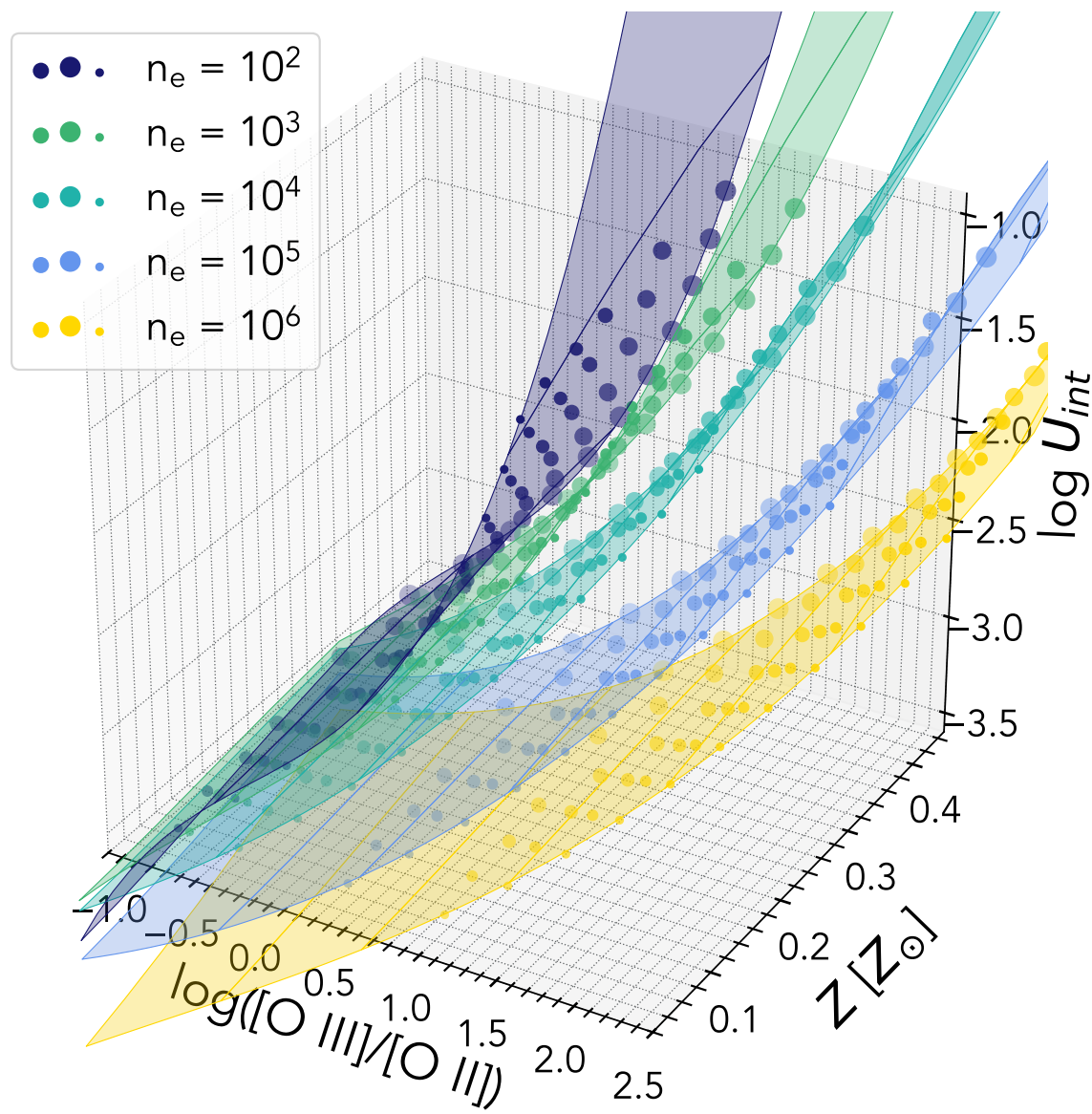


Bicubic surface fits that take O_{32} and Z to return $\log U_{int}$

Low
density
case

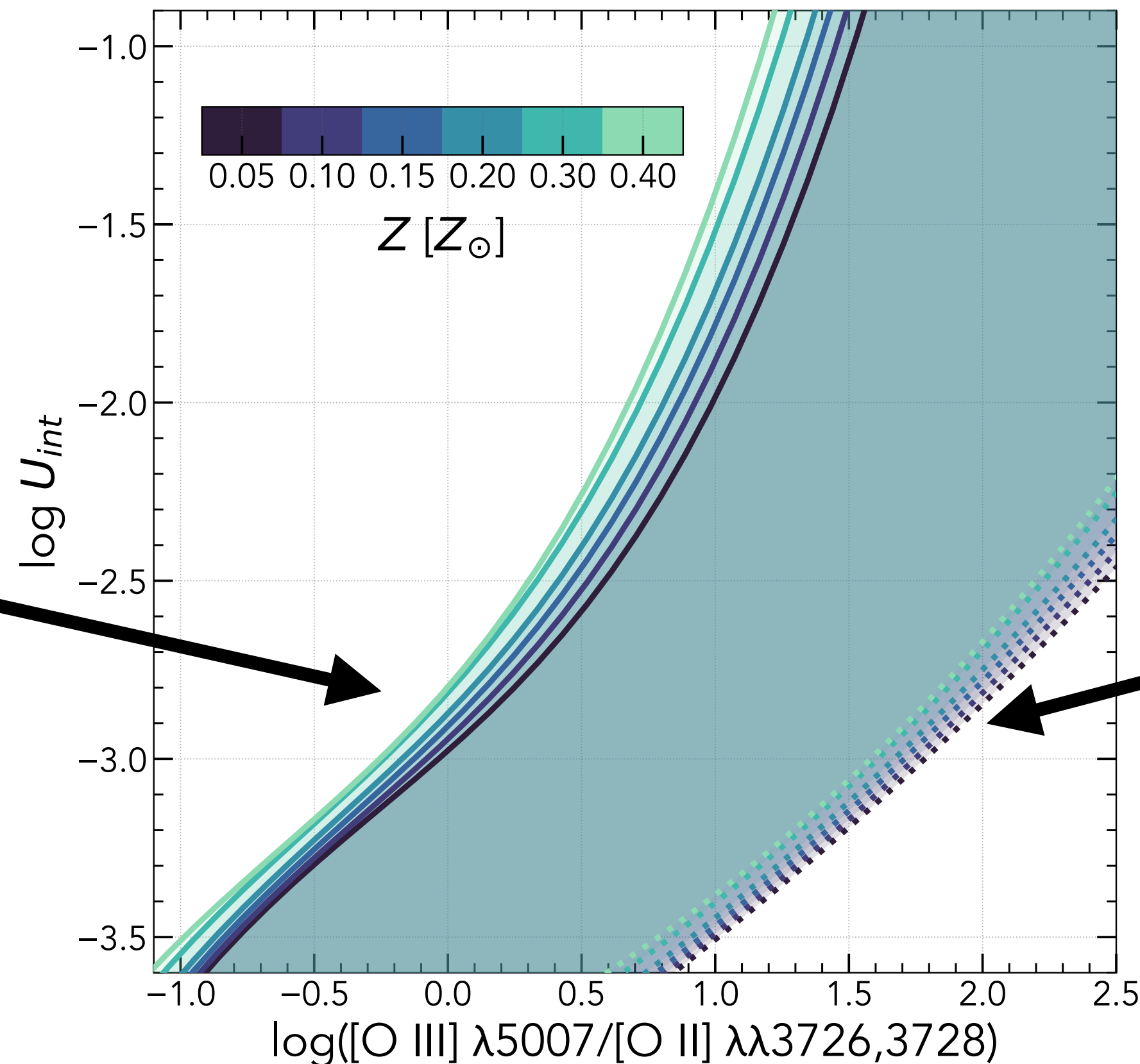


Classical
ionization
parameter
diagnostics
drastically
over predict
 $\log U_{int}$ with
increasing
density



Bicubic surface fits that take O_{32} and Z to return $\log U_{int}$

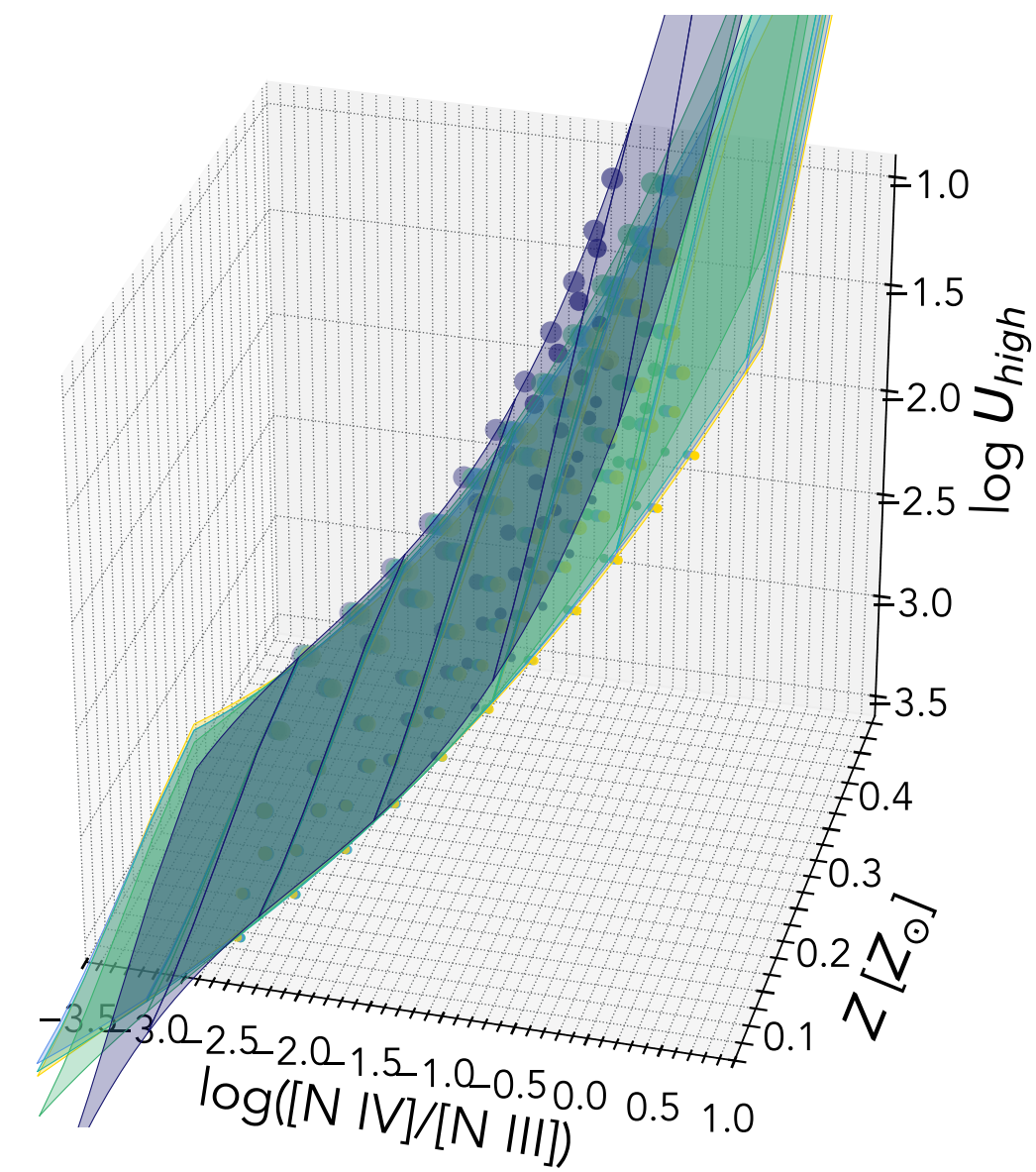
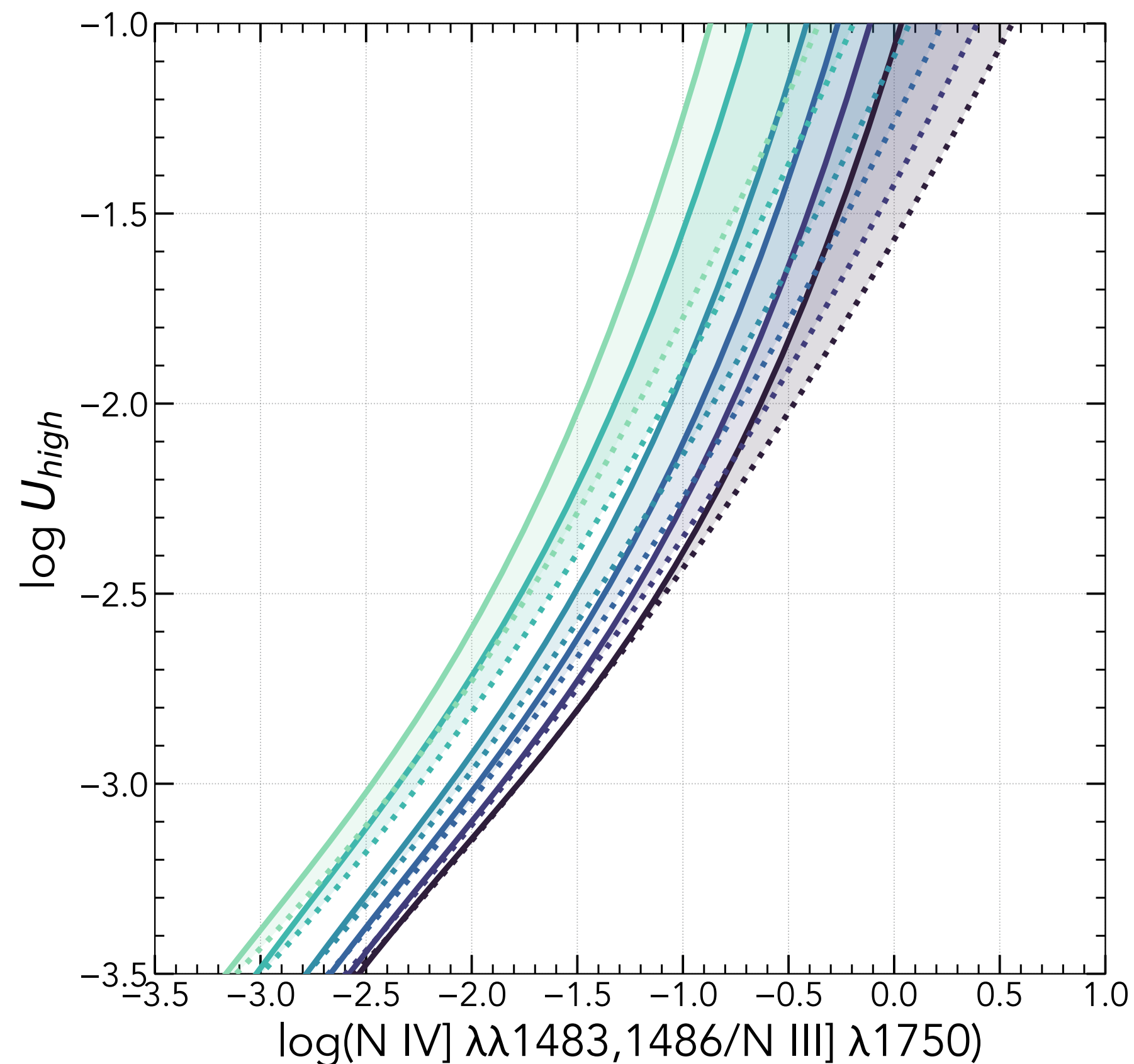
Low
density
case



High
density
case

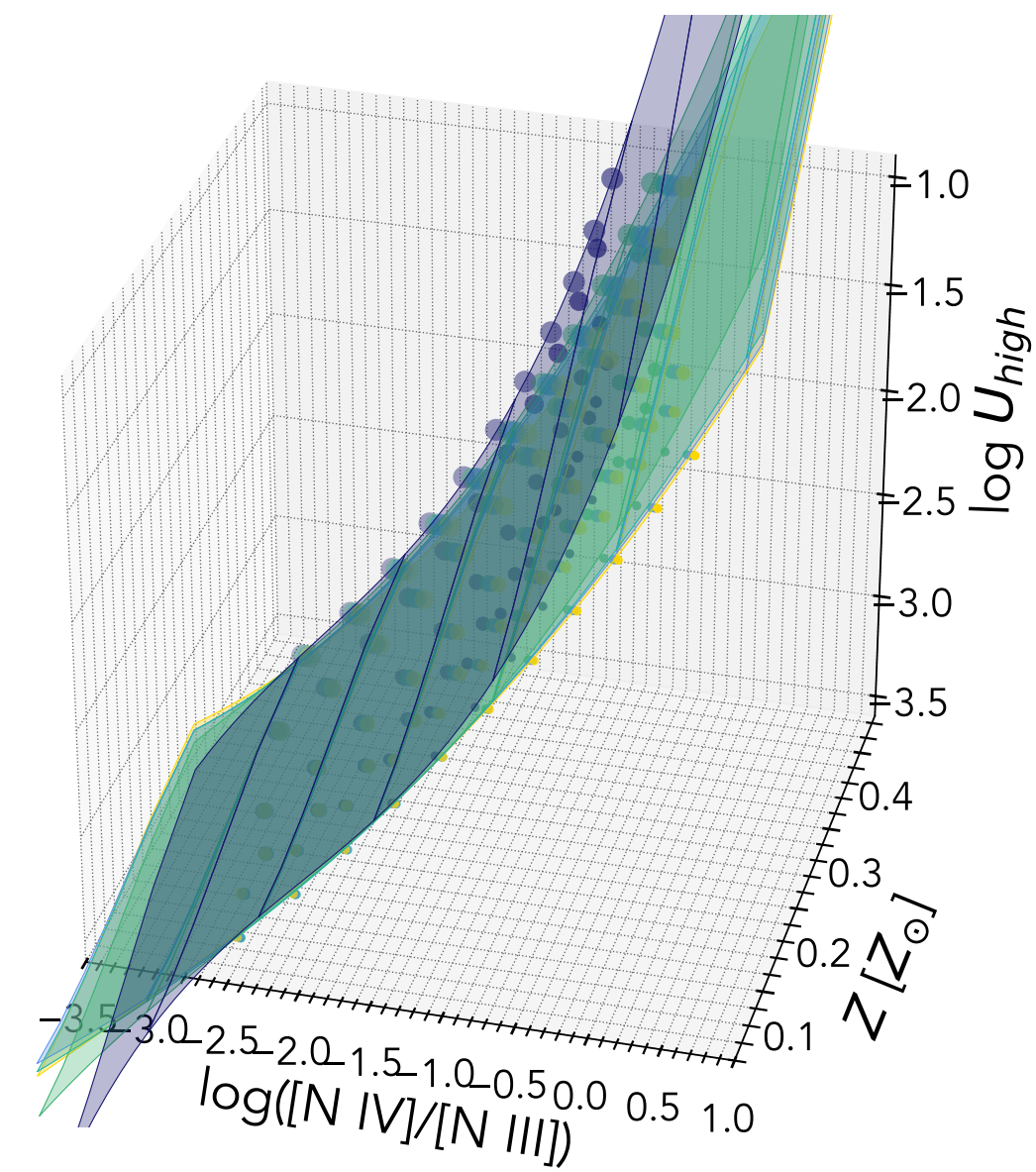
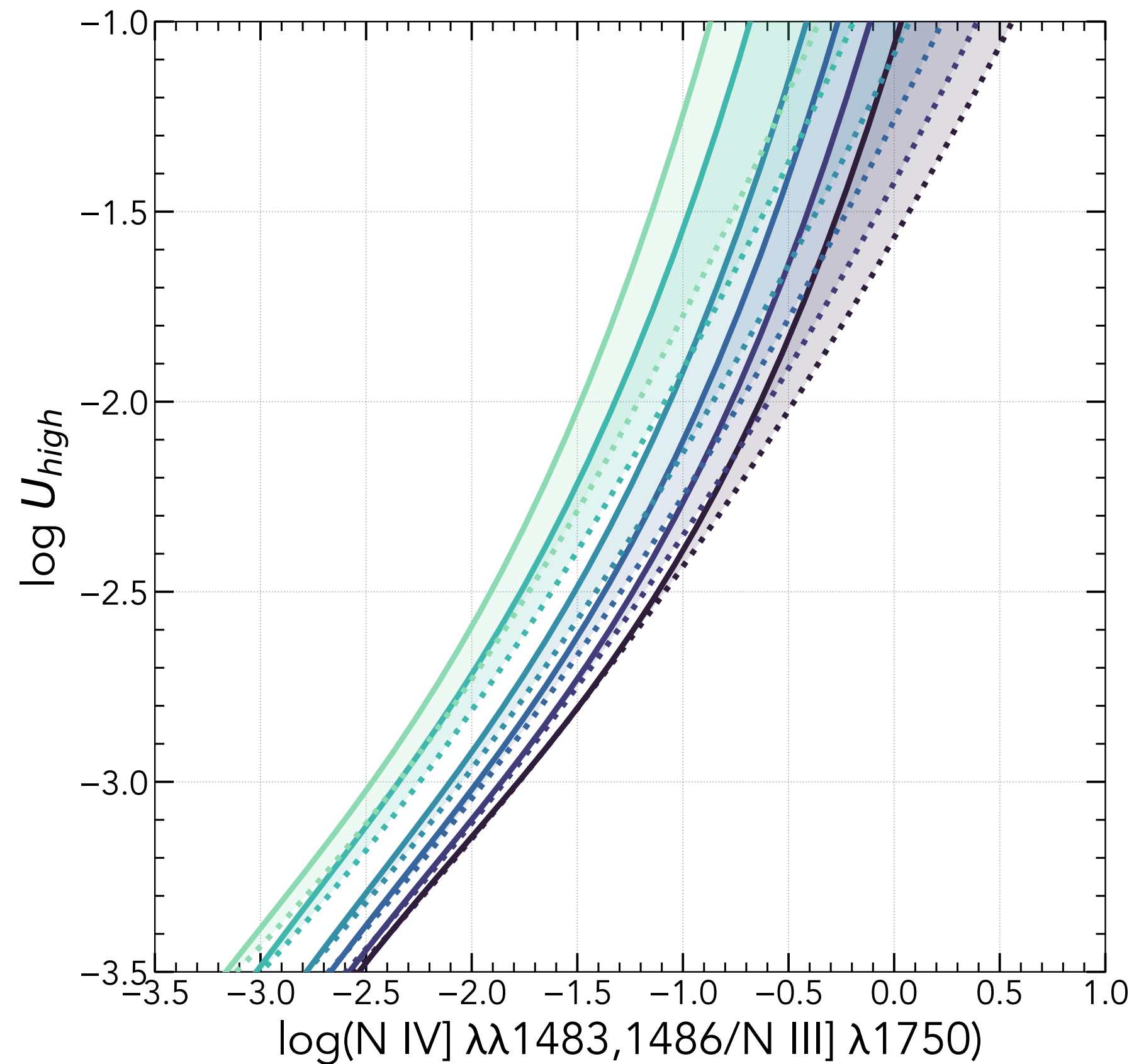
Classical
ionization
parameter
diagnostics
drastically
over predict
 $\log U_{int}$ with
increasing
density

New ionization
parameter
diagnostic, N_{43} ,
accurately
describes $\log U_{high}$
from low to high
densities



New ionization
parameter
diagnostic, N_{43} ,
accurately
describes
 $\log U_{high}$ from
low to high
densities

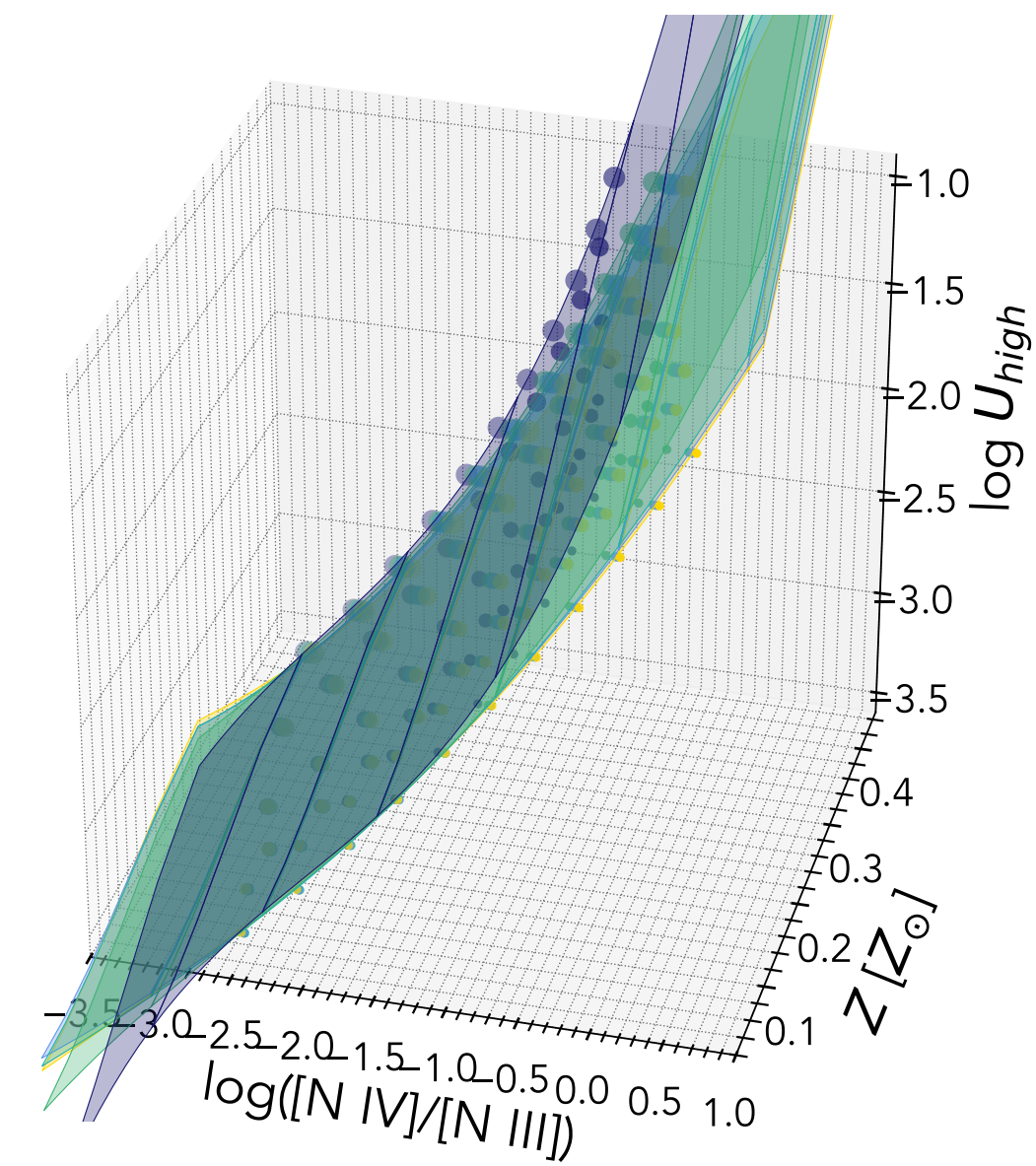
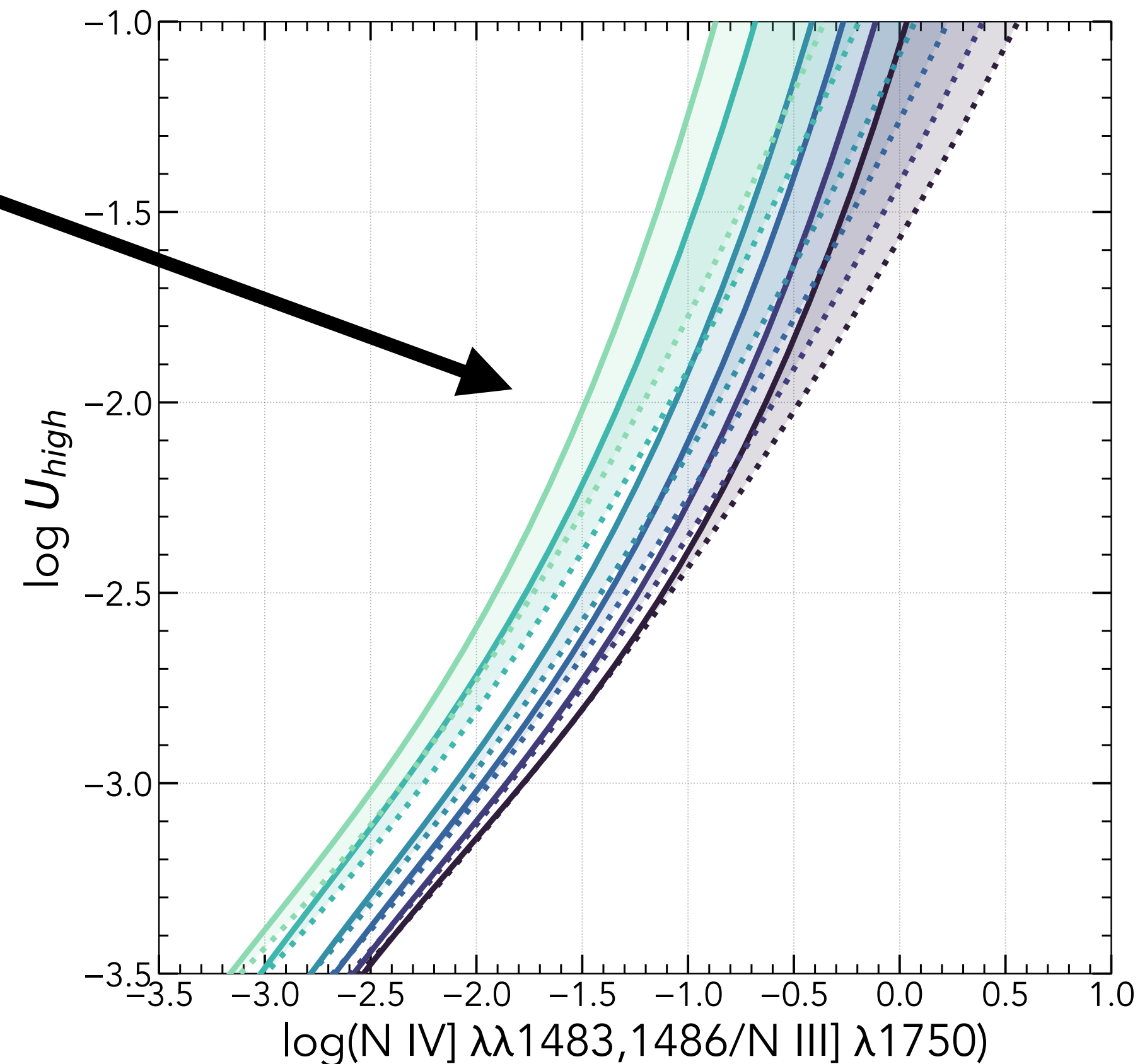
Bicubic surface fits that take
 N_{43} and Z to return $\log U_{high}$



New ionization
parameter
diagnostic, N_{43} ,
accurately
describes
 $\log U_{high}$ from
low to high
densities

Bicubic surface fits that take
 N_{43} and Z to return $\log U_{high}$

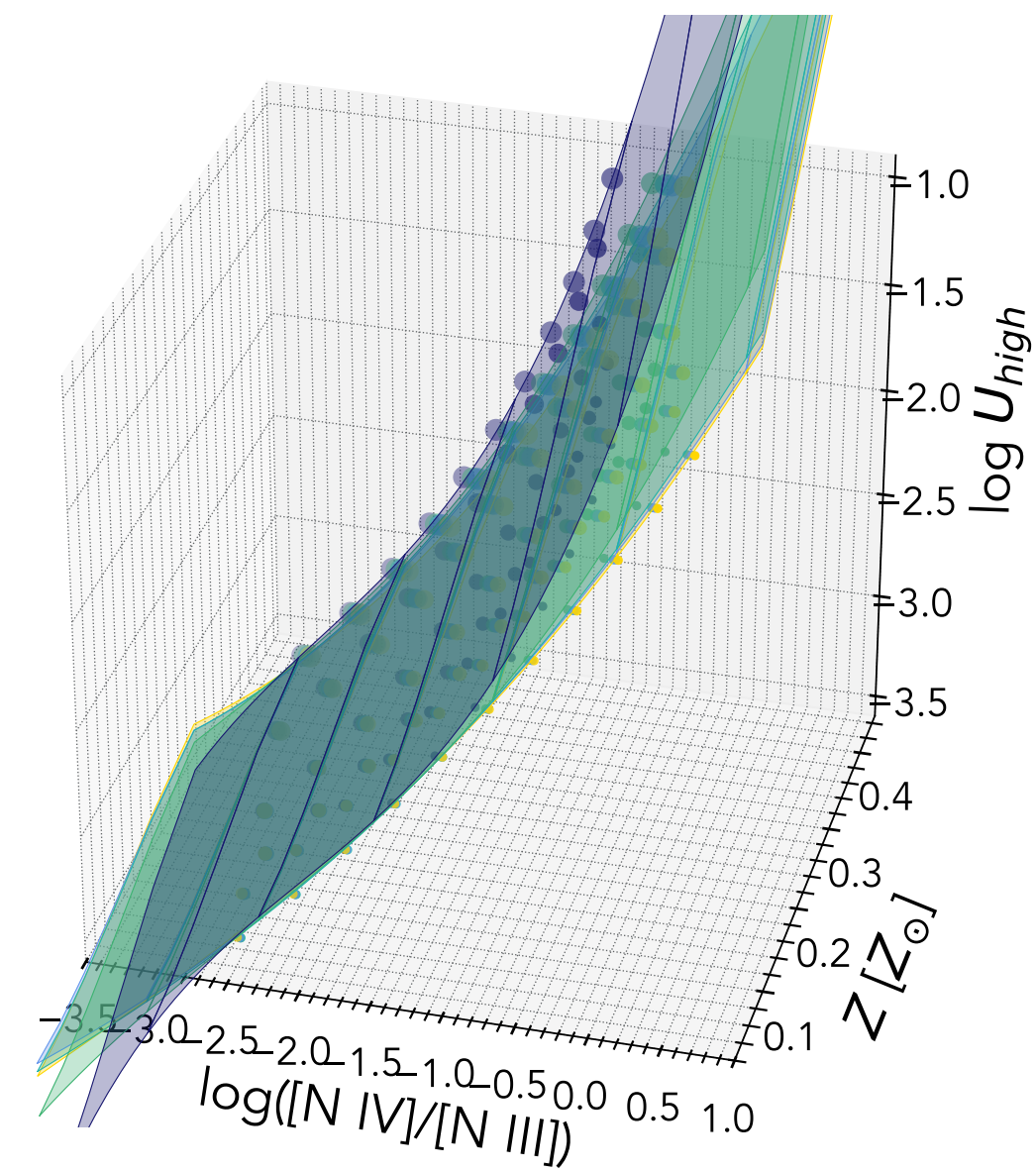
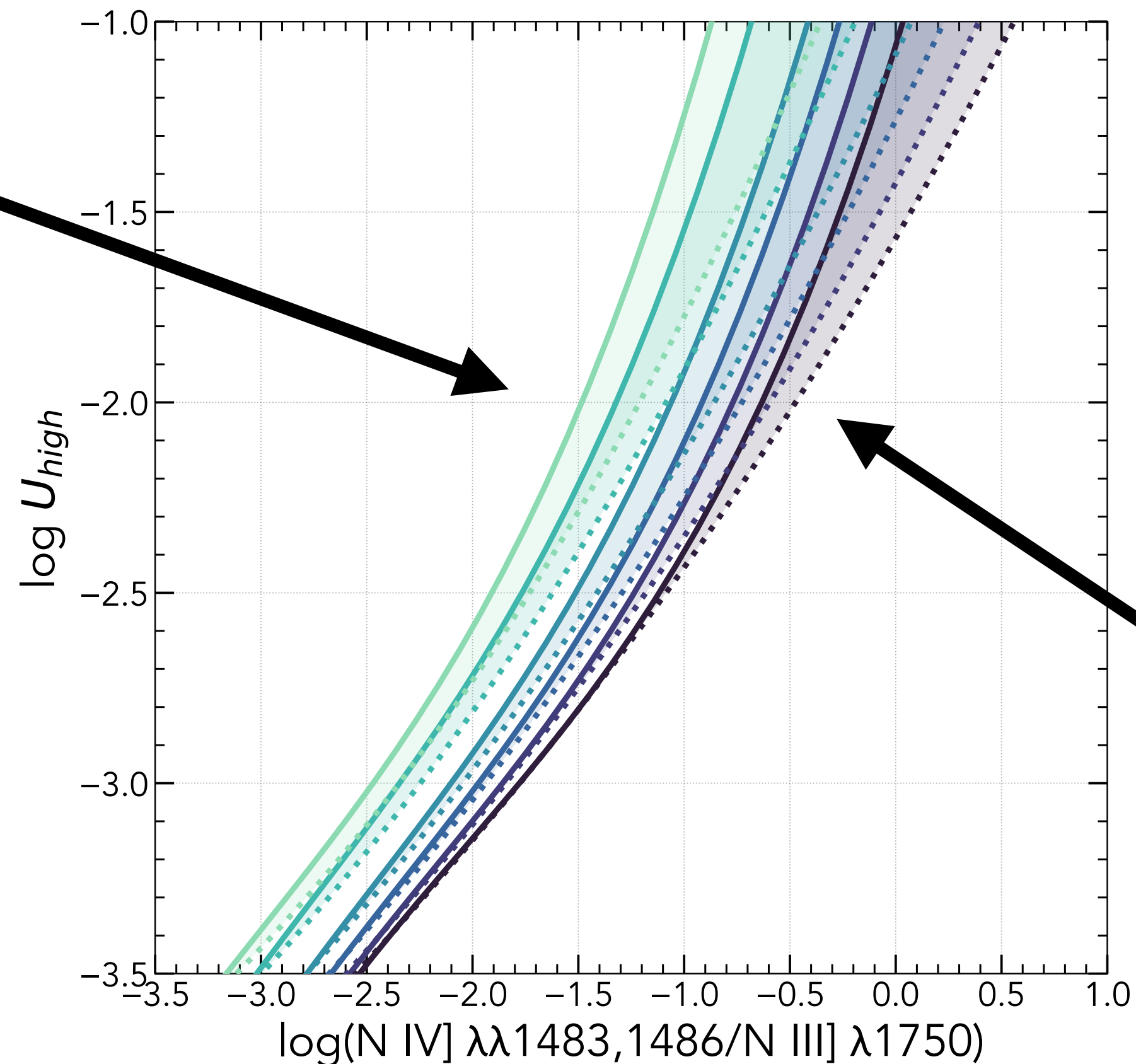
Low
density
case



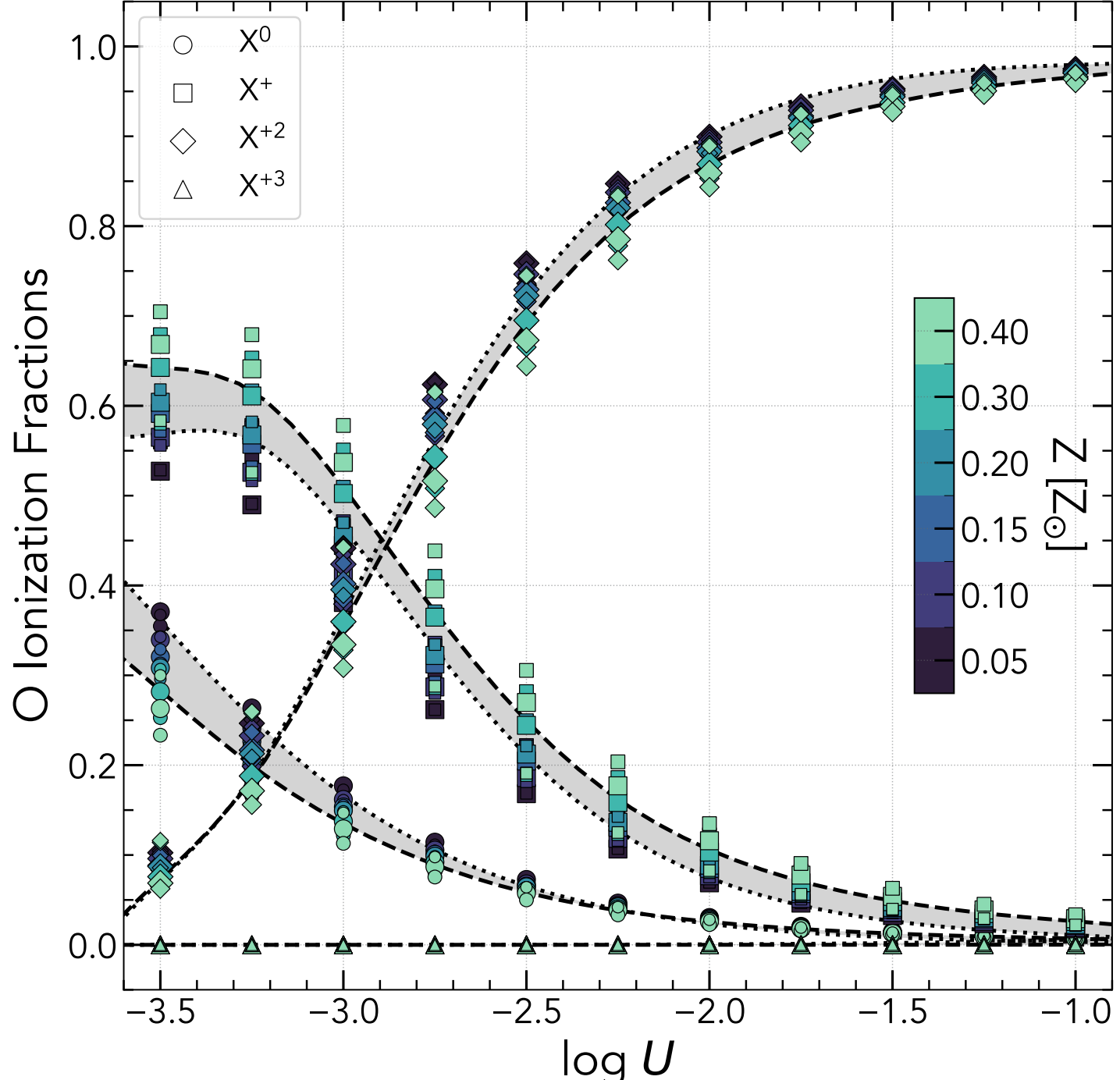
New ionization
parameter
diagnostic, N_{43} ,
accurately
describes
 $\log U_{high}$ from
low to high
densities

Bicubic surface fits that take
 N_{43} and Z to return $\log U_{high}$

Low
density
case

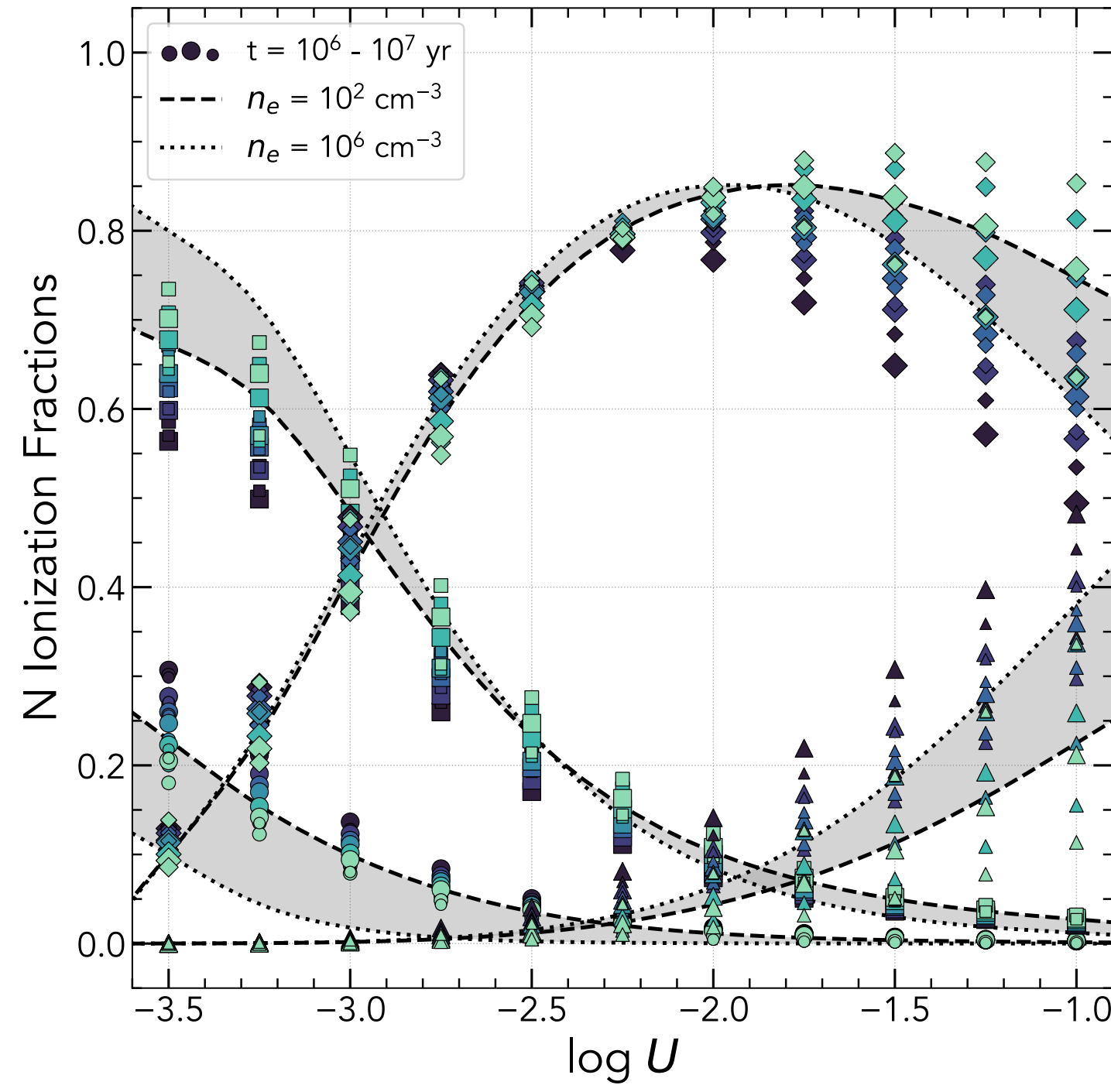


High
density
case

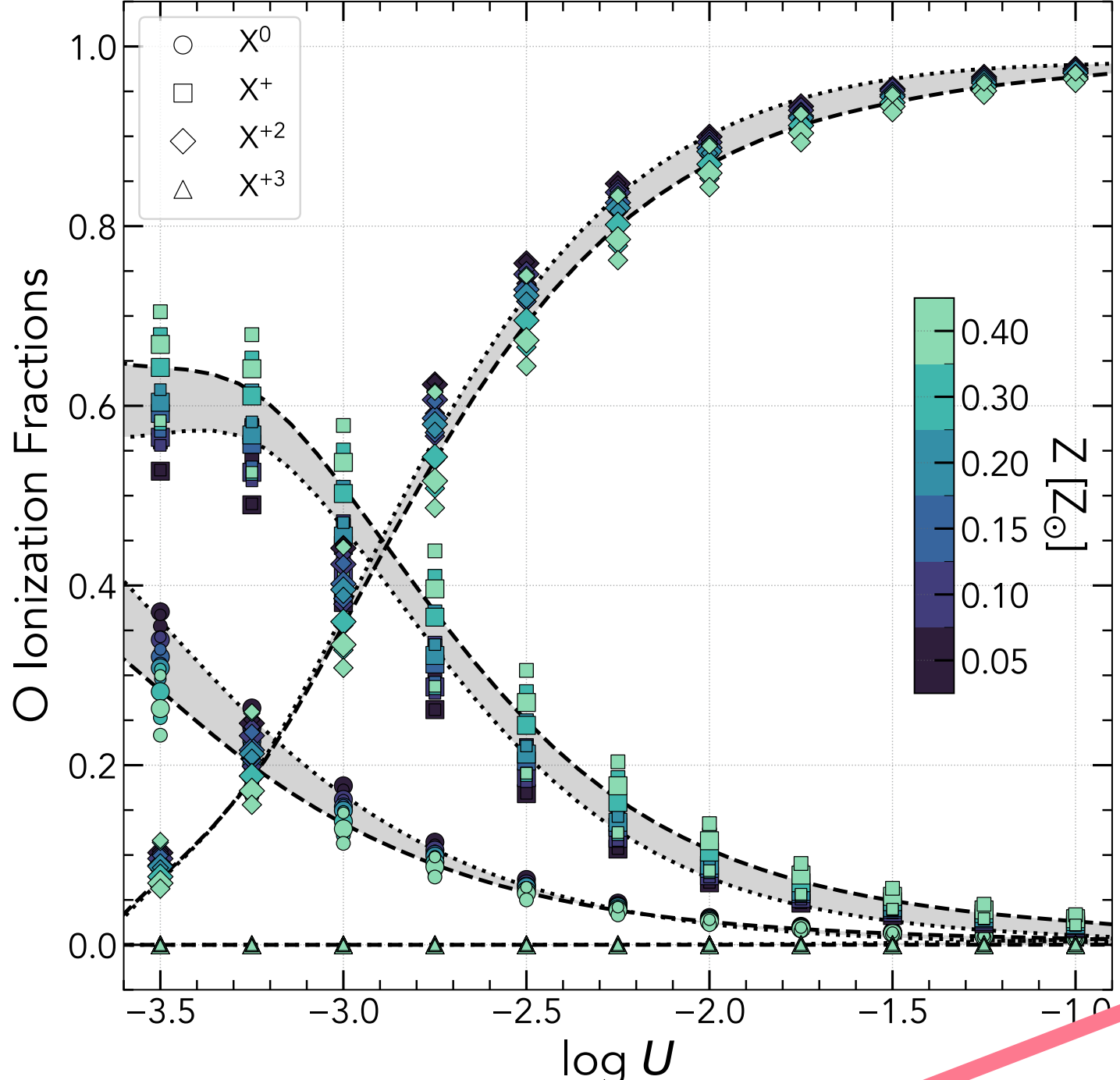


$$\frac{O}{H} = \frac{O^0}{H^+} + \frac{O^+}{H^+} + \frac{O^{+2}}{H^+} + \frac{O^{+3}}{H^+}$$

$$\frac{N}{H} = \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+}$$



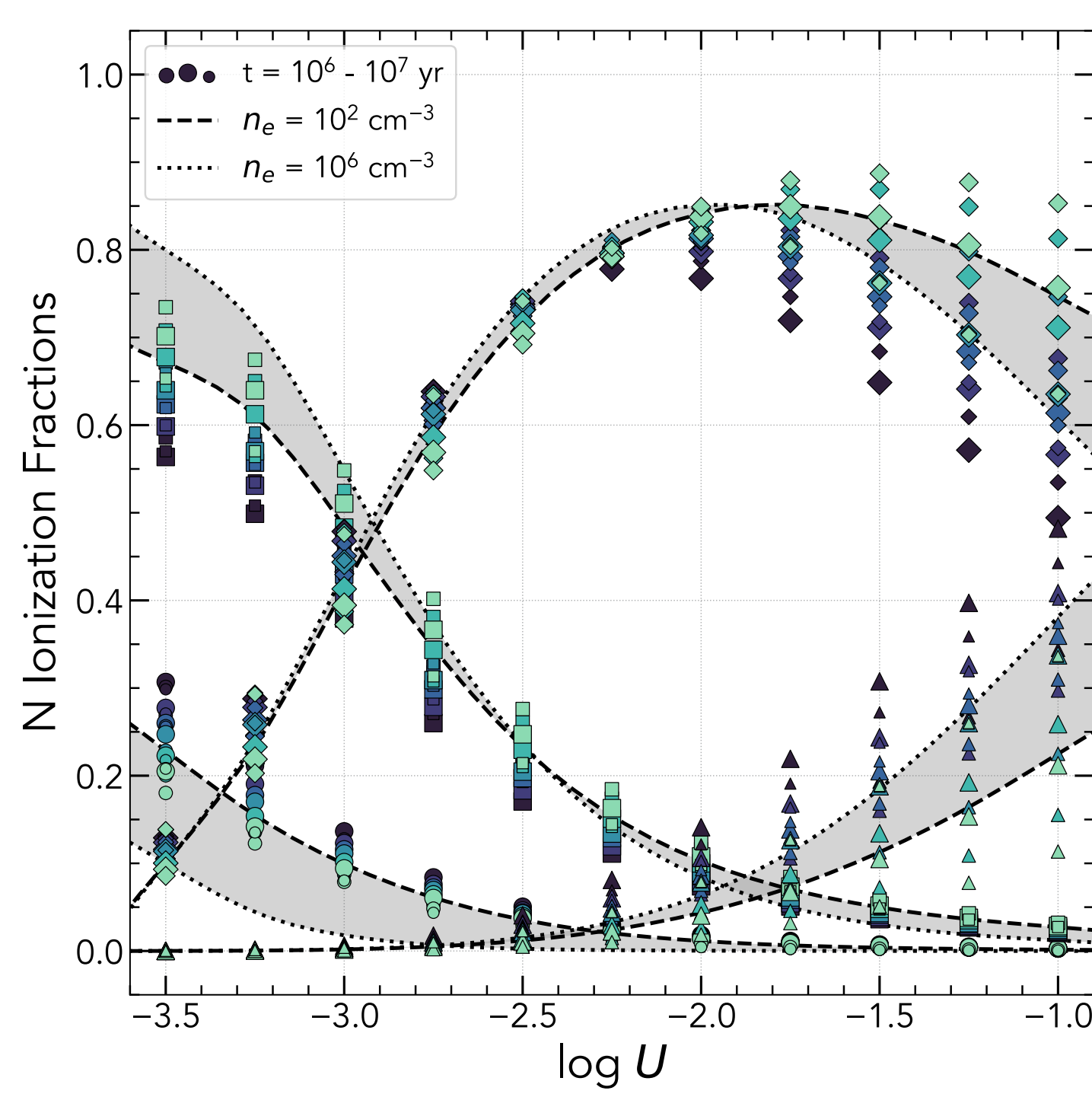
New Ionization Correction Factors (ICFs) from CLOUDY models



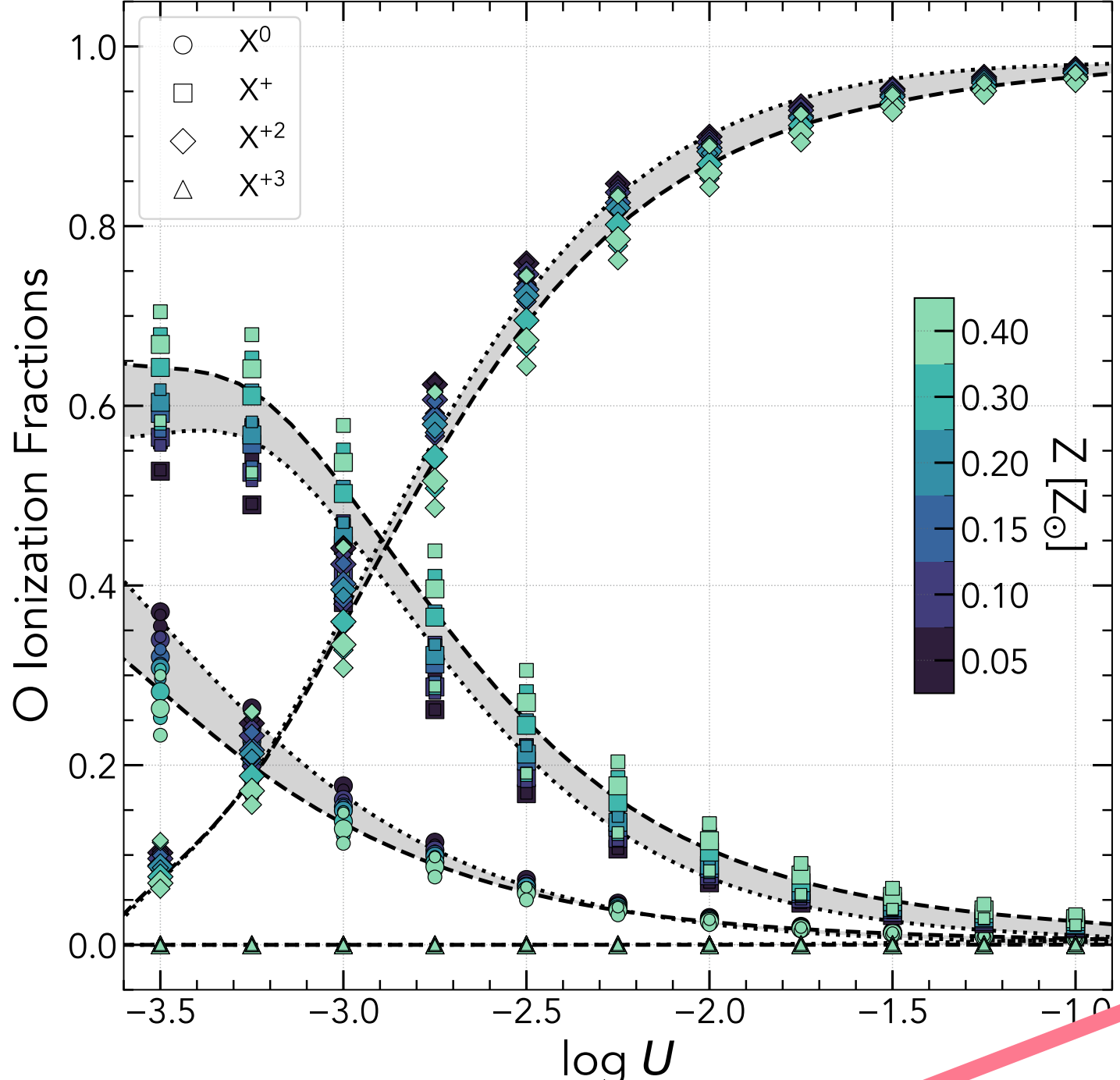
$$\frac{O}{H} = \frac{O^0}{H^+} + \frac{O^+}{H^+} + \frac{O^{+2}}{H^+} + \frac{O^{+3}}{H^+}$$

Commonly
observed

$$\frac{N}{H} = \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+}$$



New Ionization
Correction
Factors (ICFs)
from CLOUDY
models

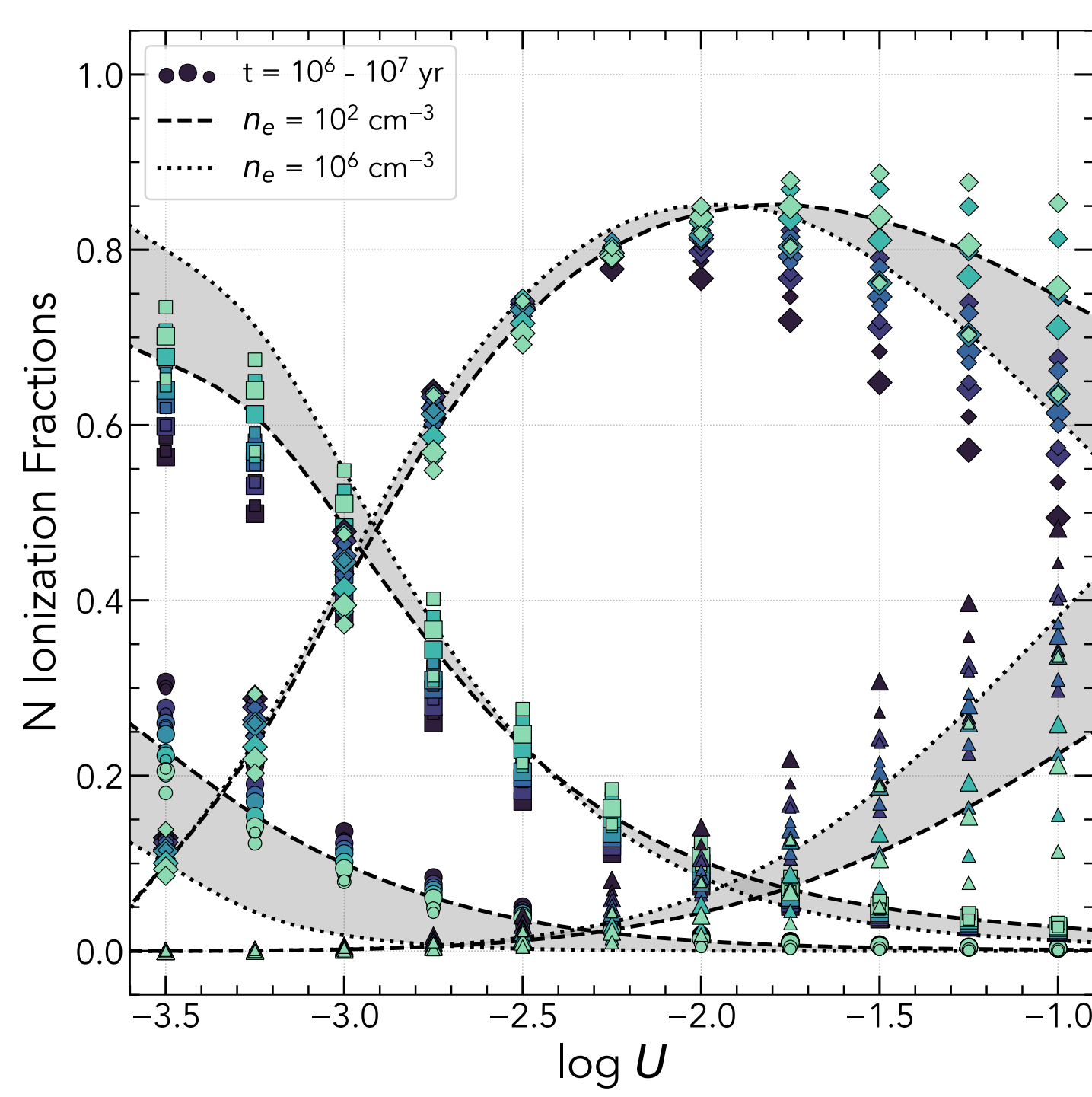


$$\frac{\text{O}}{\text{H}} = \frac{\text{O}^0}{\text{H}^+} + \frac{\text{O}^+}{\text{H}^+} + \frac{\text{O}^{+2}}{\text{H}^+} + \frac{\text{O}^{+3}}{\text{H}^+}$$

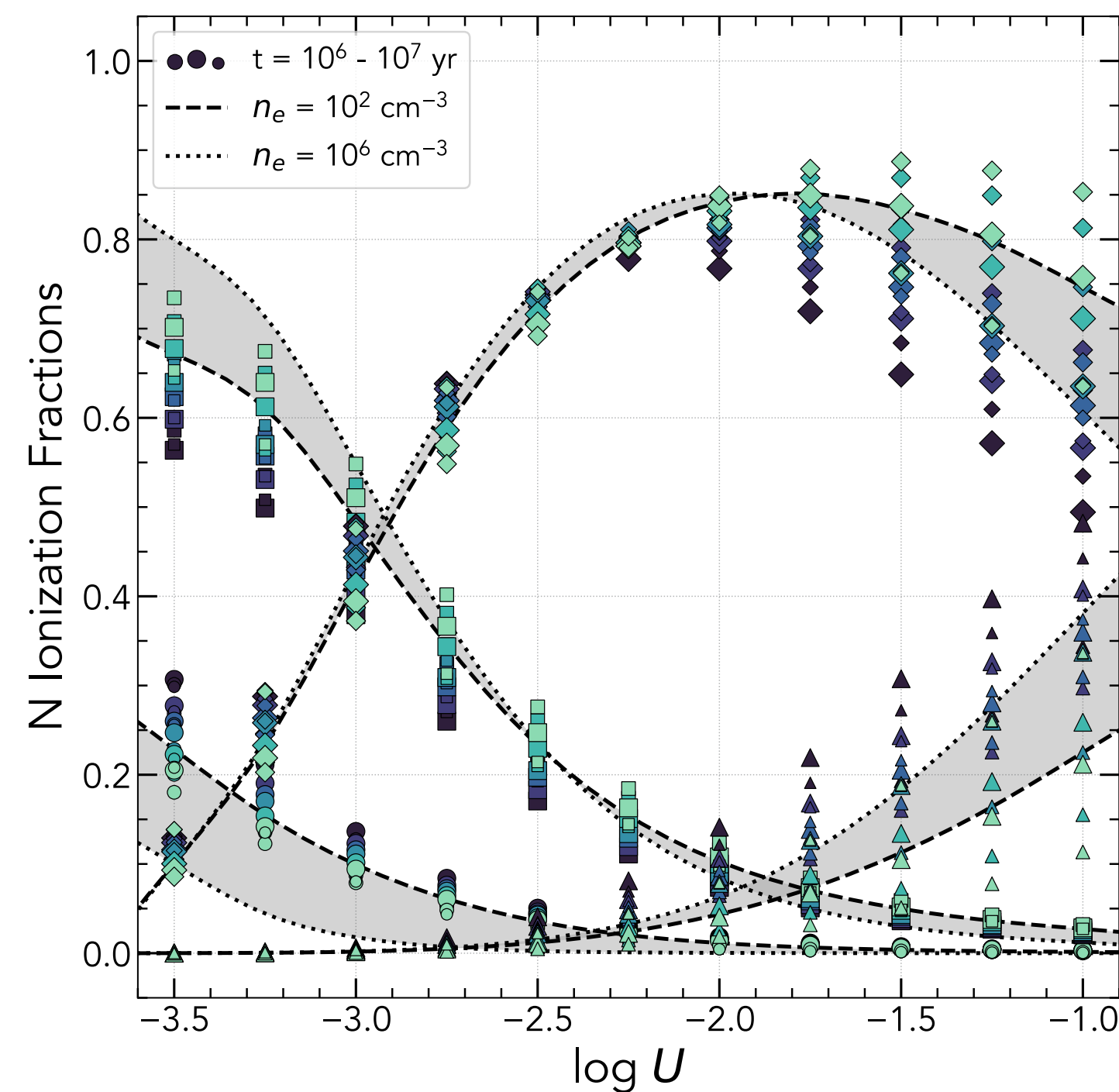
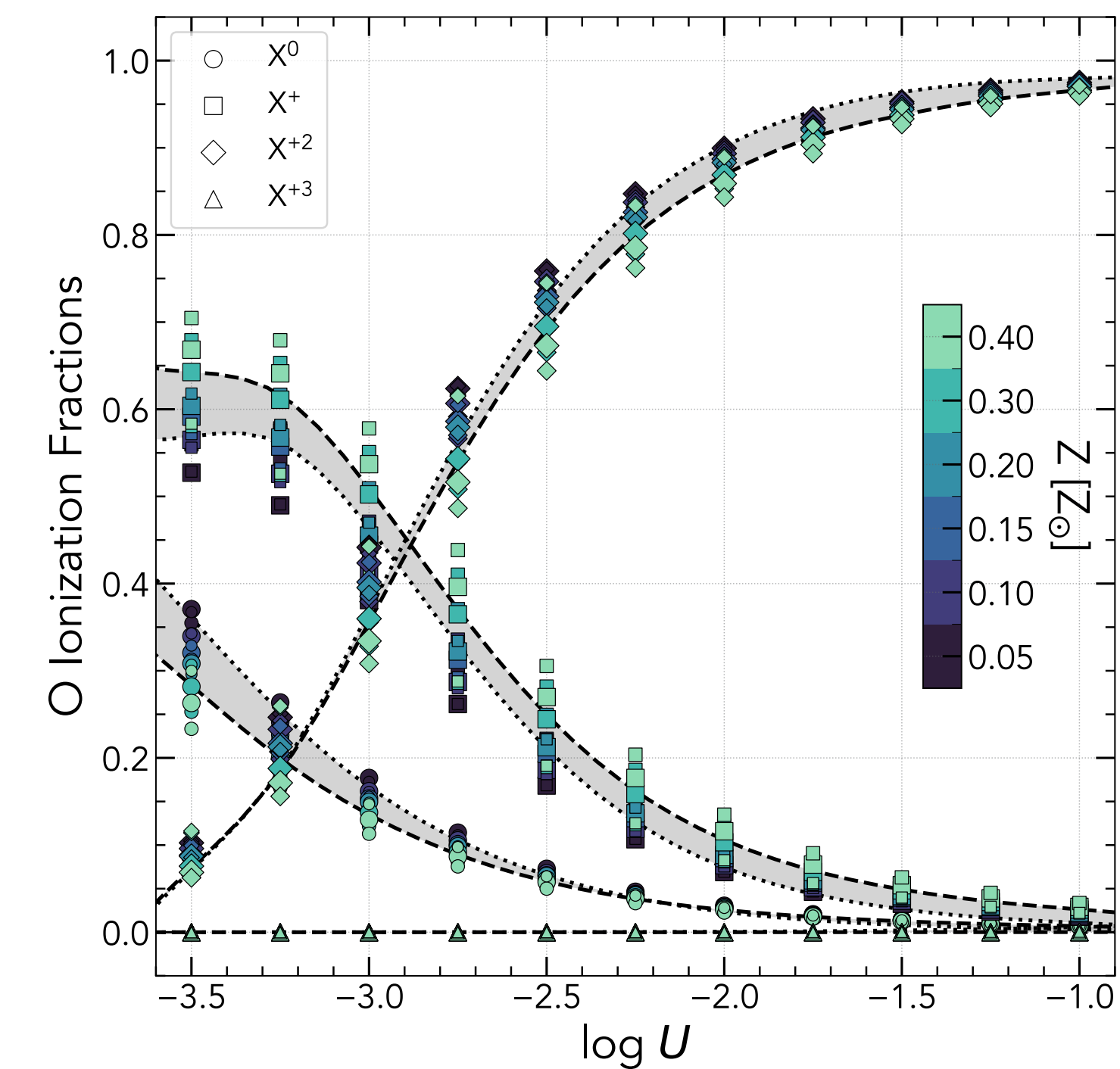
Commonly observed

$$\frac{\text{N}}{\text{H}} = \frac{\text{N}^0}{\text{H}^+} + \frac{\text{N}^+}{\text{H}^+} + \frac{\text{N}^{+2}}{\text{H}^+} + \frac{\text{N}^{+3}}{\text{H}^+}$$

Seen frequently at high-z

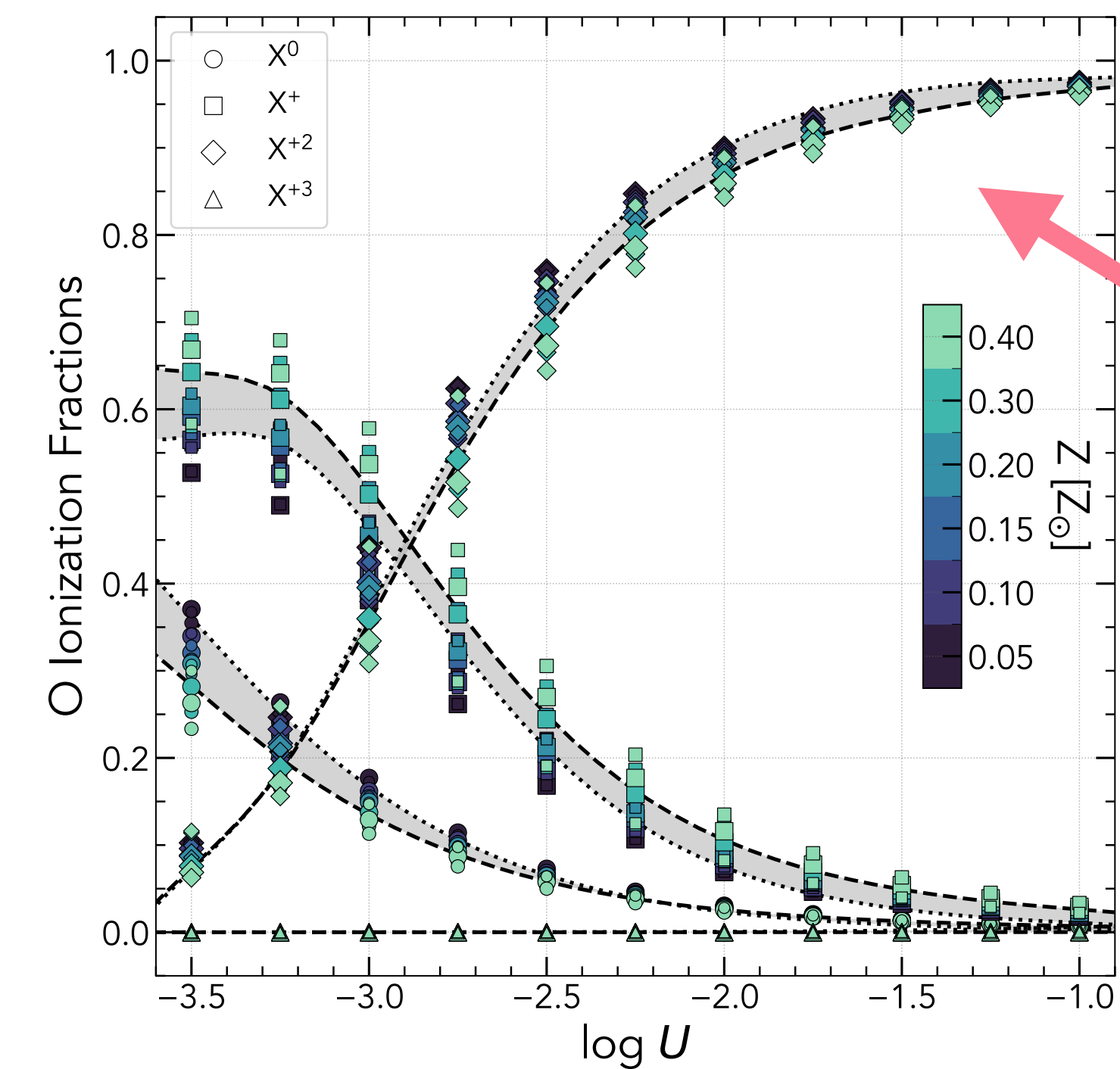


New Ionization Correction Factors (ICFs) from CLOUDY models



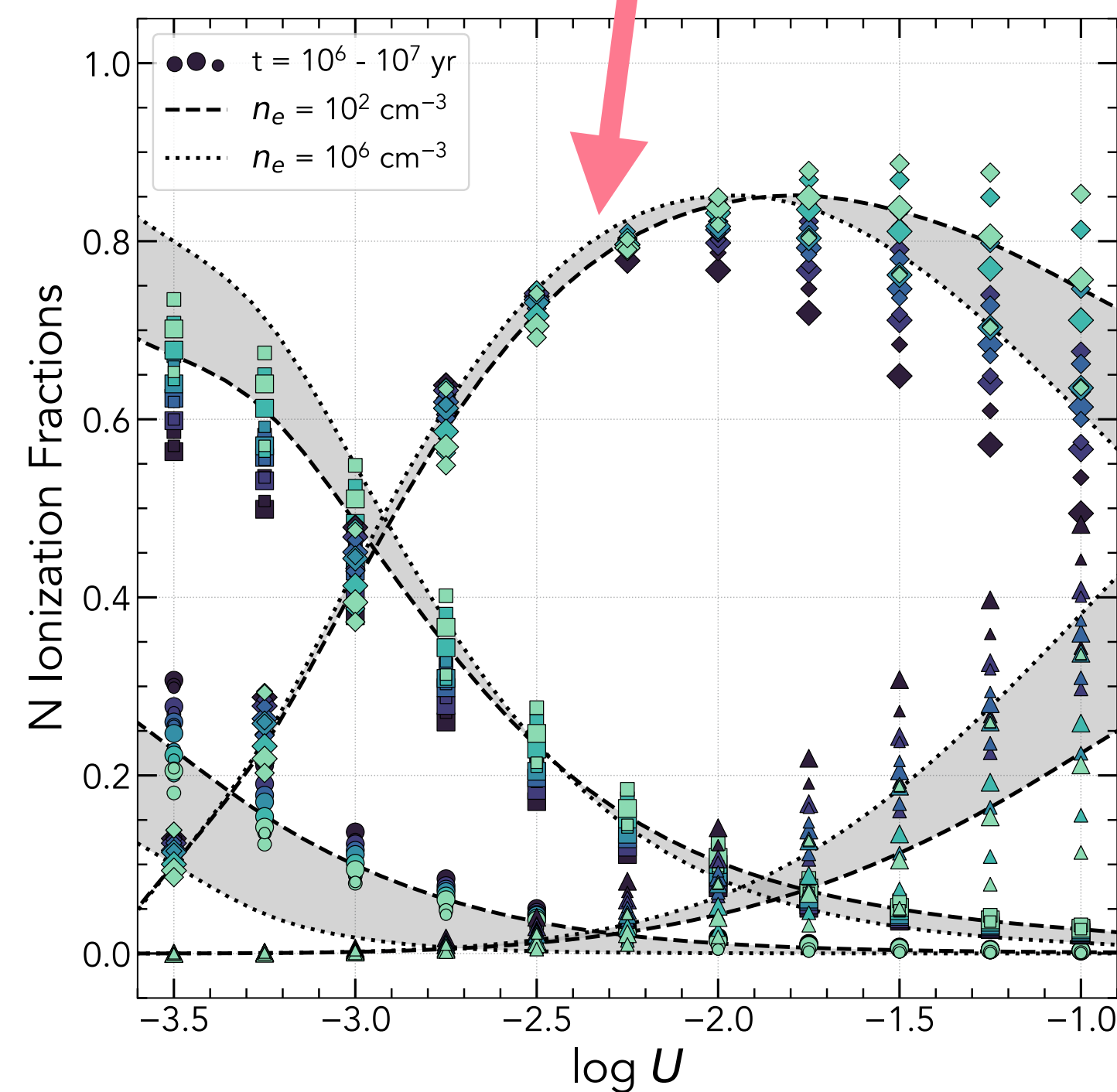
Neither low-
ionization
optical
emission or
high-ionization
UV emission
will fully
account for the
total
abundance of
an element

**New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models**

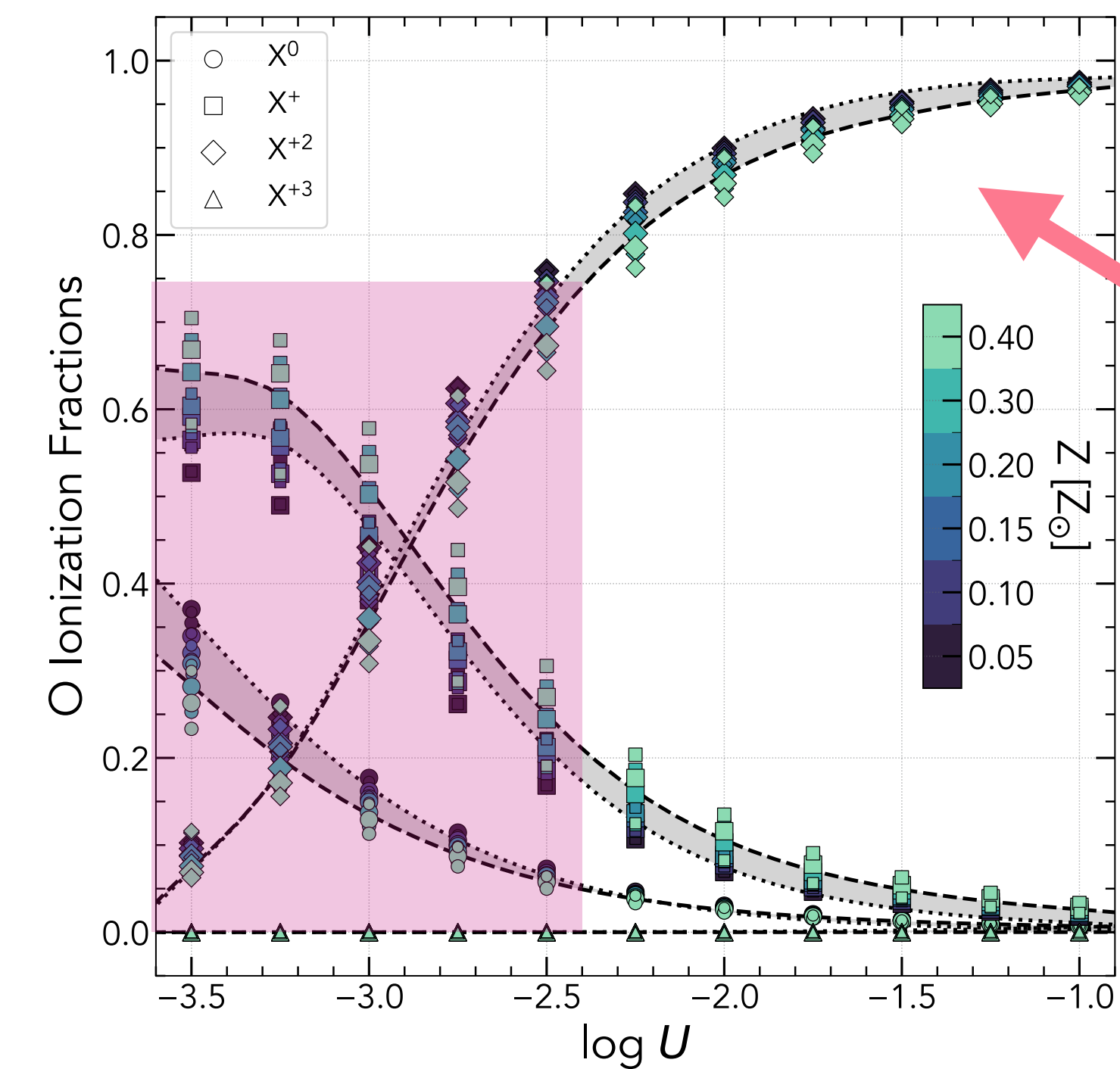


N^{+2} and O^{+2} are
dominate species
of N and O

Neither low-
ionization
optical
emission or
high-ionization
UV emission
will fully
account for the
total
abundance of
an element



**New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models**

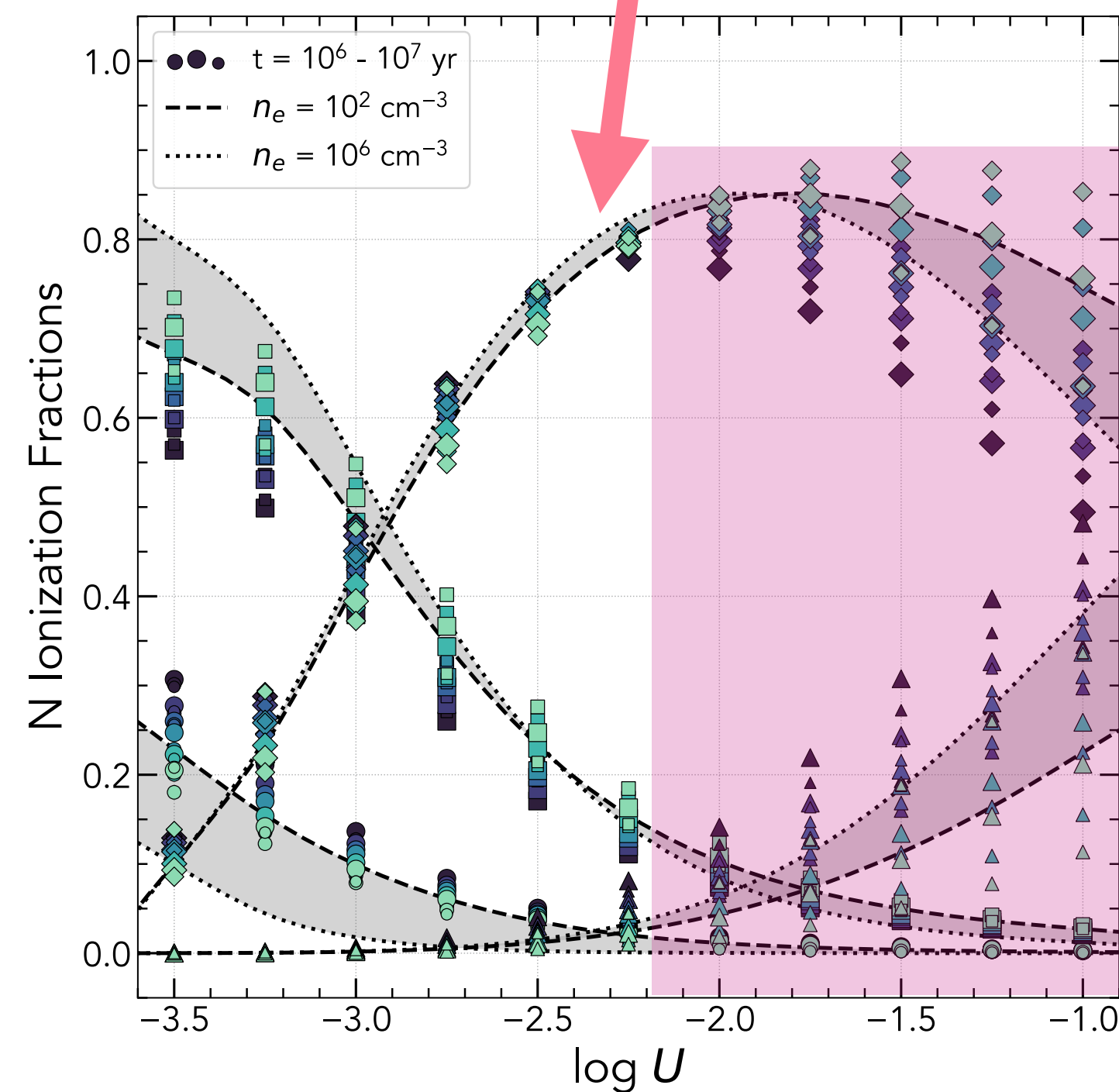


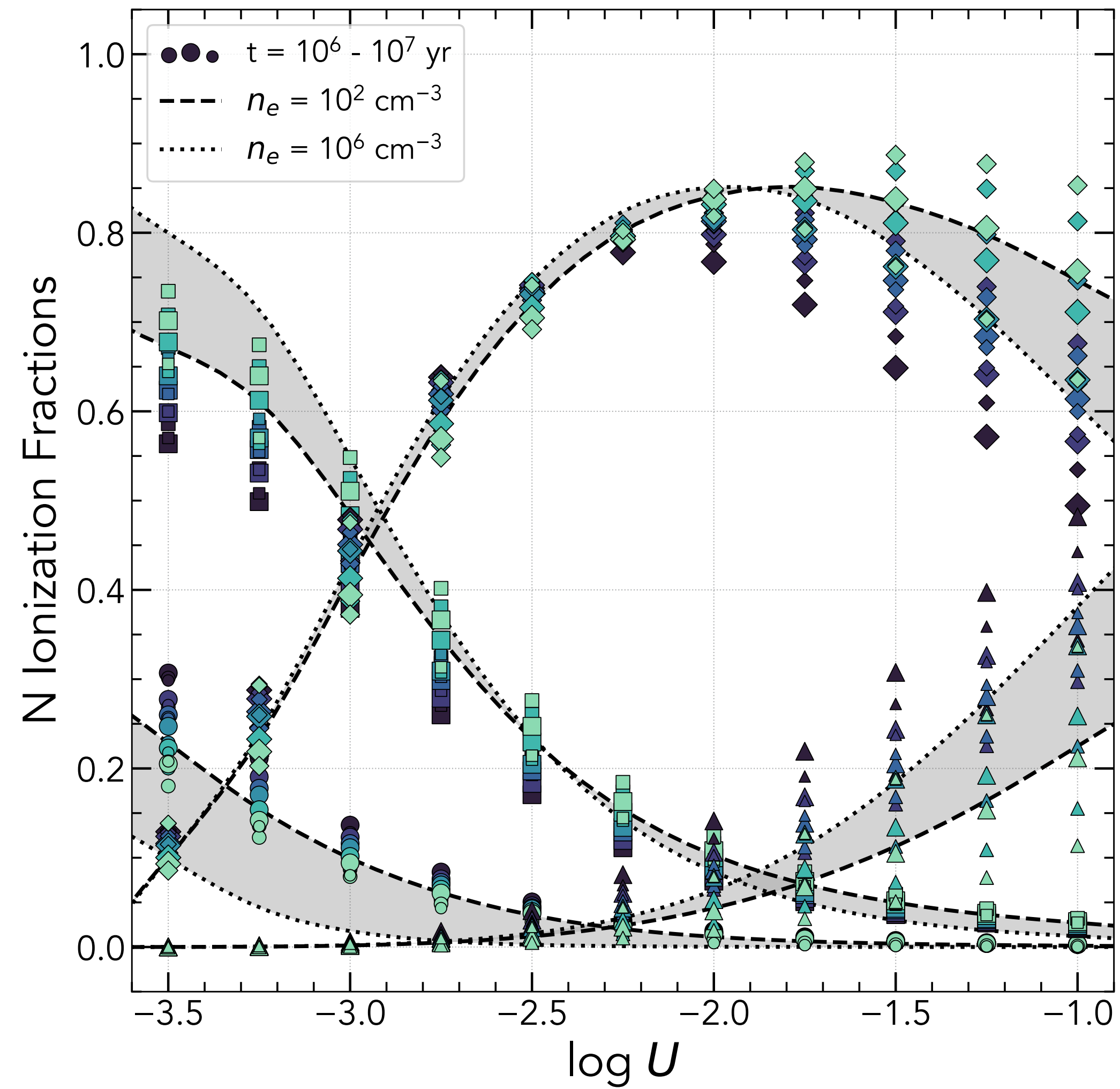
N^{+2} and O^{+2} are
dominate species
of N and O

Neither low-
ionization
optical
emission or
high-ionization
UV emission
will fully
account for the
total
abundance of
an element

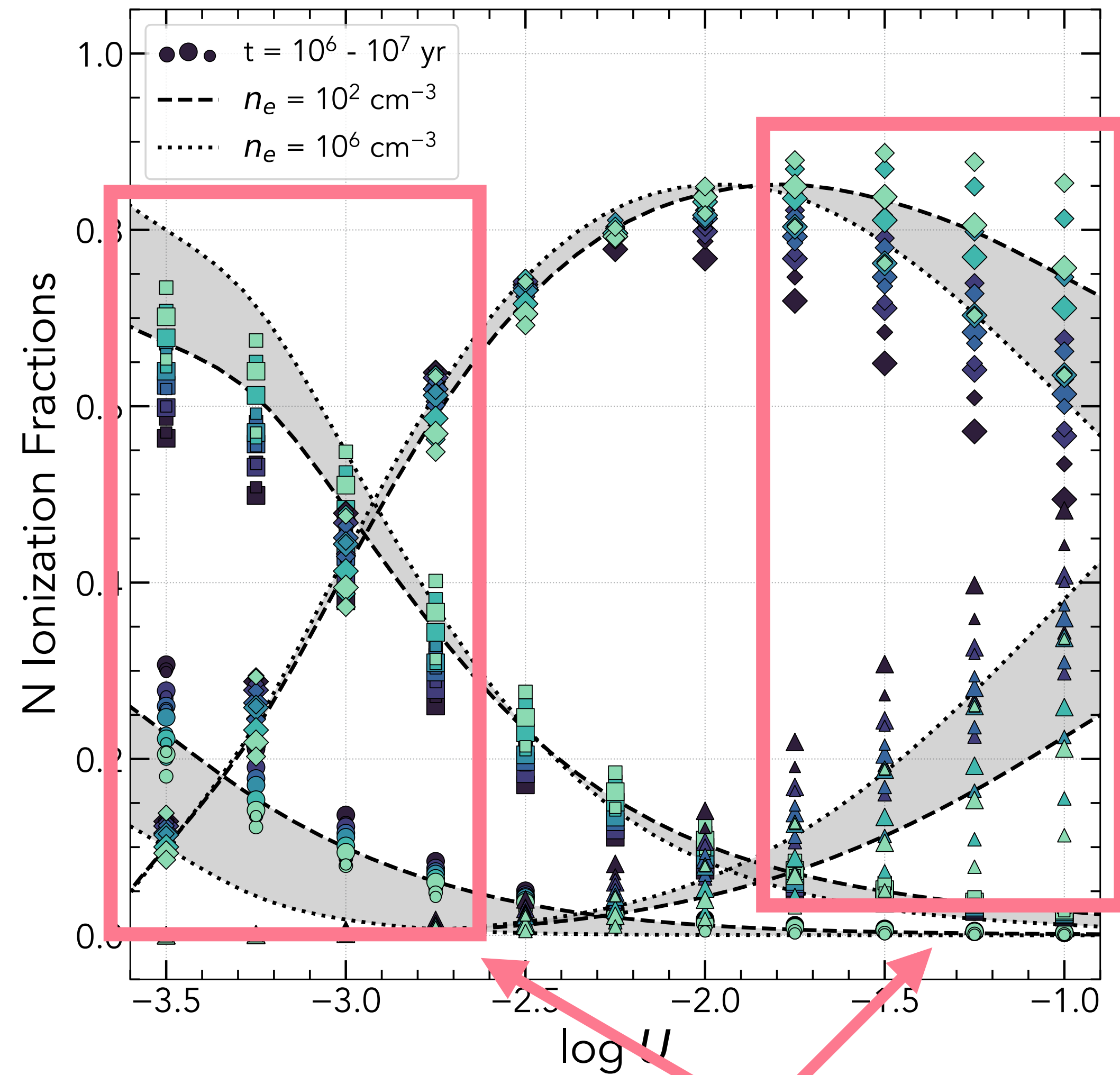
New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

Ionization fractions
can be very **sensitive**
to burst age across
ionization parameters



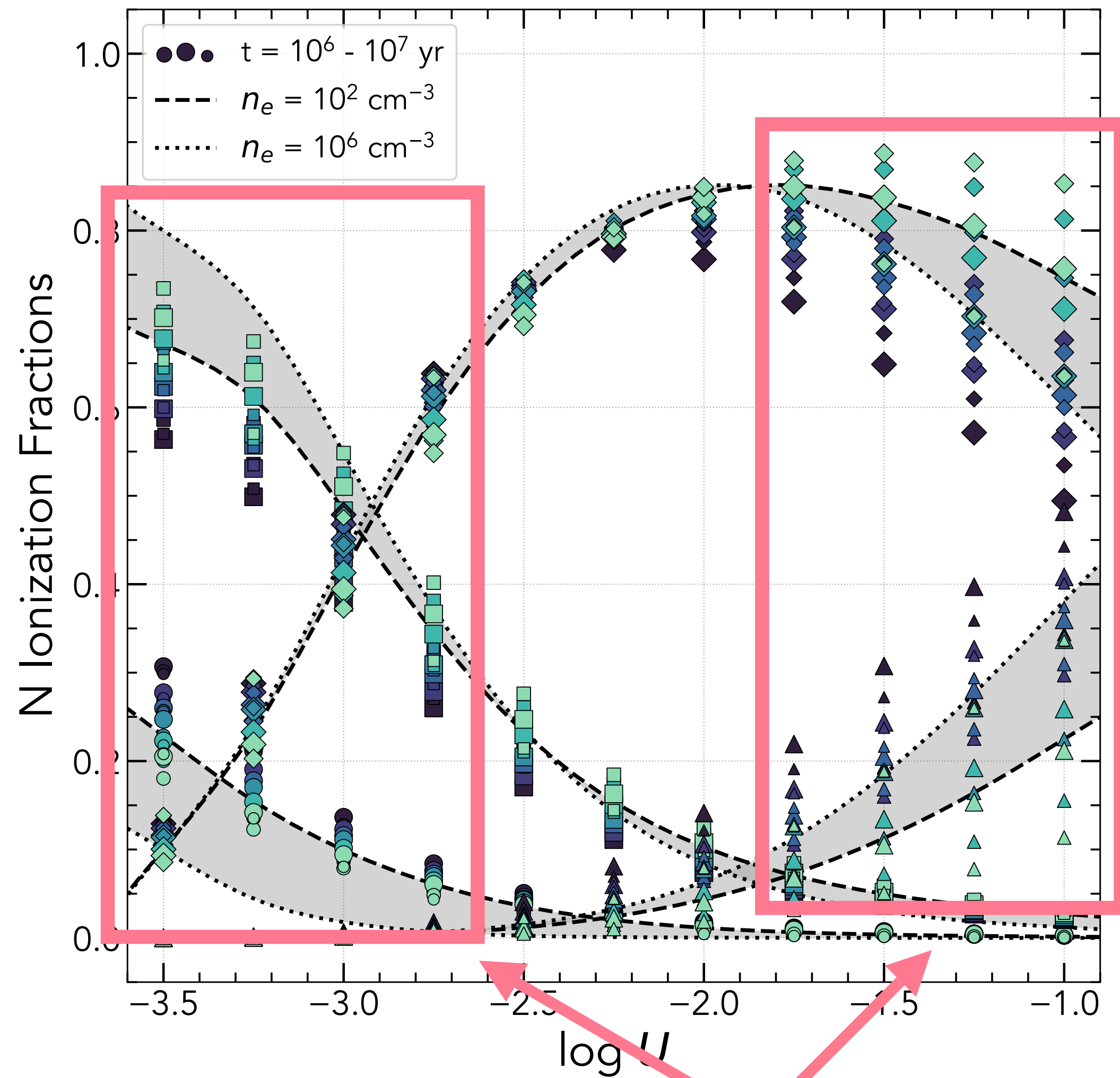


New
 Ionization
 Correction
 Factors (ICFs)
 with CLOUDY
 models



N ionization fractions are **especially sensitive to density** at extreme ionization parameters

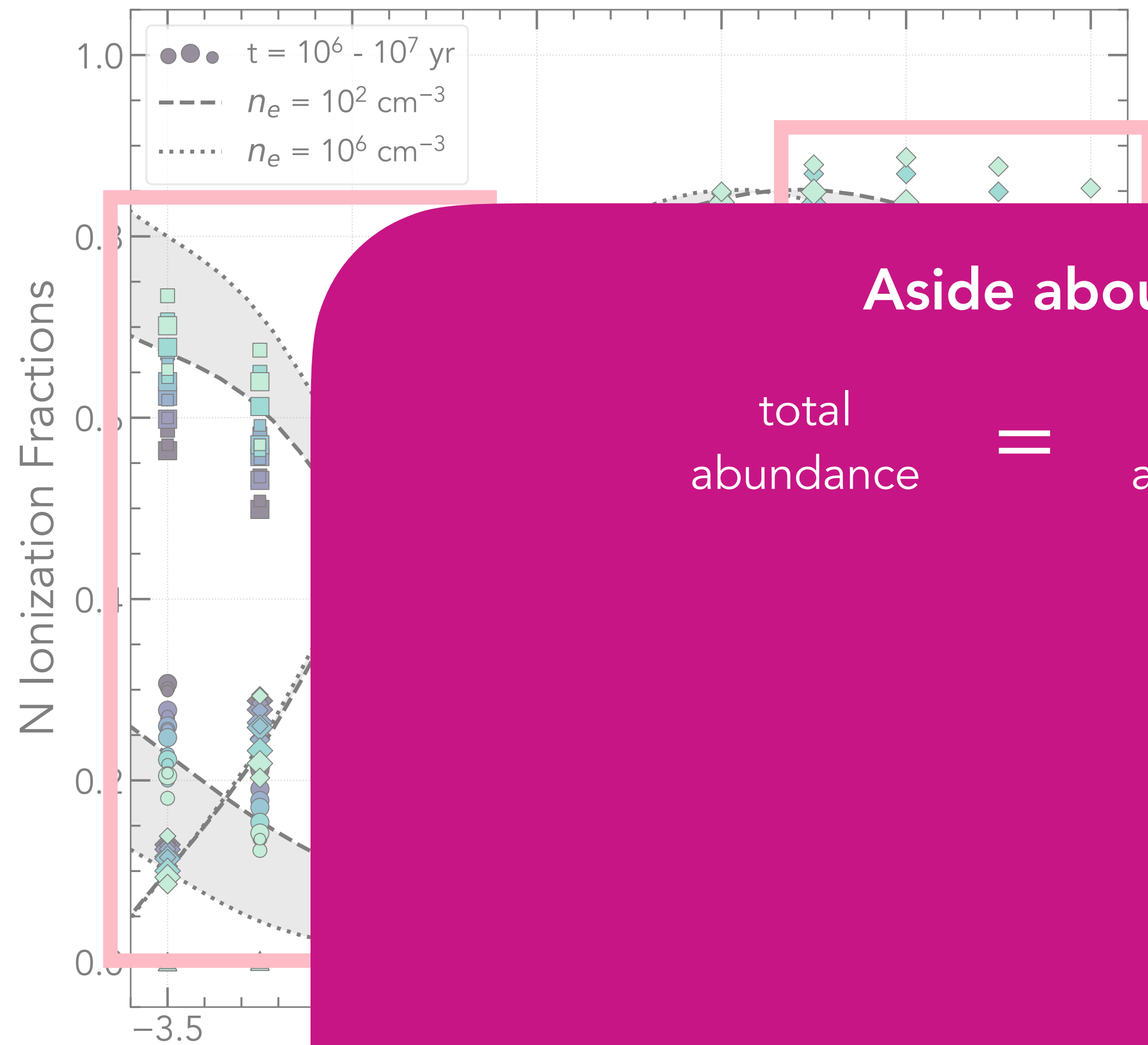
New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models



ICFs that **take into consideration metallicity, burst age, and density** will be essential to accurate N/O abundances

New Ionization Correction Factors (ICFs) with CLOUDY models

N ionization fractions are **especially sensitive to density** at extreme ionization parameters

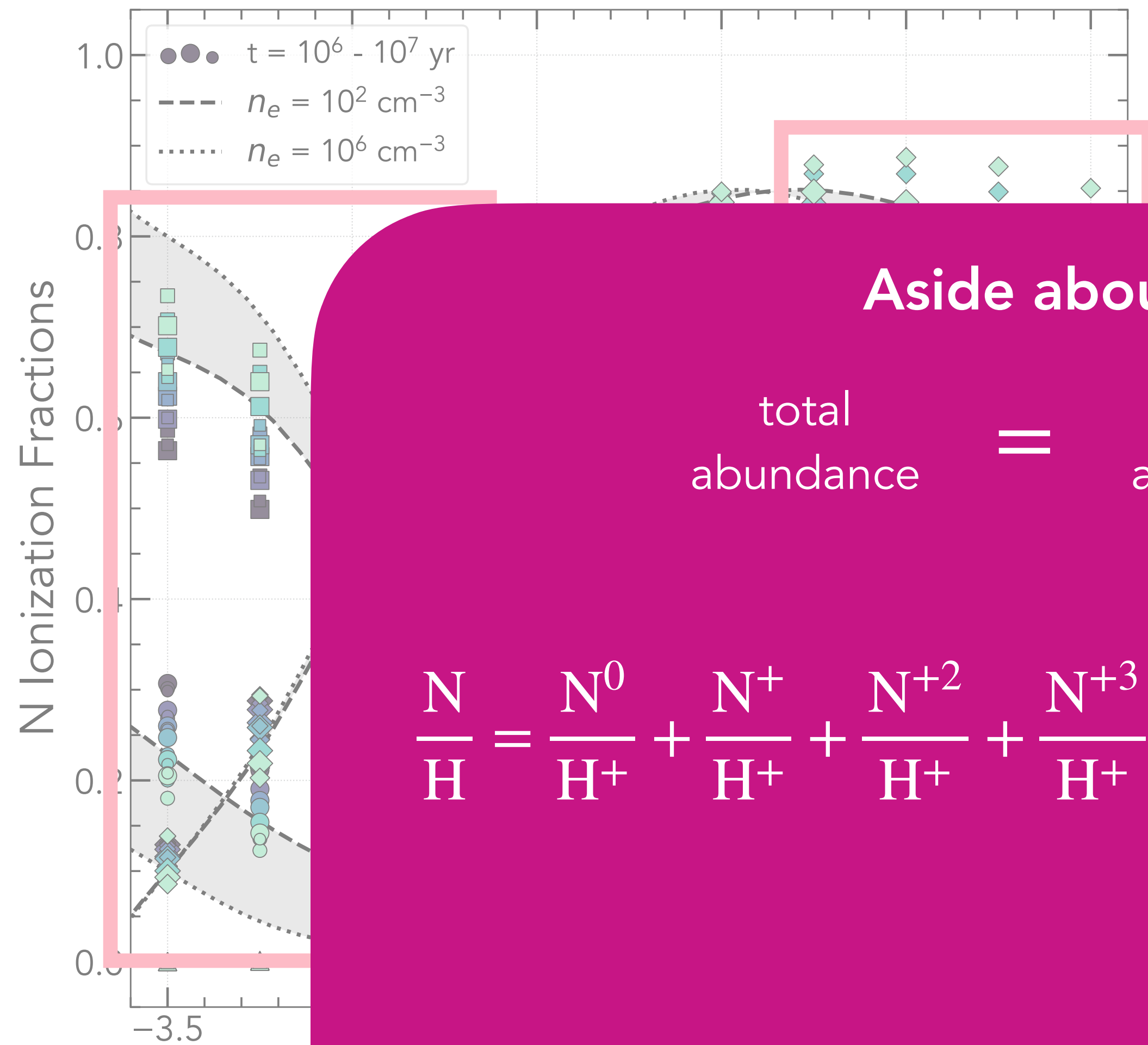


Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

ser



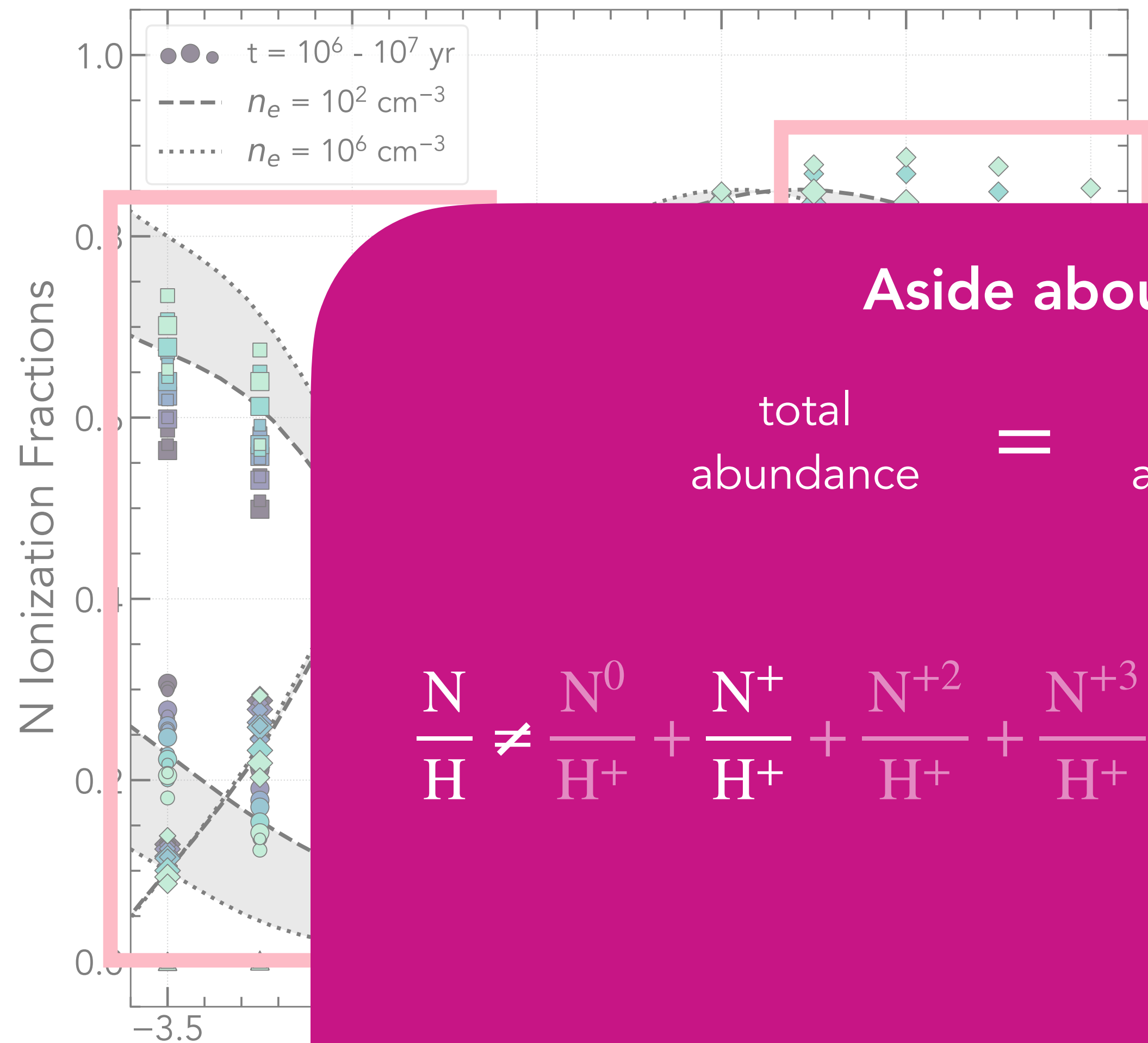
Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

$$\frac{N}{H} = \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+}$$

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

ser



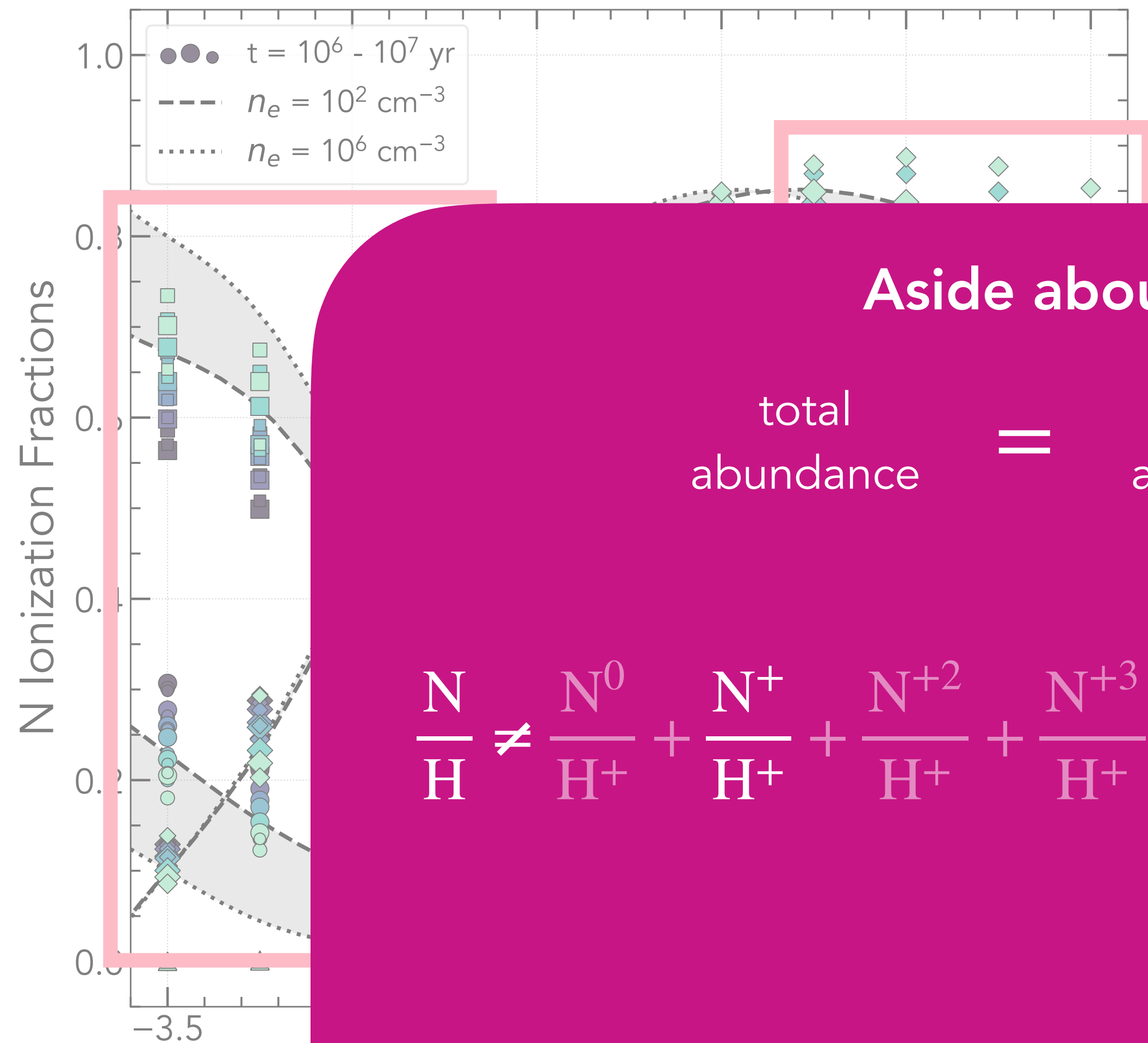
Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

$$\frac{N}{H} \neq \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+}$$

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

ser



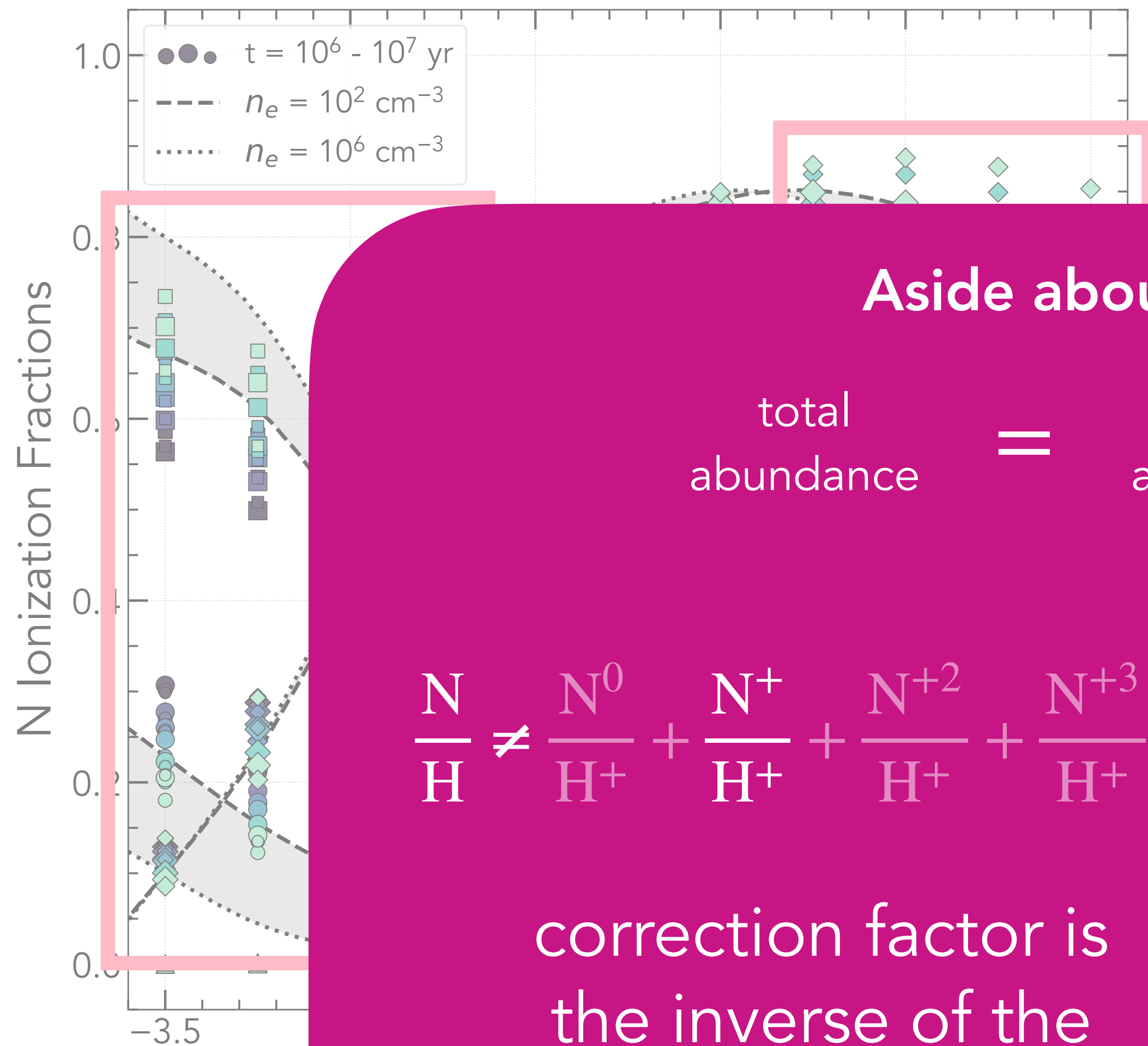
Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

$$\frac{N}{H} \neq \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+} \quad \rightarrow \quad X(N^{+i}) = \frac{N^{+i}}{N_{\text{tot}}}$$

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

ser



Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

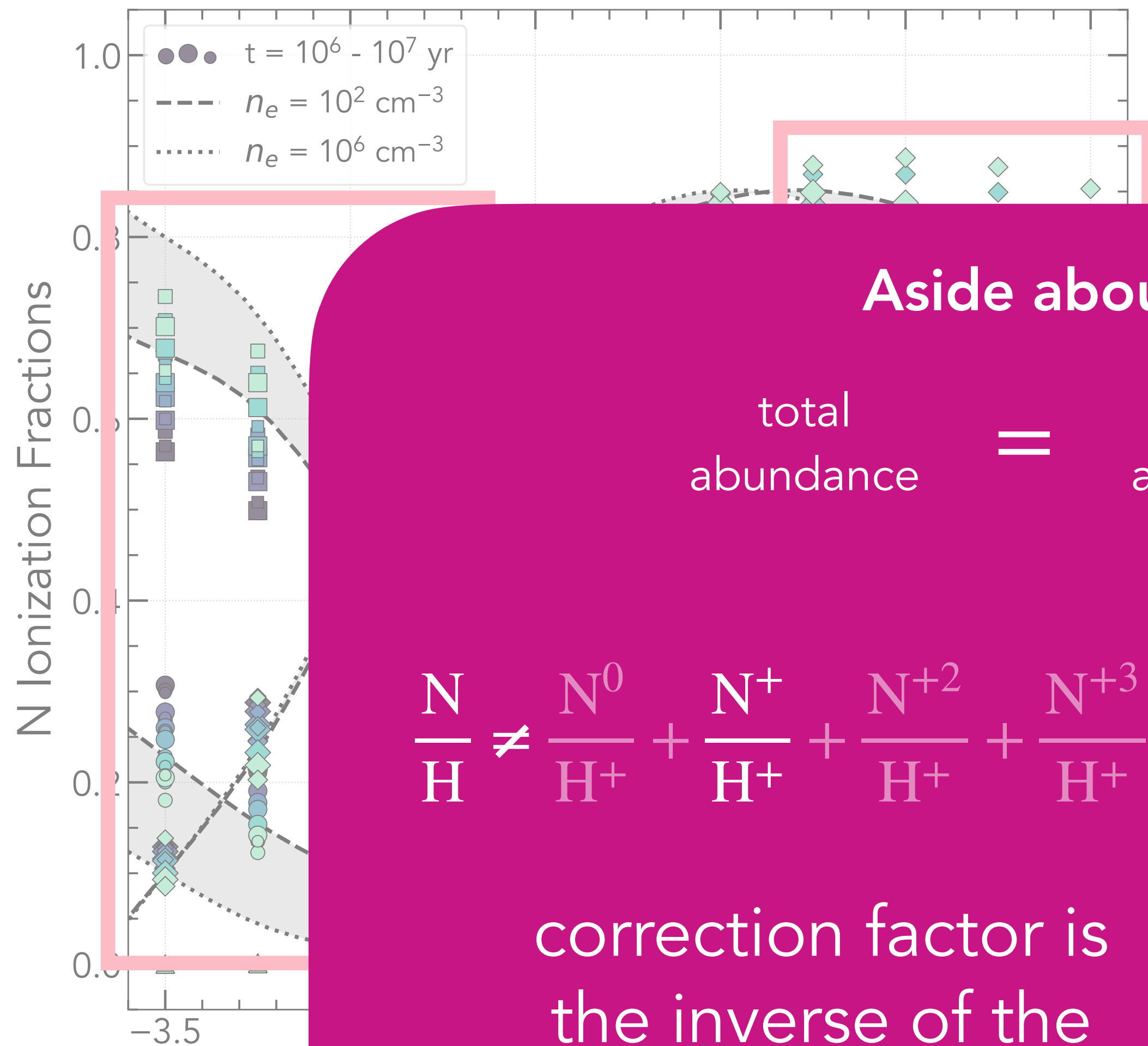
$$\frac{N}{H} \neq \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+} \quad \rightarrow \quad X(N^{+i}) = \frac{N^{+i}}{N_{\text{tot}}}$$

correction factor is
the inverse of the
ionization fraction

$$[X(N^{+i})]^{-1} = \frac{N_{\text{tot}}}{N^{+i}} = N \text{ ICF}$$

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

ser



Aside about applying ICFs...

$$\text{total abundance} = \text{ionic abundance} \times \frac{1}{\text{ionic fraction}}$$

$$\frac{N}{H} \neq \frac{N^0}{H^+} + \frac{N^+}{H^+} + \frac{N^{+2}}{H^+} + \frac{N^{+3}}{H^+} \quad \rightarrow \quad X(N^{+i}) = \frac{N^{+i}}{N_{\text{tot}}}$$

correction factor is
the inverse of the
ionization fraction

$$[X(N^{+i})]^{-1} = \frac{N_{\text{tot}}}{N^{+i}} = N \text{ ICF}$$

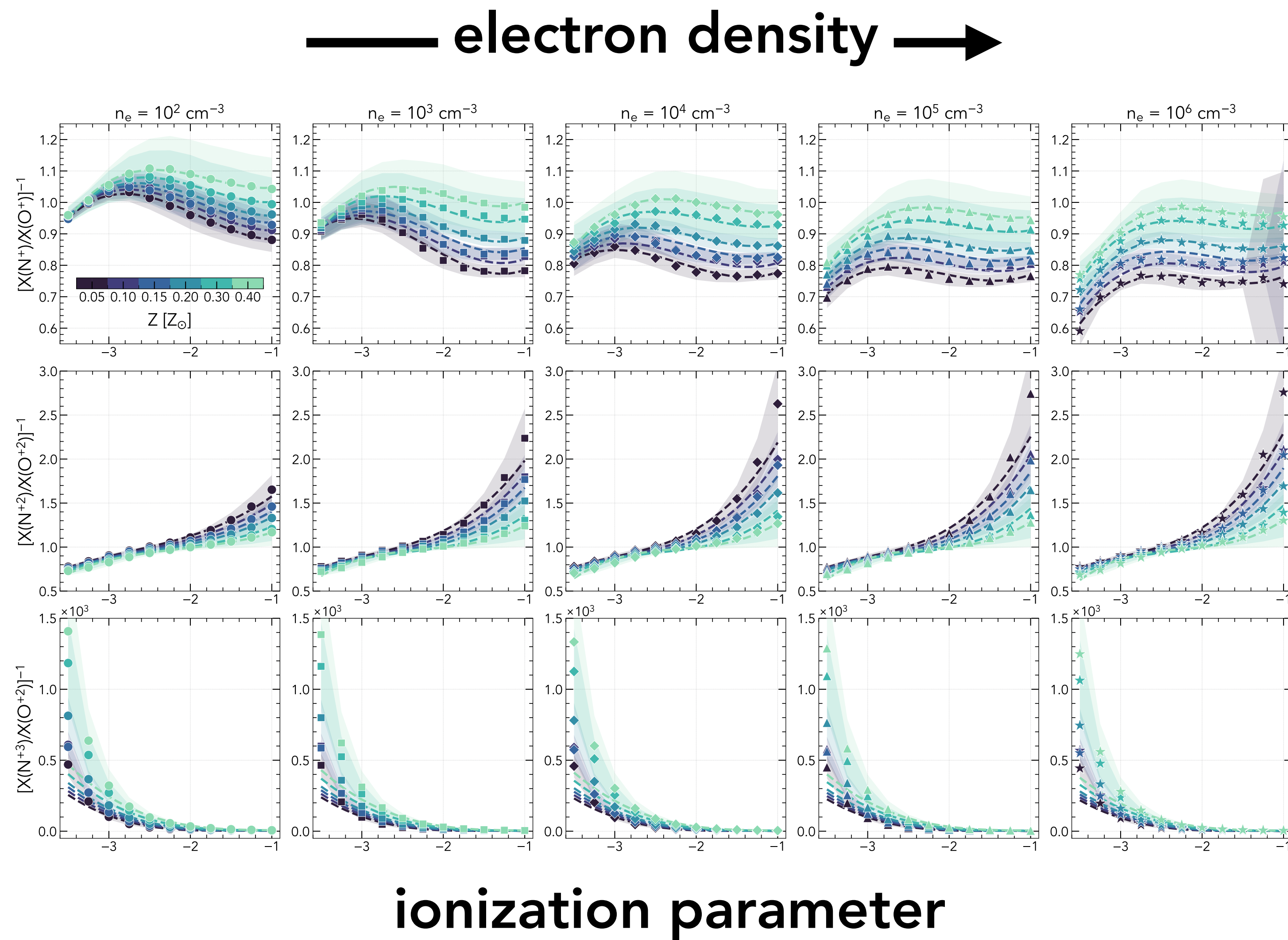
$$\frac{N}{H} = \frac{N^{+i}}{H^+} \times N \text{ ICF}$$

apply to the observed
ionic abundance

New
Ionization
Correction
Factors (ICFs)
with CLOUDY
models

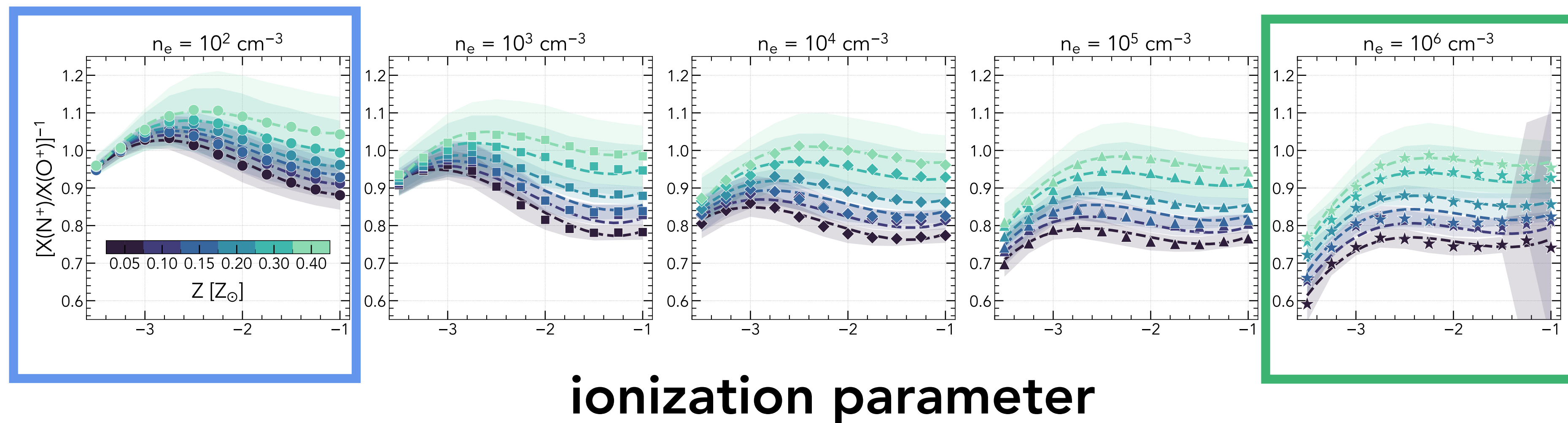
ser

Density changes
each ICF in a
unique way



Density changes
each ICF in a
unique way

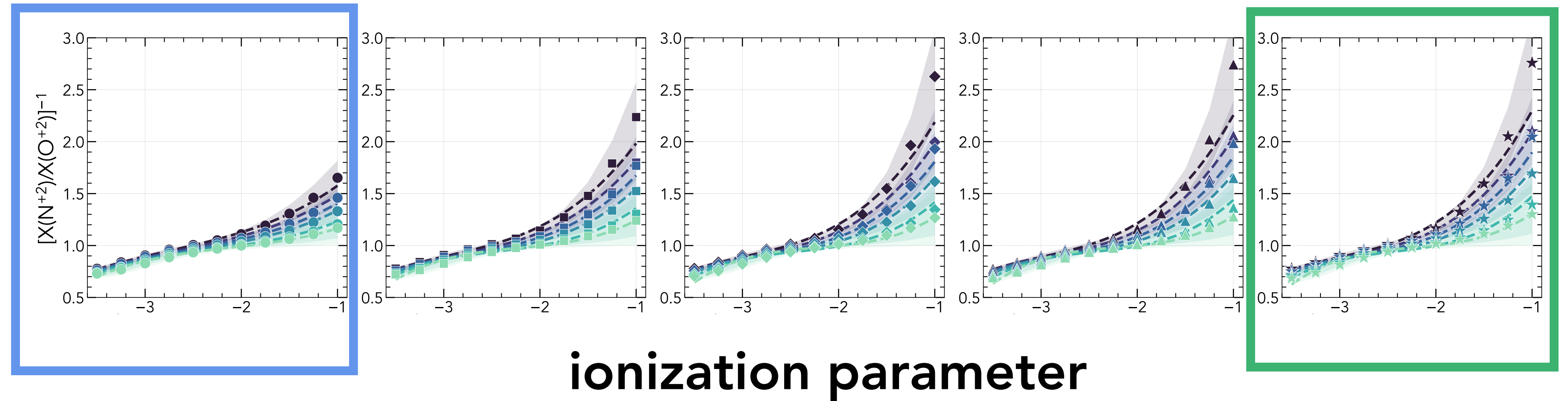
———— electron density ———→



N^+/O^+ goes from being **within 10% of the total N/O**
to **overestimating the N/O by up to 40%** for some
 $\log U$ and Z

Density changes
each ICF in a
unique way

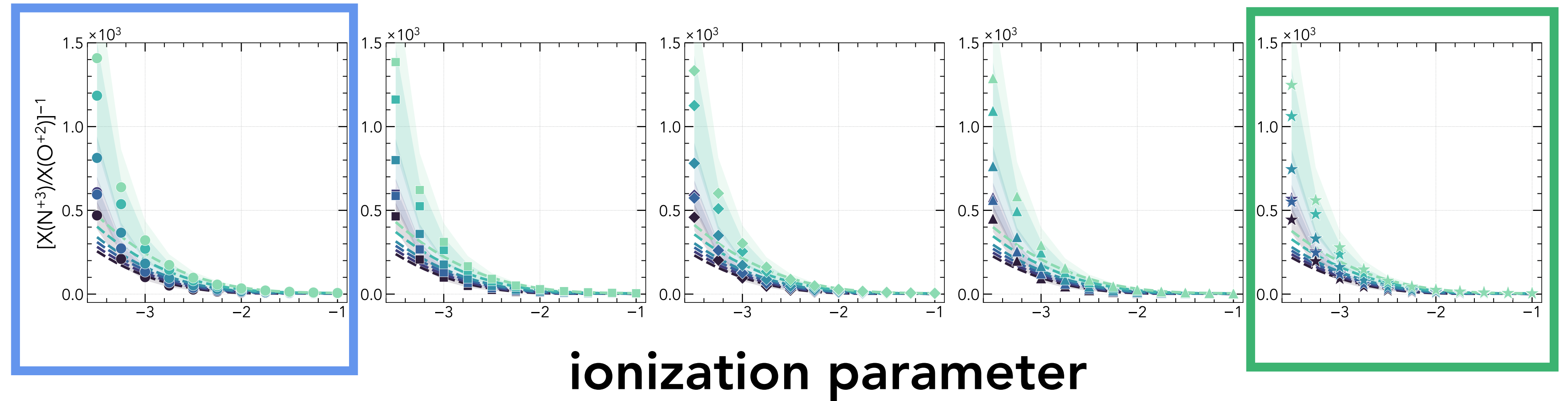
—— electron density →



N^{+2}/O^{+2} maintains its shape but goes from being
within ~50% of the total N/O to under estimating
the N/O by up to 300% for at high $\log U$ and low Z

Density changes
each ICF in a
unique way

—— electron density ——→



N^{+3}/O^{+2} maintains consistency in that it
underestimates total N/O by as much 100 -
15000% and continues underestimating the total
N/O by roughly the same amount

**When $n_e = 10^5 \text{ cm}^{-3}$, but you
assume $n_e = 10^2 \text{ cm}^{-3}$...**

$\log U$ will be overestimated by 1.13 dex

[OIII] $\lambda 4363/\lambda 5007$ temperatures will
be overestimated by 2200 K

$12 + \log(\text{O}/\text{H})$ will be underestimated
by 0.38 dex

**Assuming the
low-density limit
in a high-density
case will
significantly bias
our measurements**

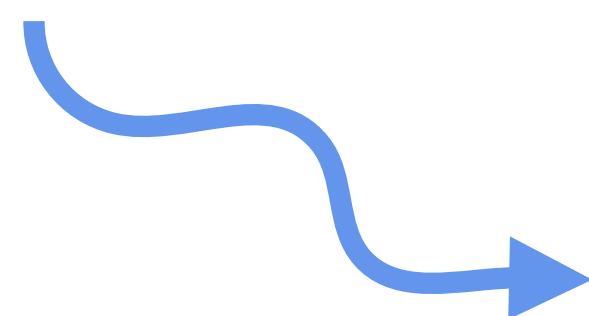
Are high densities
biasing observed
galaxy properties?

yes!

How much does
the interpretation
of those
properties change
at high densities?

Photoionization Modeling

extreme galaxies require
extreme models



models covering large
ranges of densities and
chemical properties



create tools
to interpret
high-z
sources

Modeling + Observational Study

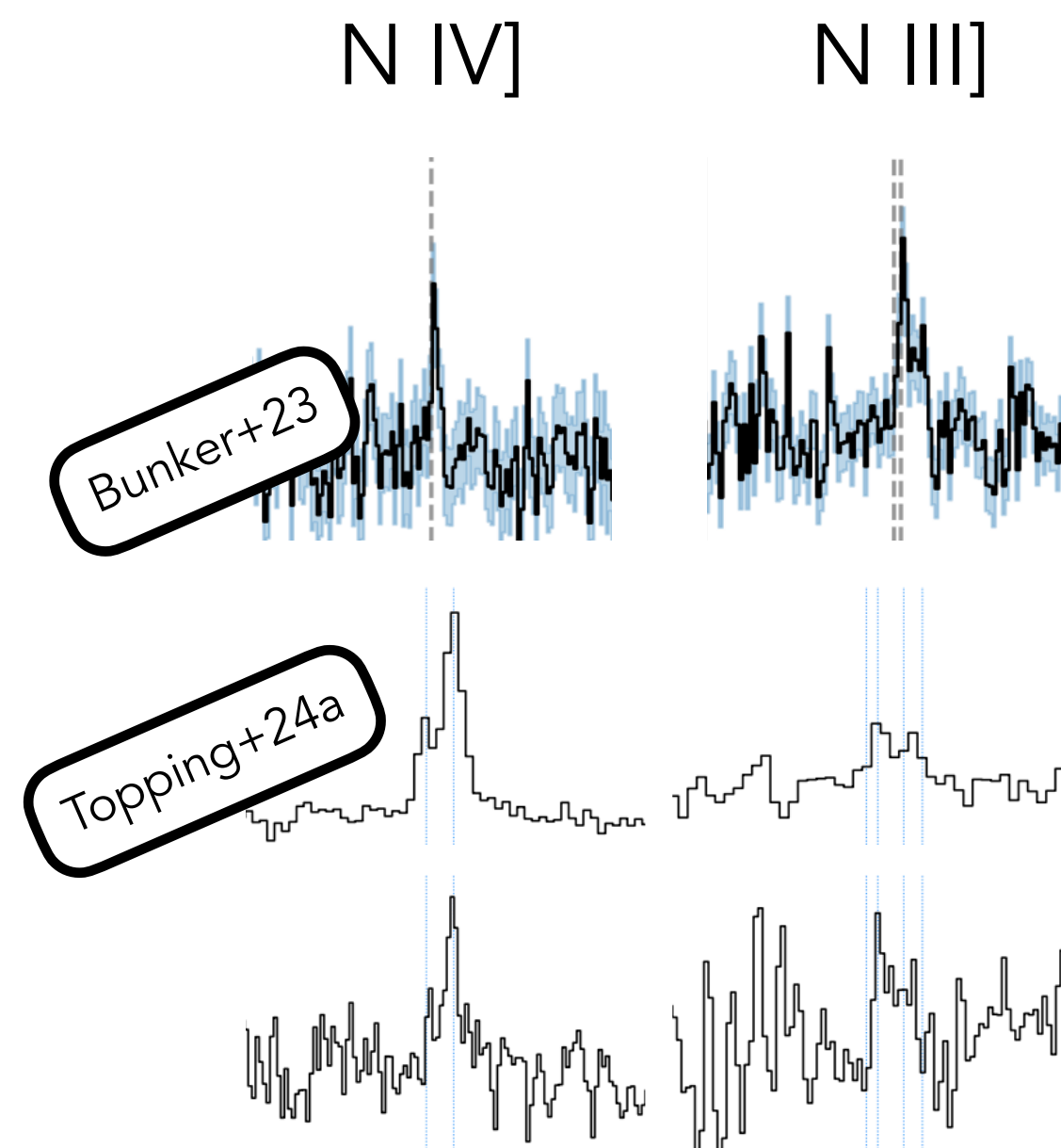
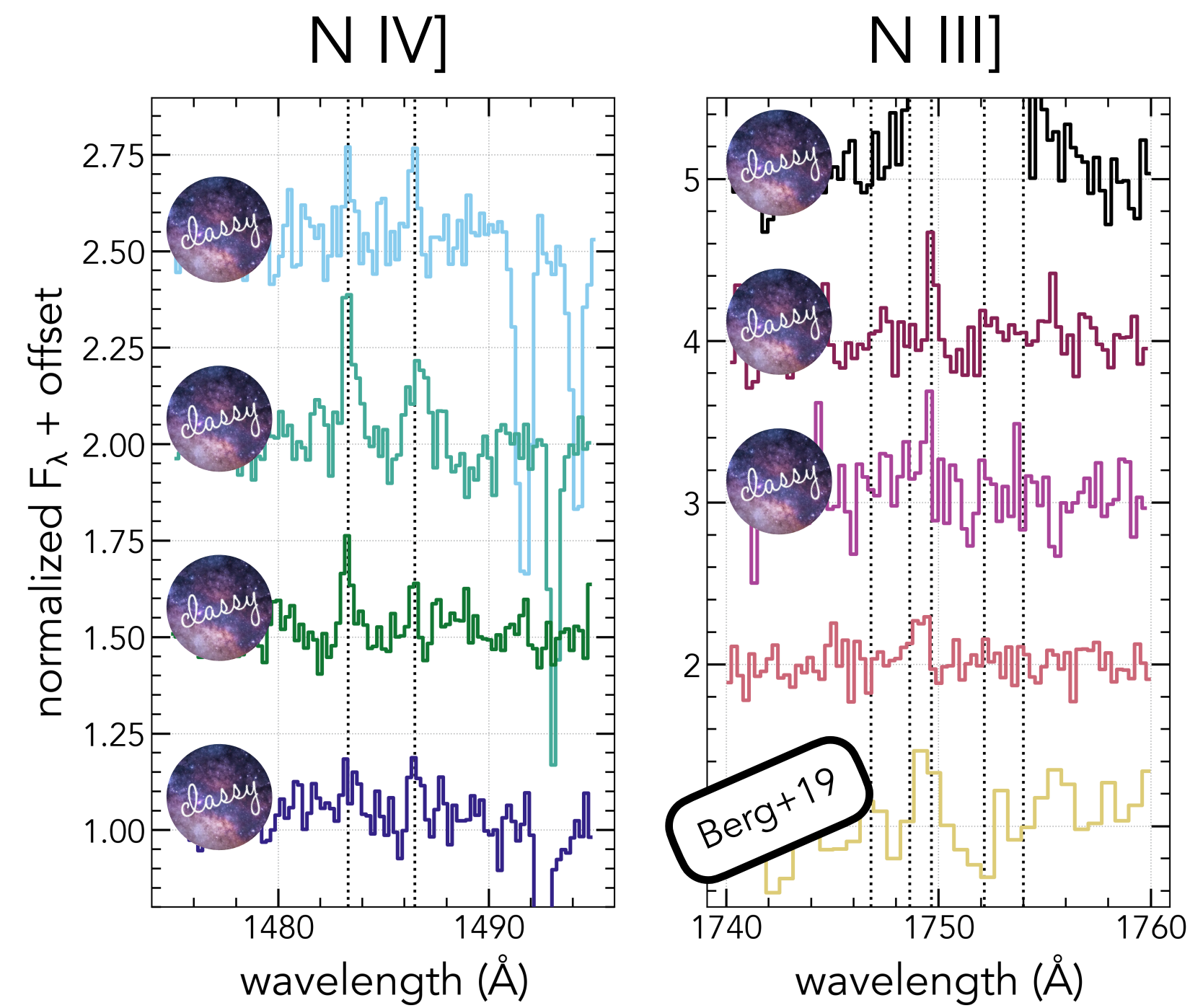
UV N-emitting sample
from $0 < z < 11$



uniform abundance
determination



test varied
approaches to
the density
structure



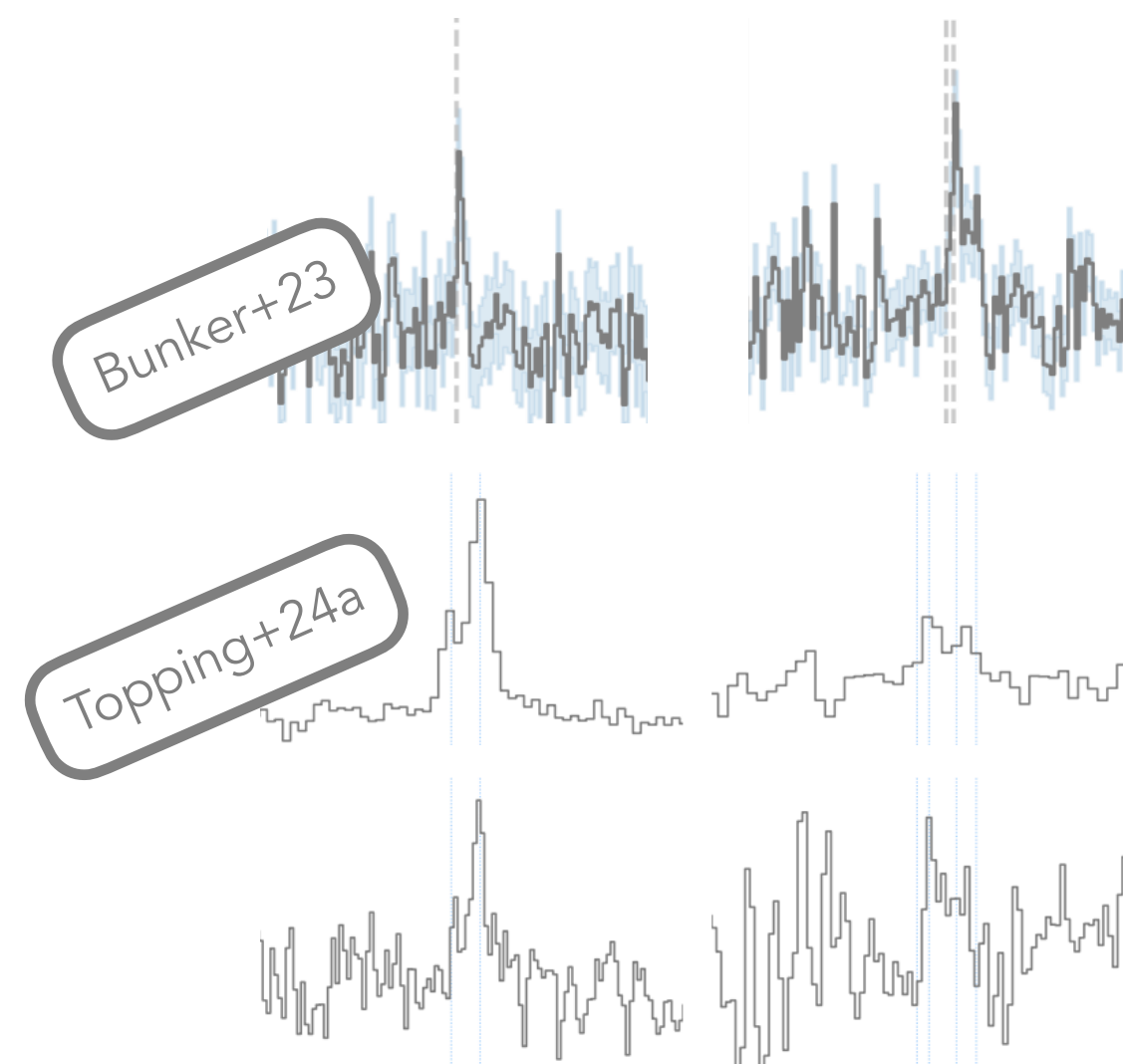
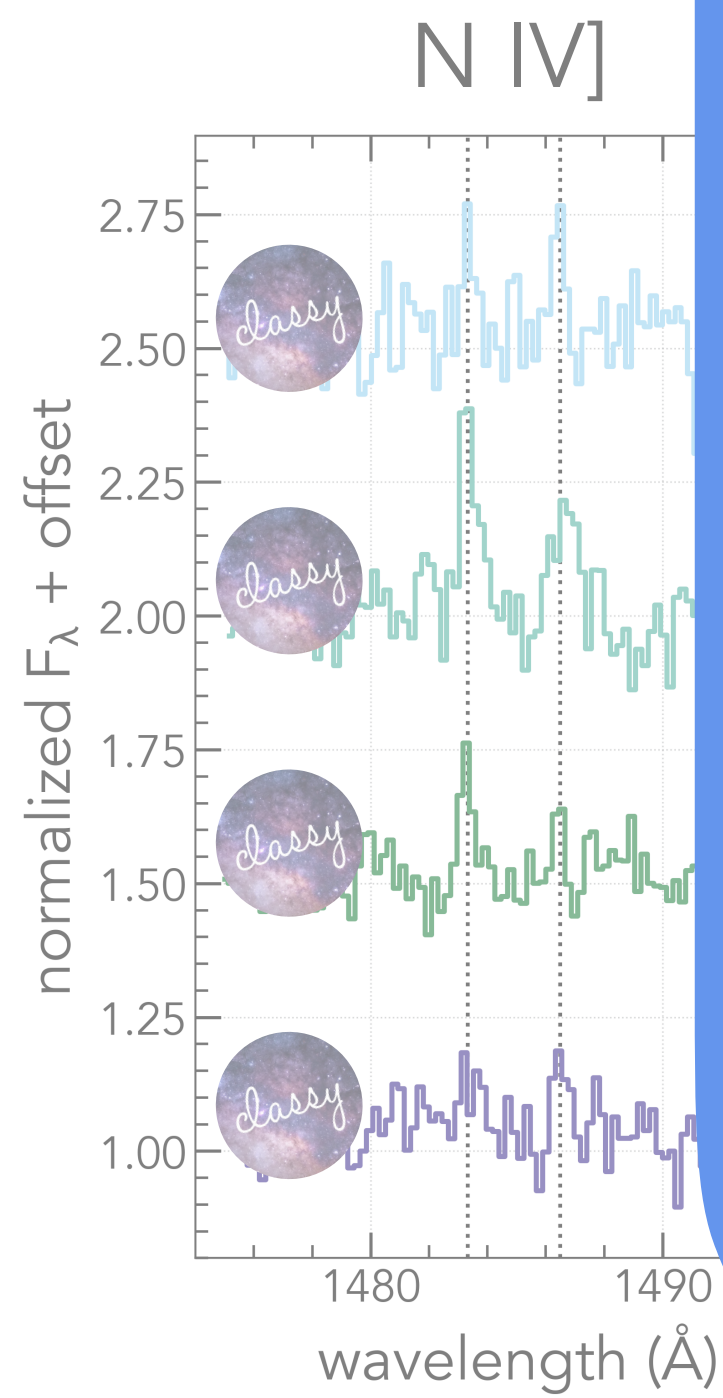
+ more

Sample of UV N Emitters across Cosmic Time



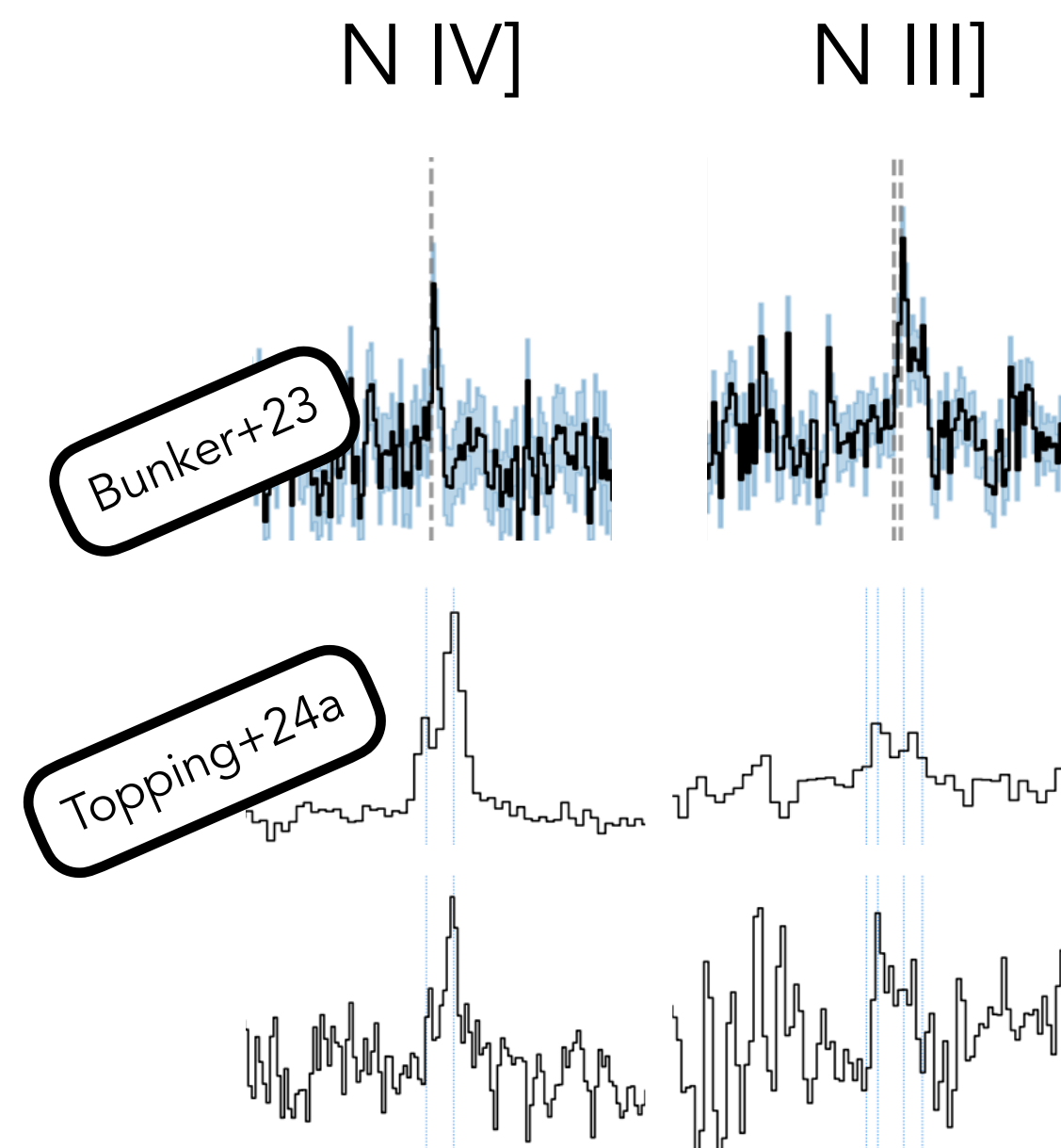
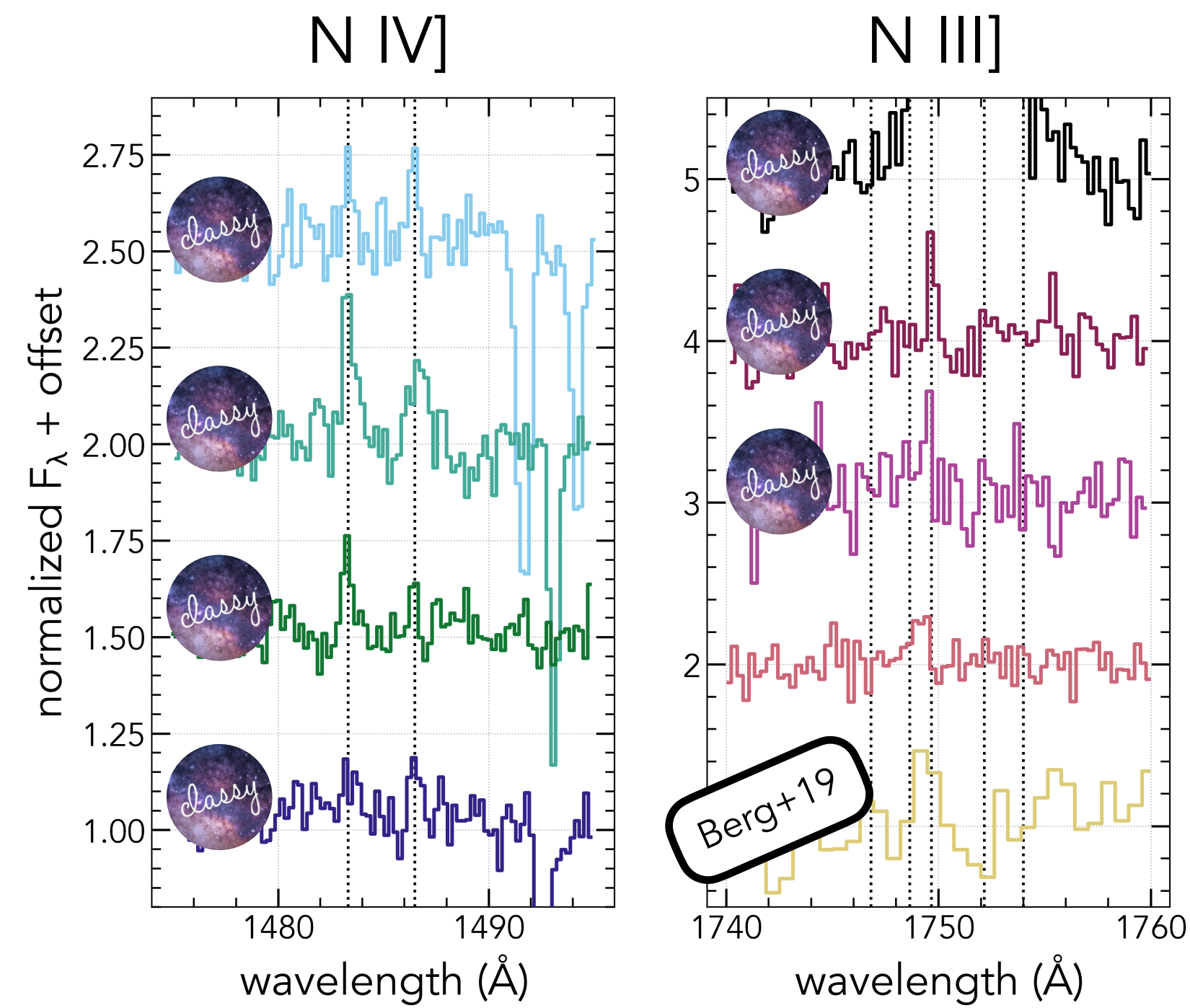
COS Legacy Archive Spectroscopic SurveyY (CLASSY)

- 45 star-forming galaxies with enhanced star-formation rates
- Rest-frame far-UV spectra enabling detailed study of nebular emission
- High-resolution, high-signal-to-noise spectral atlas with coverage from 1150 - 2000 Å



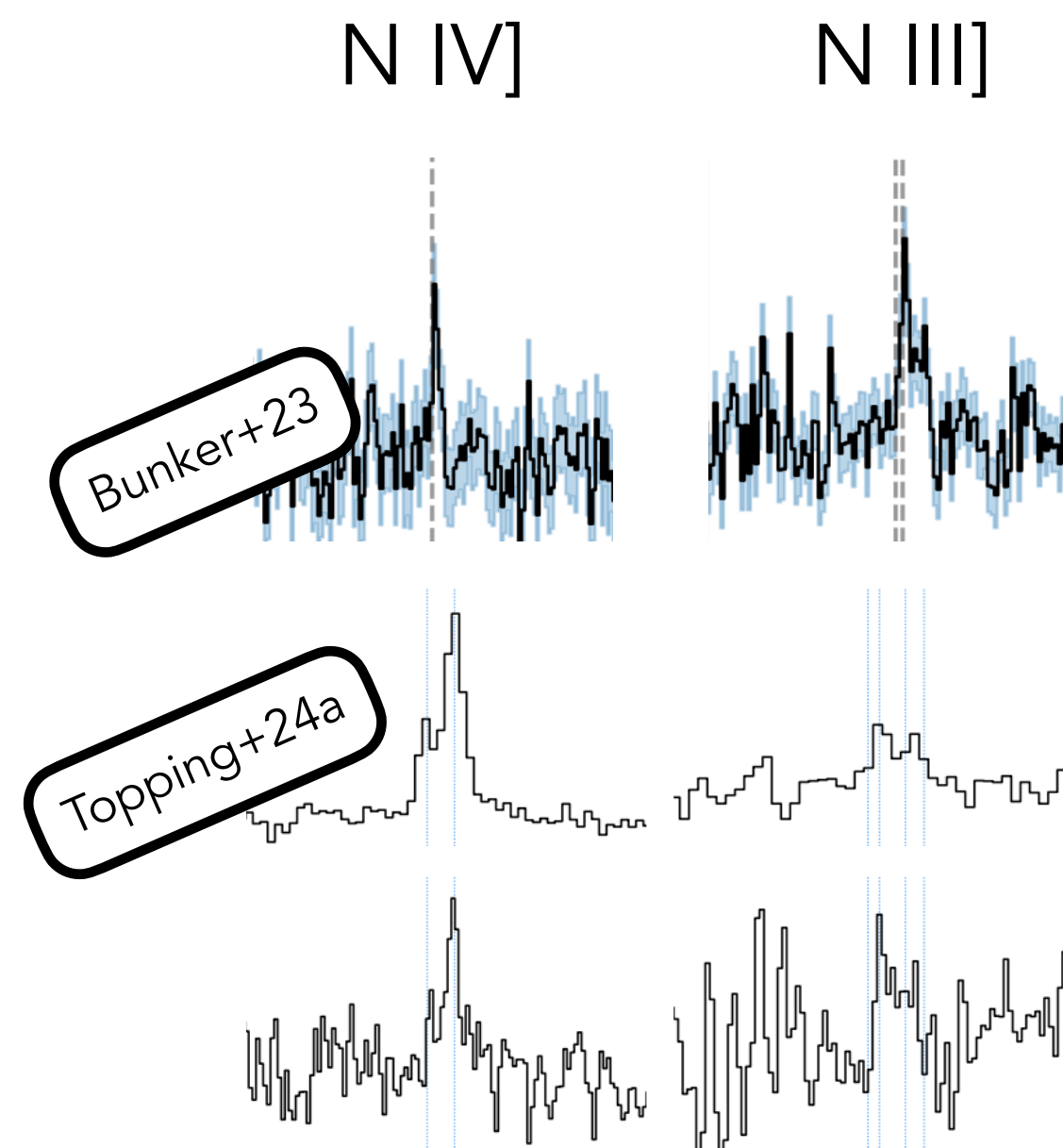
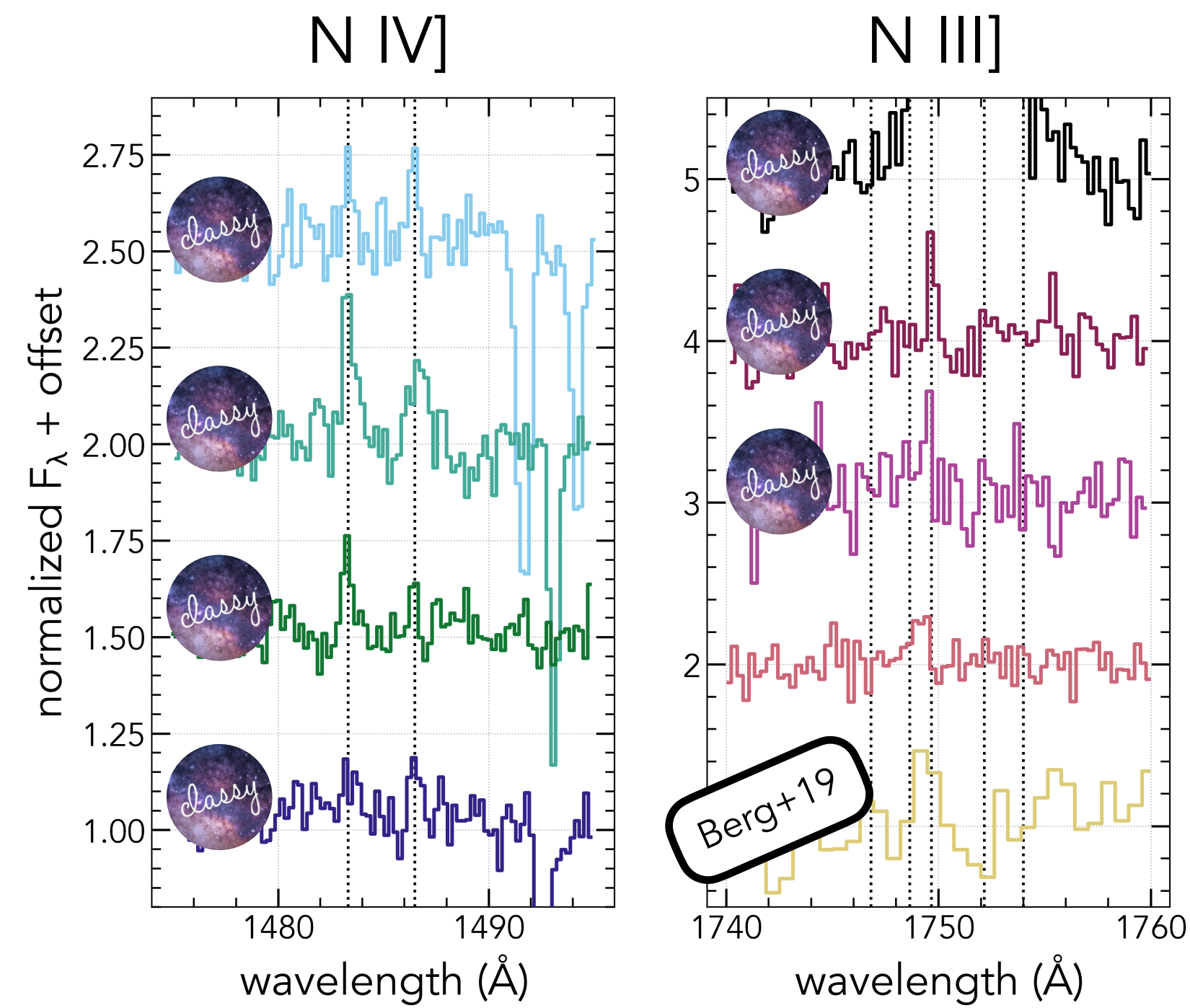
+ more

across Cosmic
Time



+ more

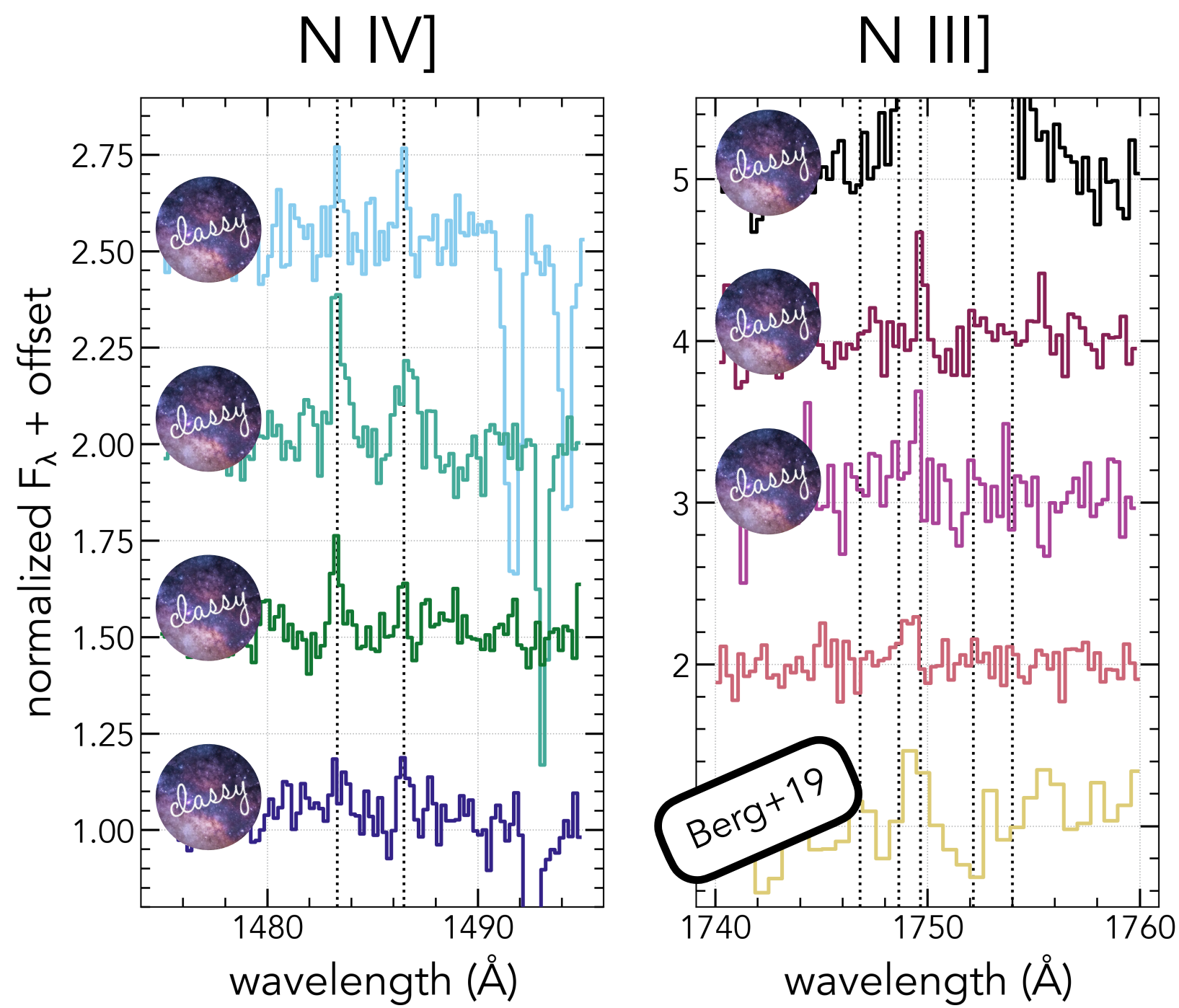
Sample of UV N Emitters across Cosmic Time



+ more

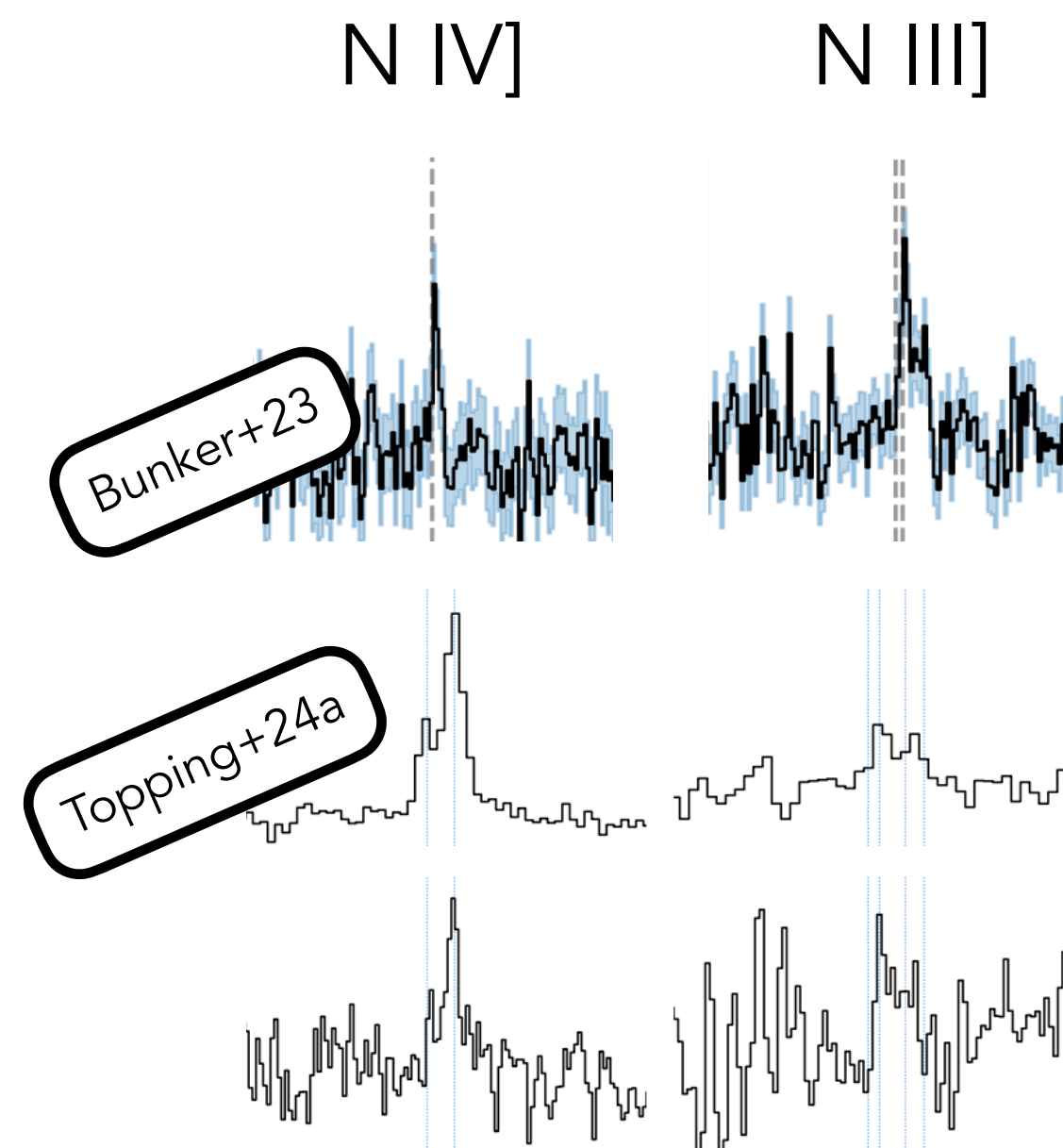
Our sample emphasizes attempting to understand UV N emission and how it may evolve with redshift

Sample of UV N Emitters across Cosmic Time



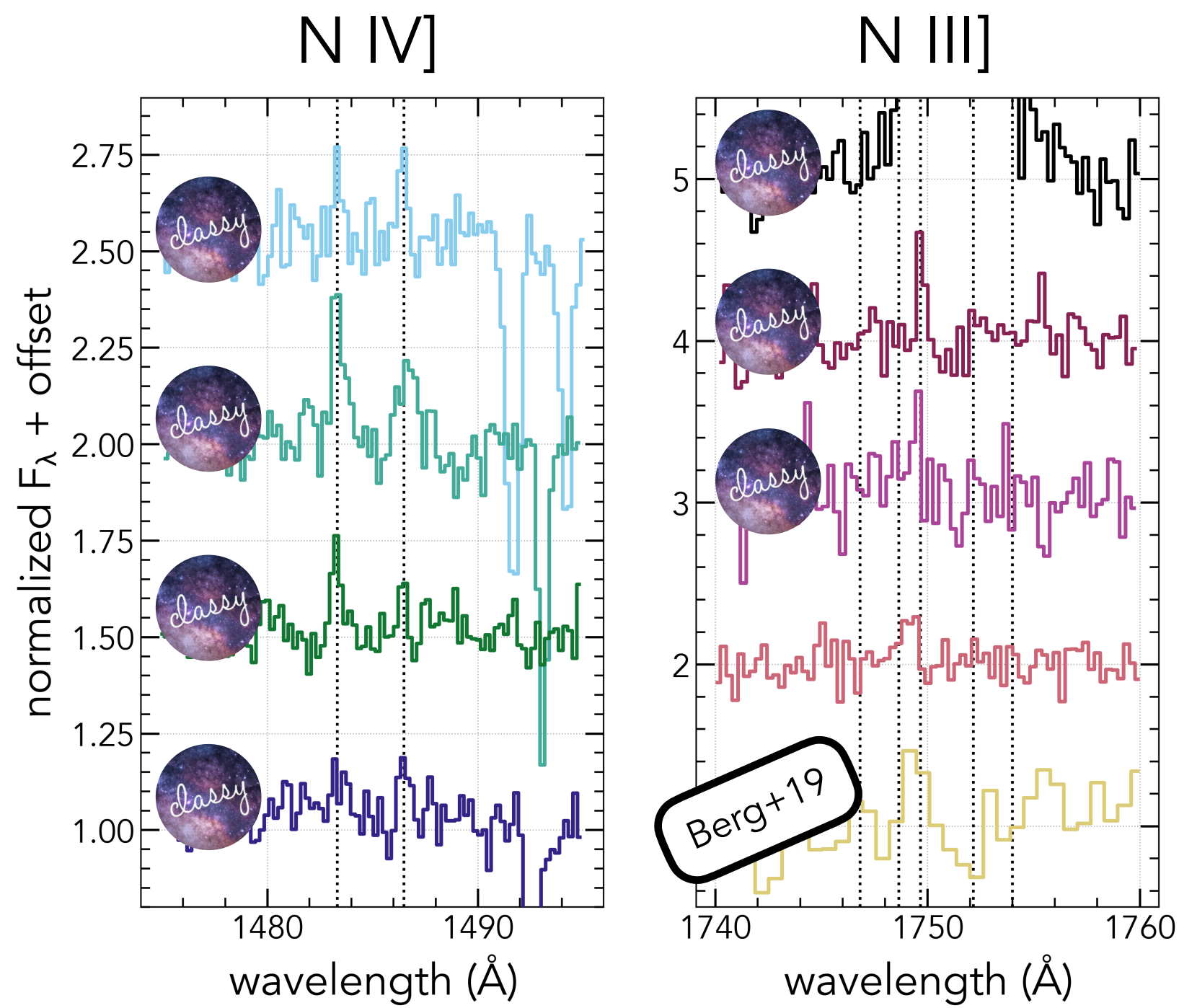
Low-z sample
are **either** N III]
or N IV]
emitters

Our sample
emphasizes
attempting to
understand
UV N
emission and
how it may
evolve with
redshift



+ more

Sample of
UV N
Emitters
across
Cosmic Time

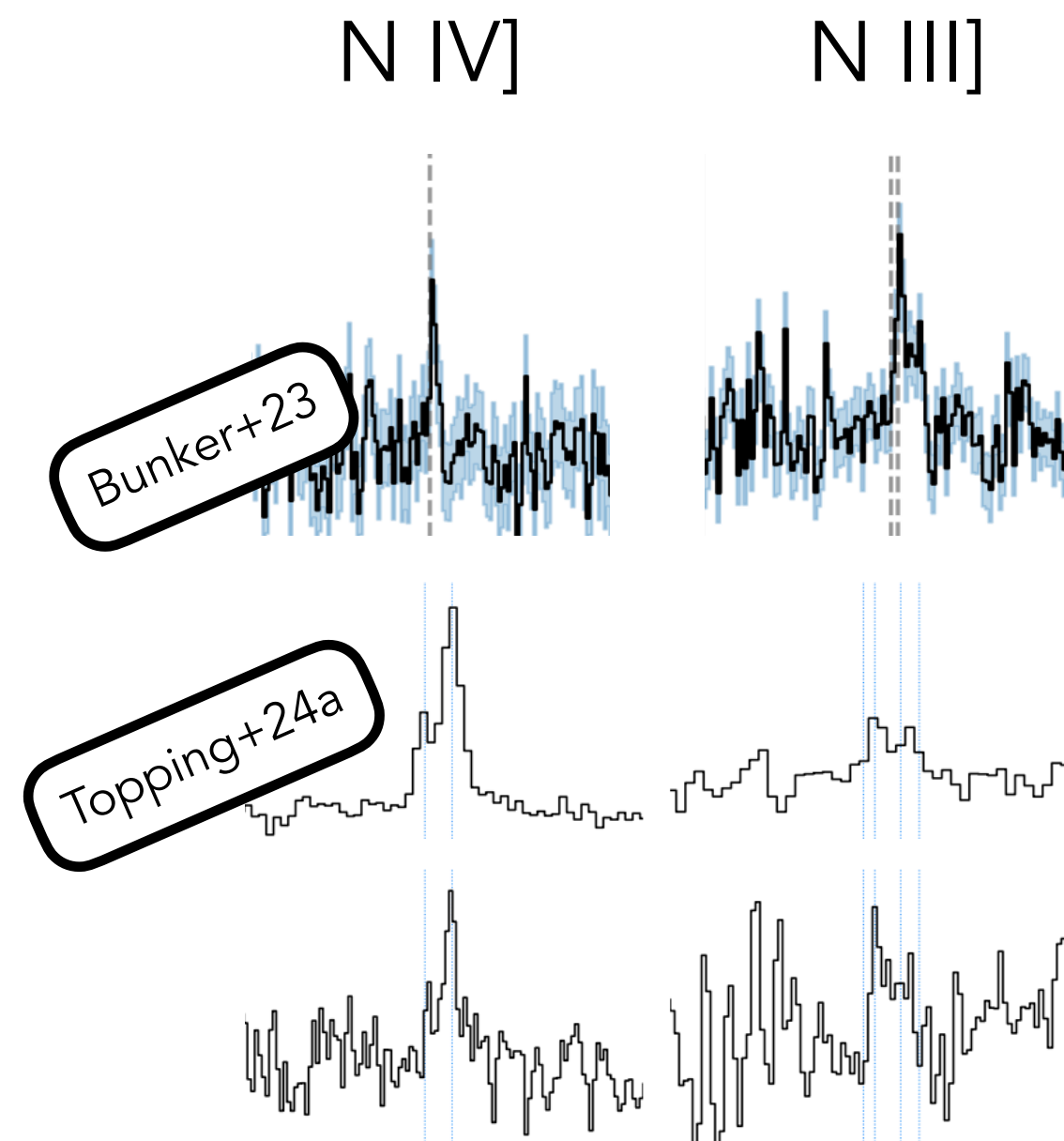


Low-z sample
are **either** N III]
or N IV]
emitters

Our sample
emphasizes
attempting to
understand
UV N
emission and
how it may
evolve with
redshift

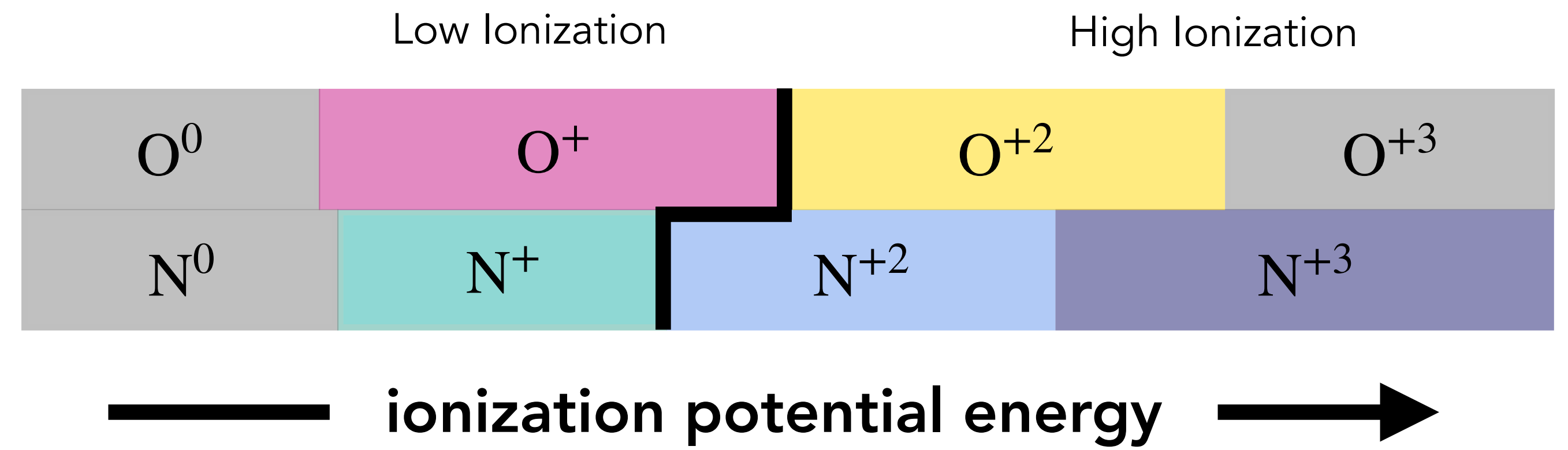
Sample of
UV N
Emitters
across
Cosmic Time

high-z sample
are **often both**
N III] and N IV]
emitters



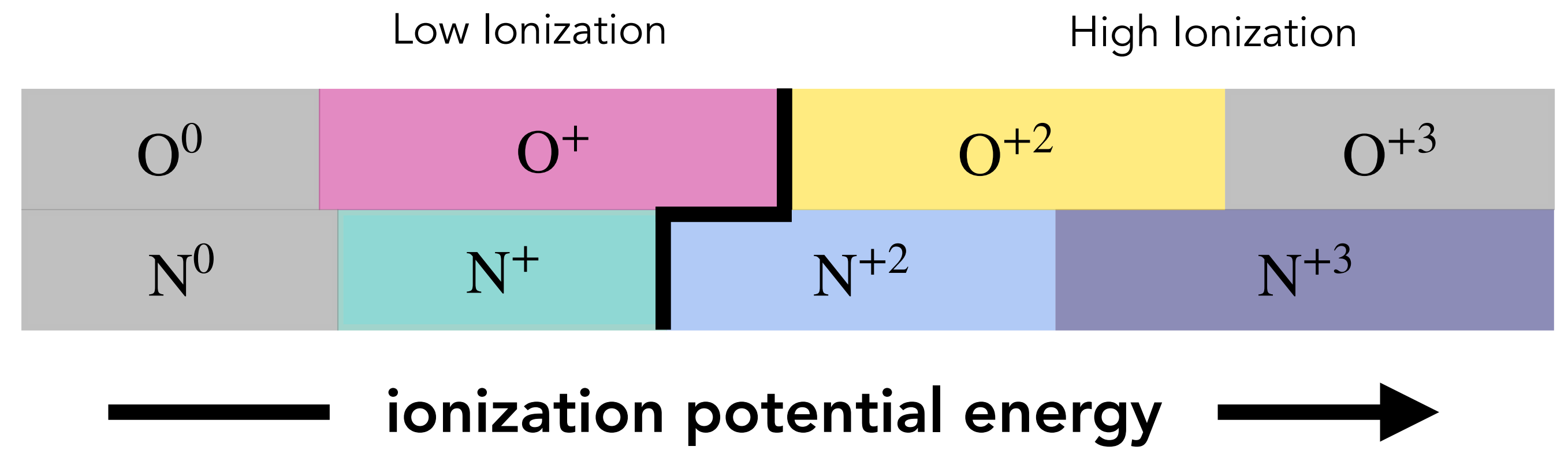
+ more

2-Zone Ionization Model



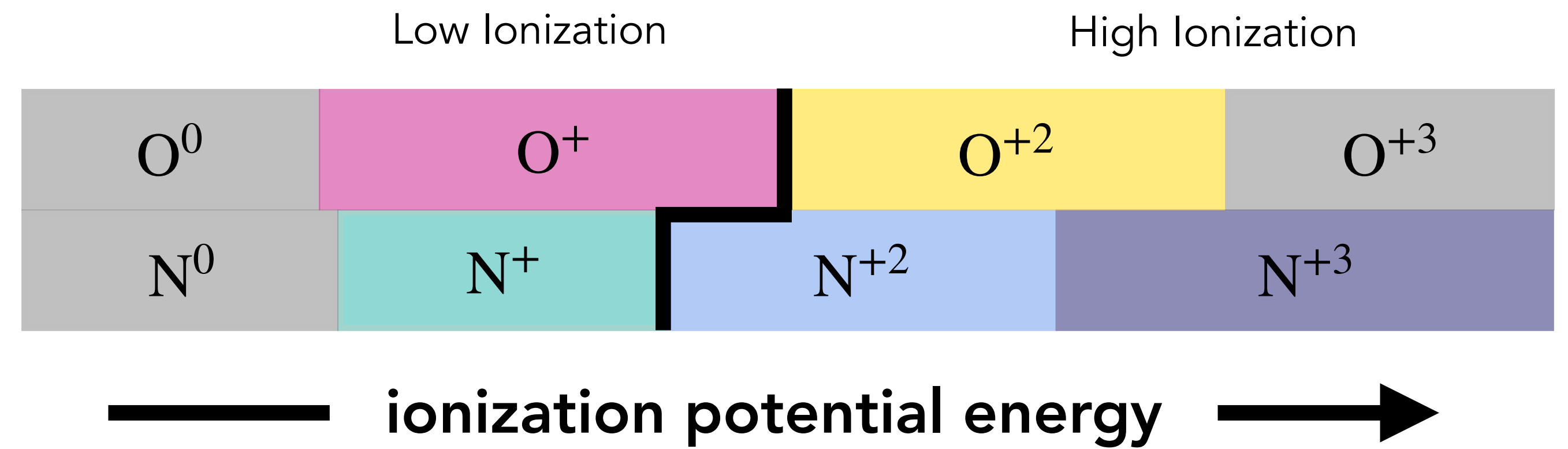
2-Zone Ionization Model

Different
emission lines
will probe
various zones of
the ionization
structure



2-Zone Ionization Model

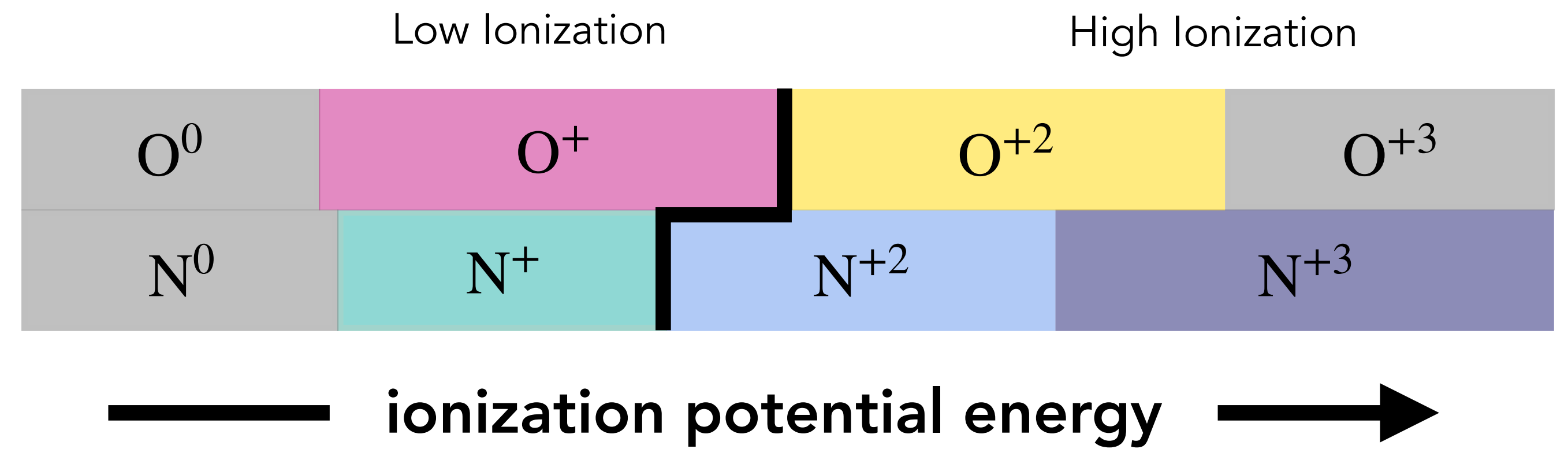
Different
emission lines
will probe
various zones of
the ionization
structure



Our first measurements take the simplest
approach to inferring abundances

2-Zone Ionization Model

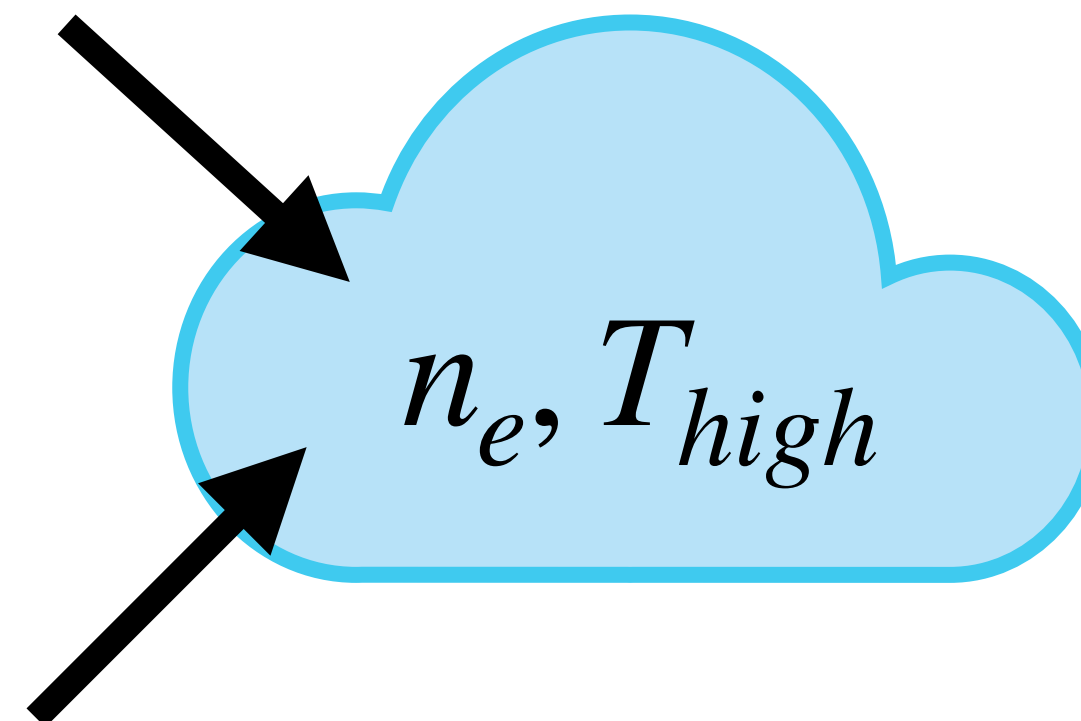
Different emission lines will probe various zones of the ionization structure



Our first measurements take the simplest approach to inferring abundances

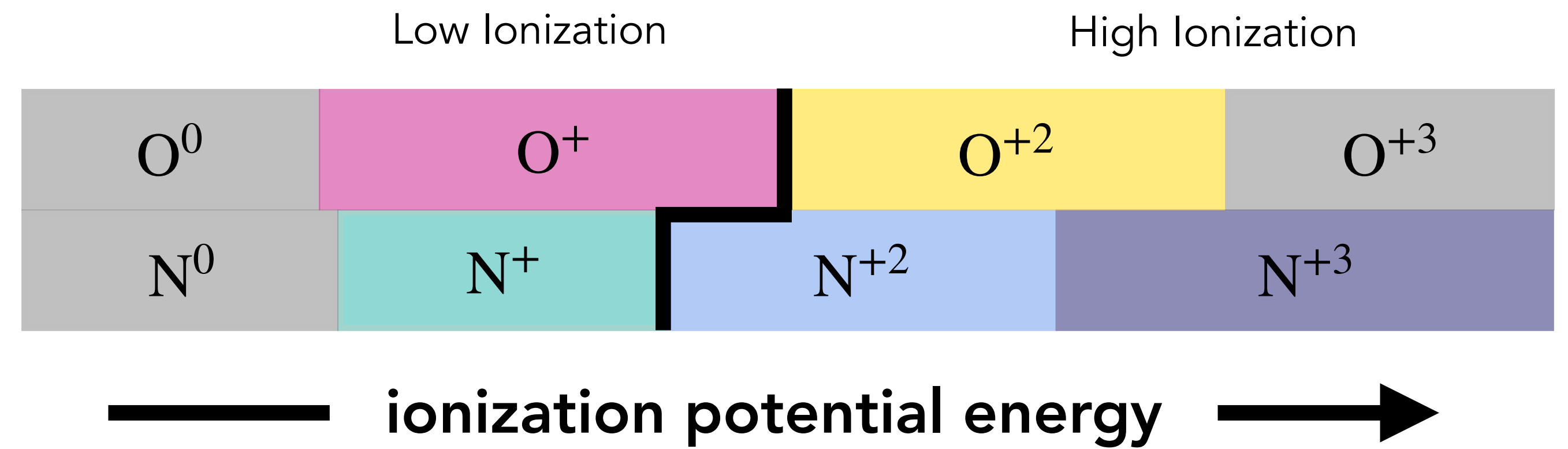
low-ionization
S⁺ and O⁺
emission

high-ionization
O⁺² emission



2-Zone Ionization Model

Different emission lines will probe various zones of the ionization structure

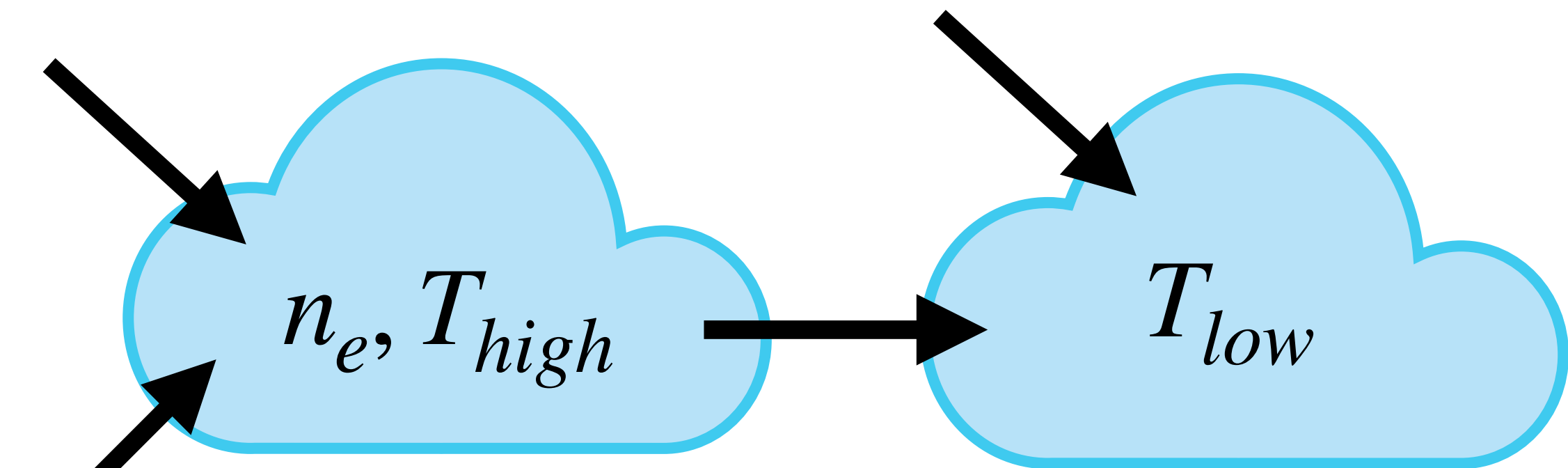


Our first measurements take the simplest approach to inferring abundances

low-ionization
S⁺ and O⁺
emission

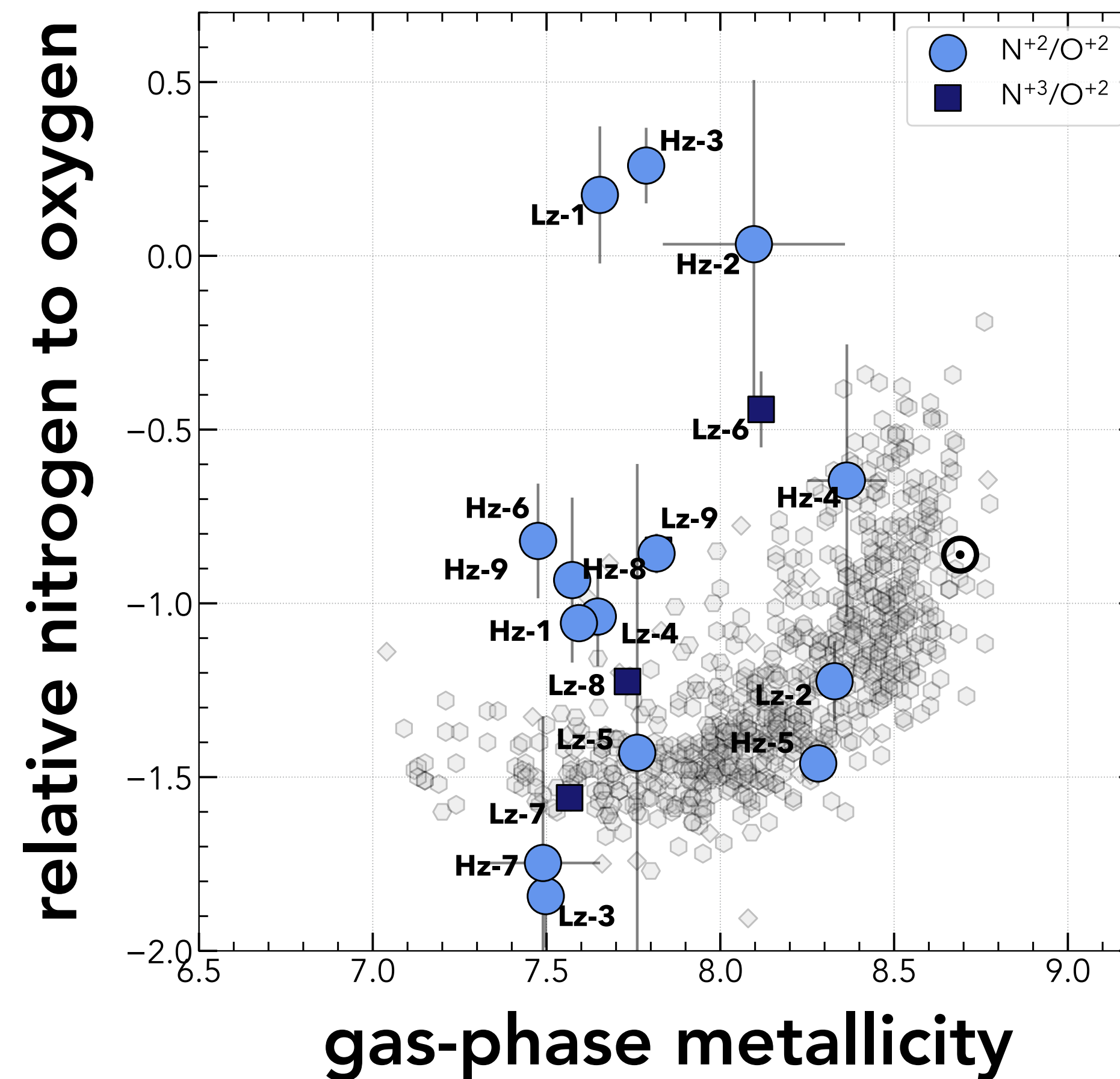
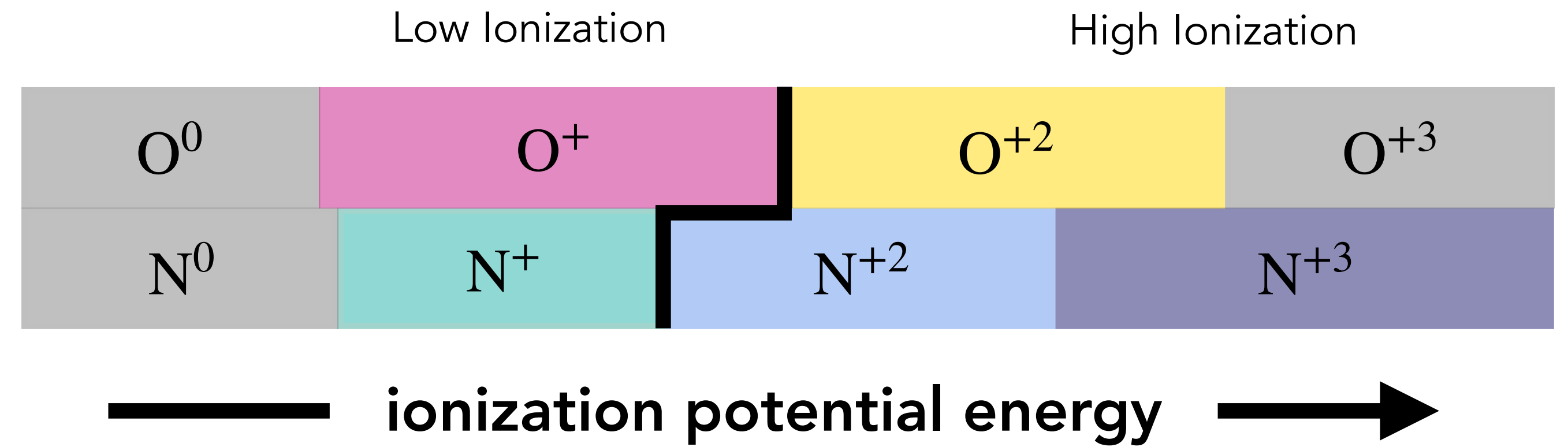
Garnett+92
 $T_e - T_e$ relation

high-ionization
O⁺² emission



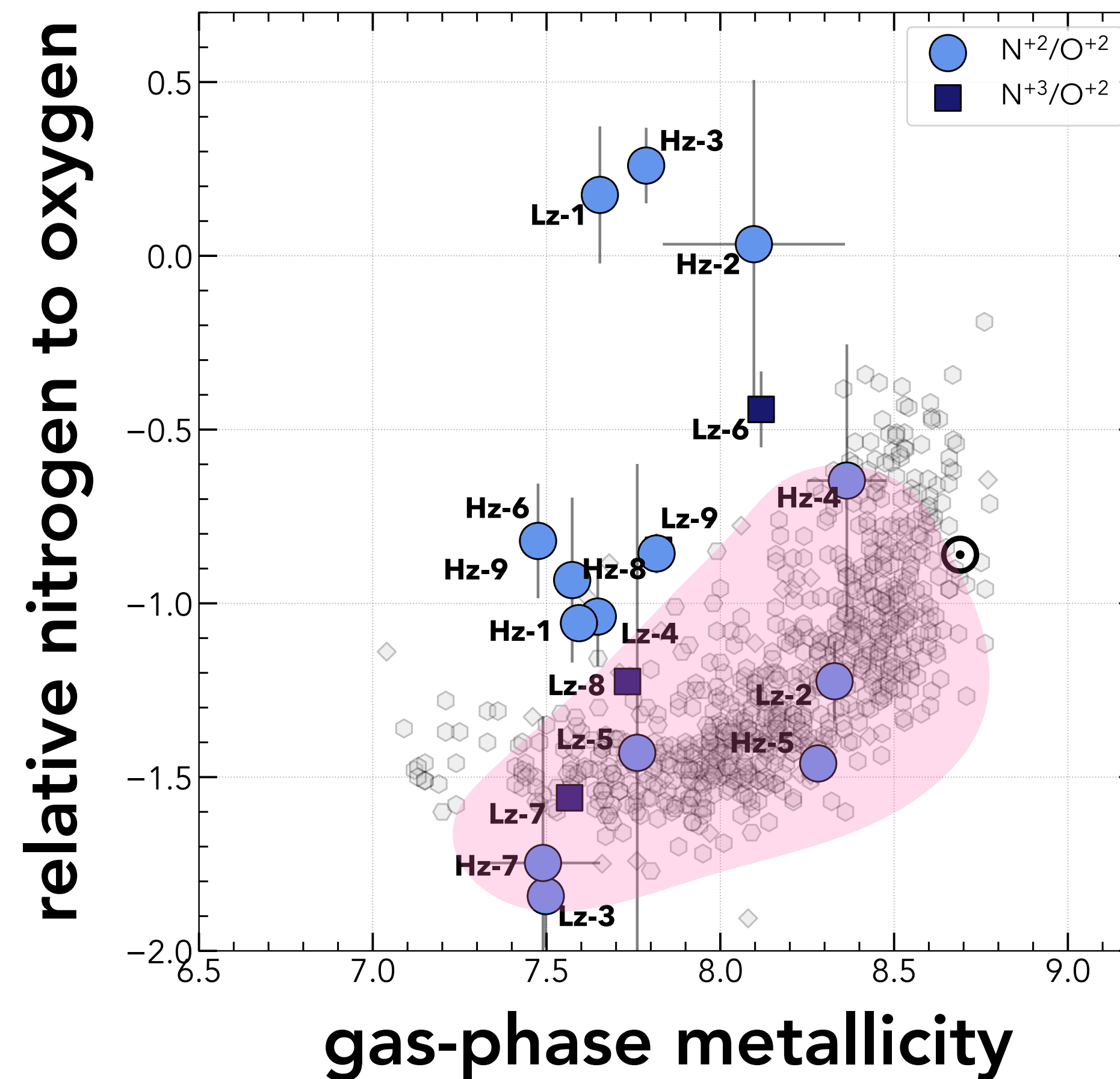
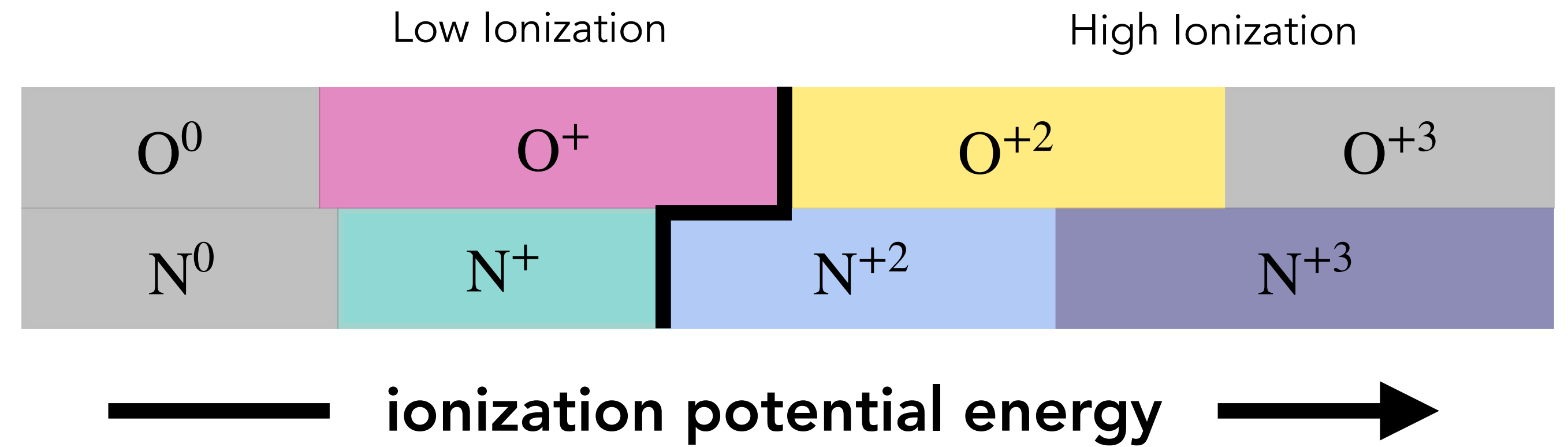
2-Zone Ionization Model

How does this
model stack up
after applying
ICFs?



2-Zone Ionization Model

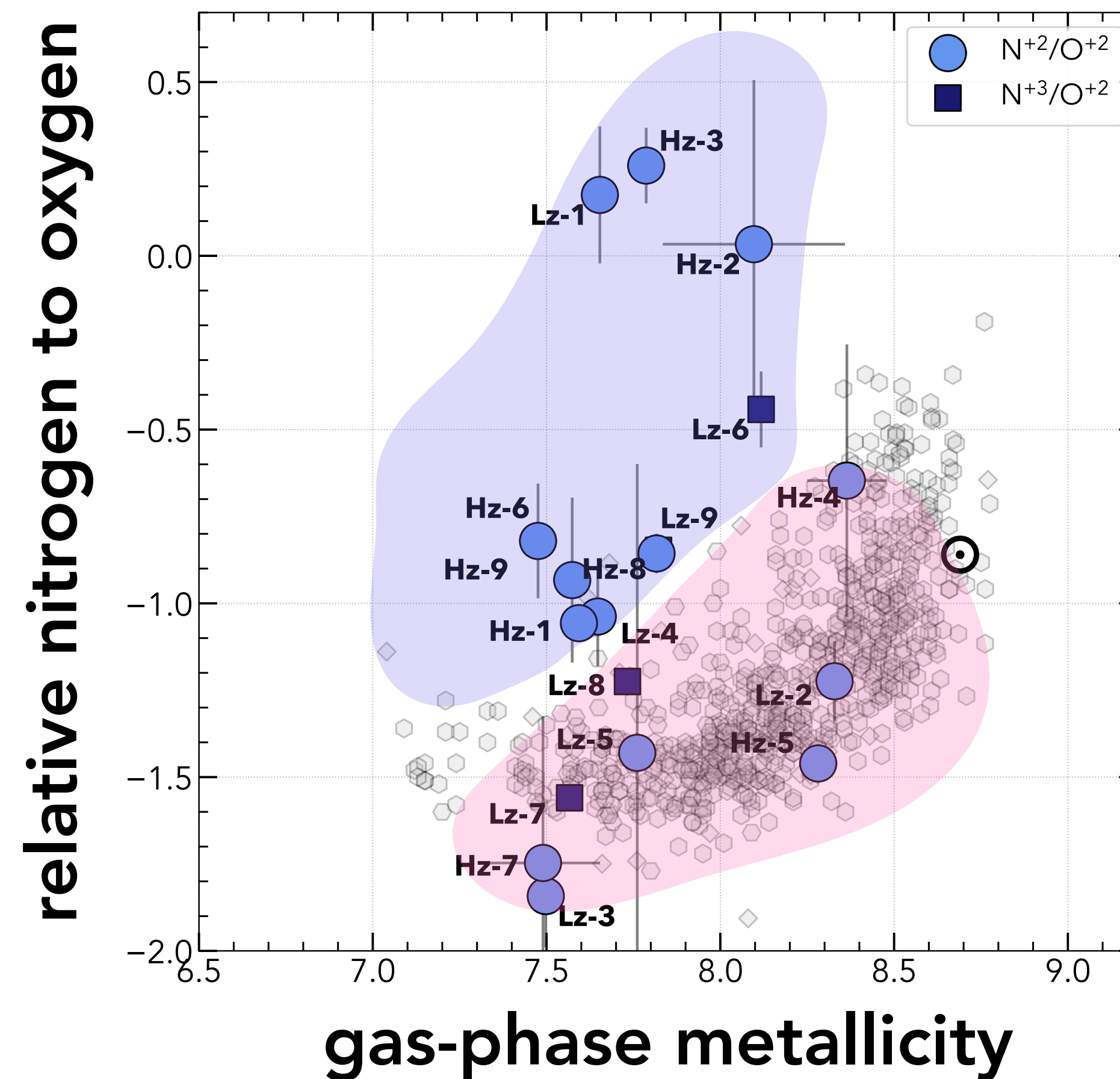
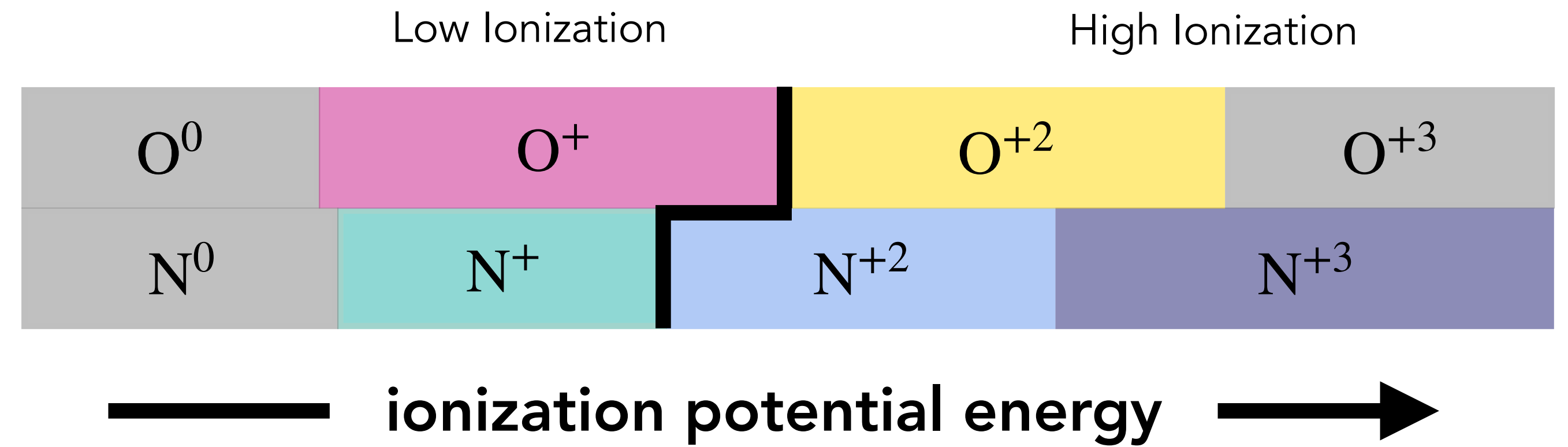
How does this model stack up after applying ICFs?



Low-z sample is **consistent** with low-ionization scatter

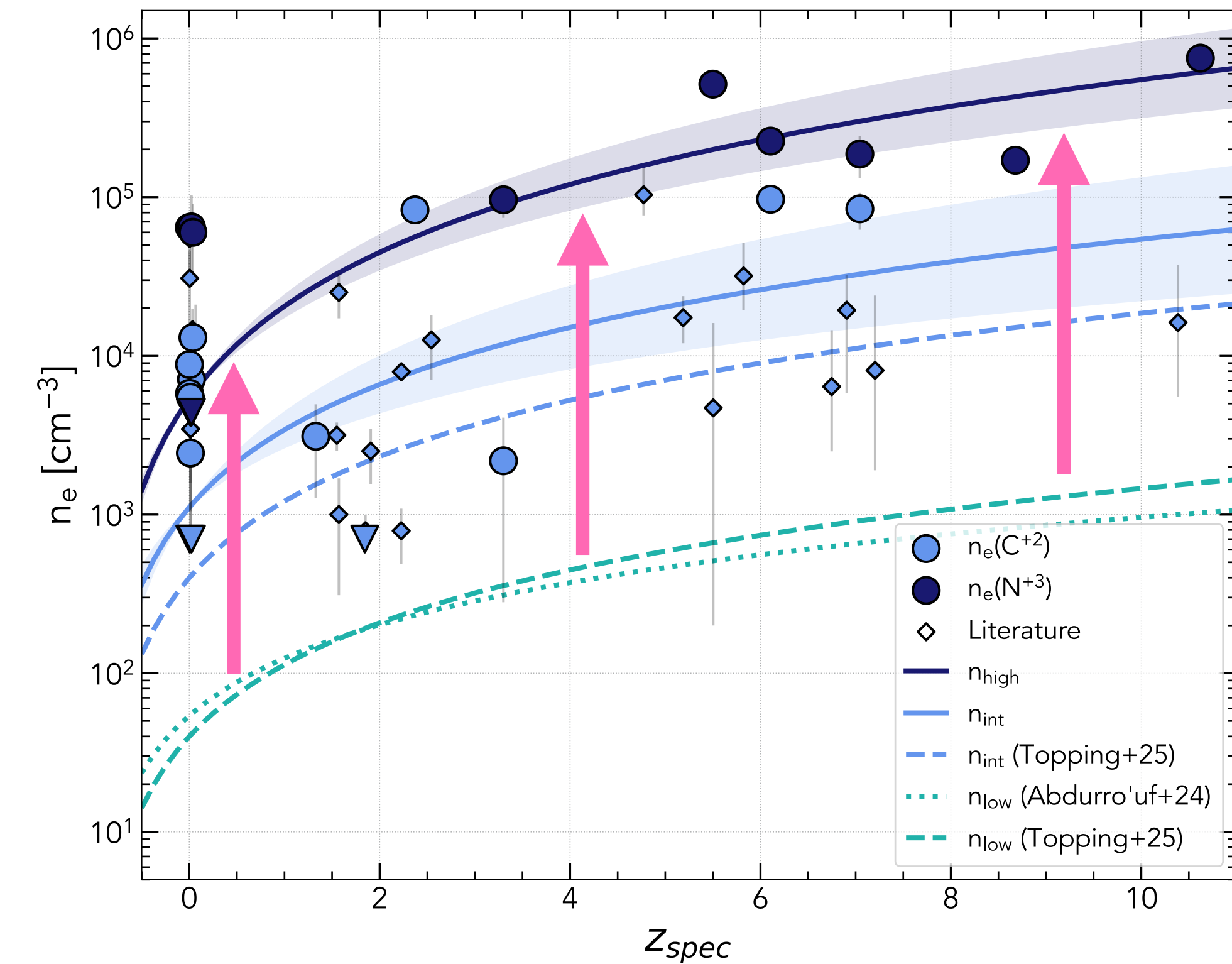
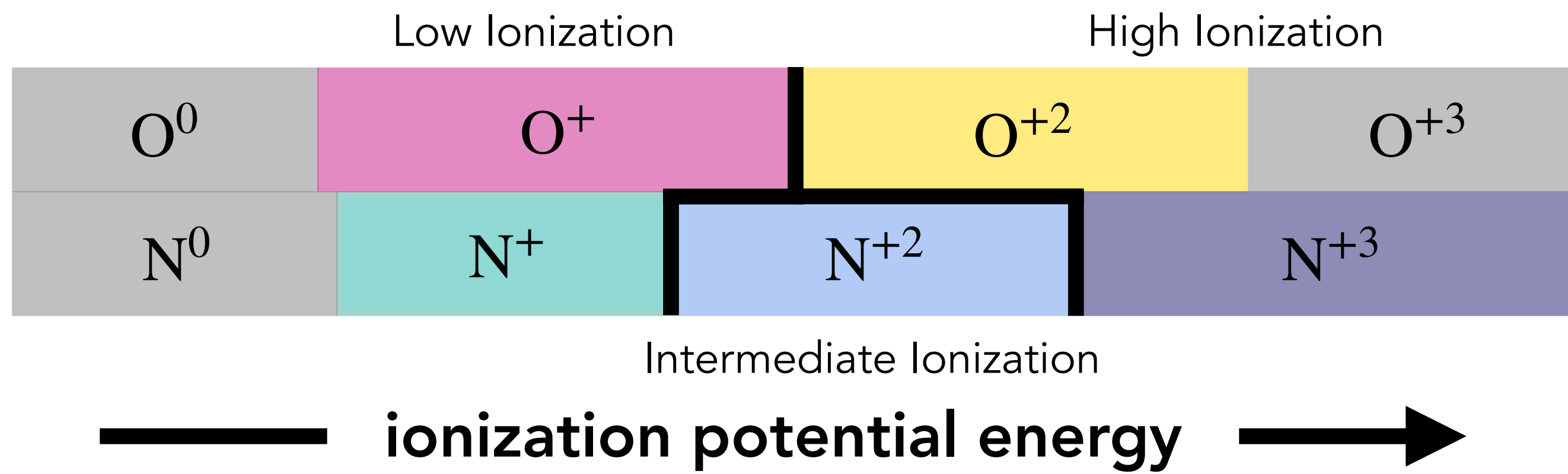
2-Zone Ionization Model

How does this model stack up after applying ICFs?

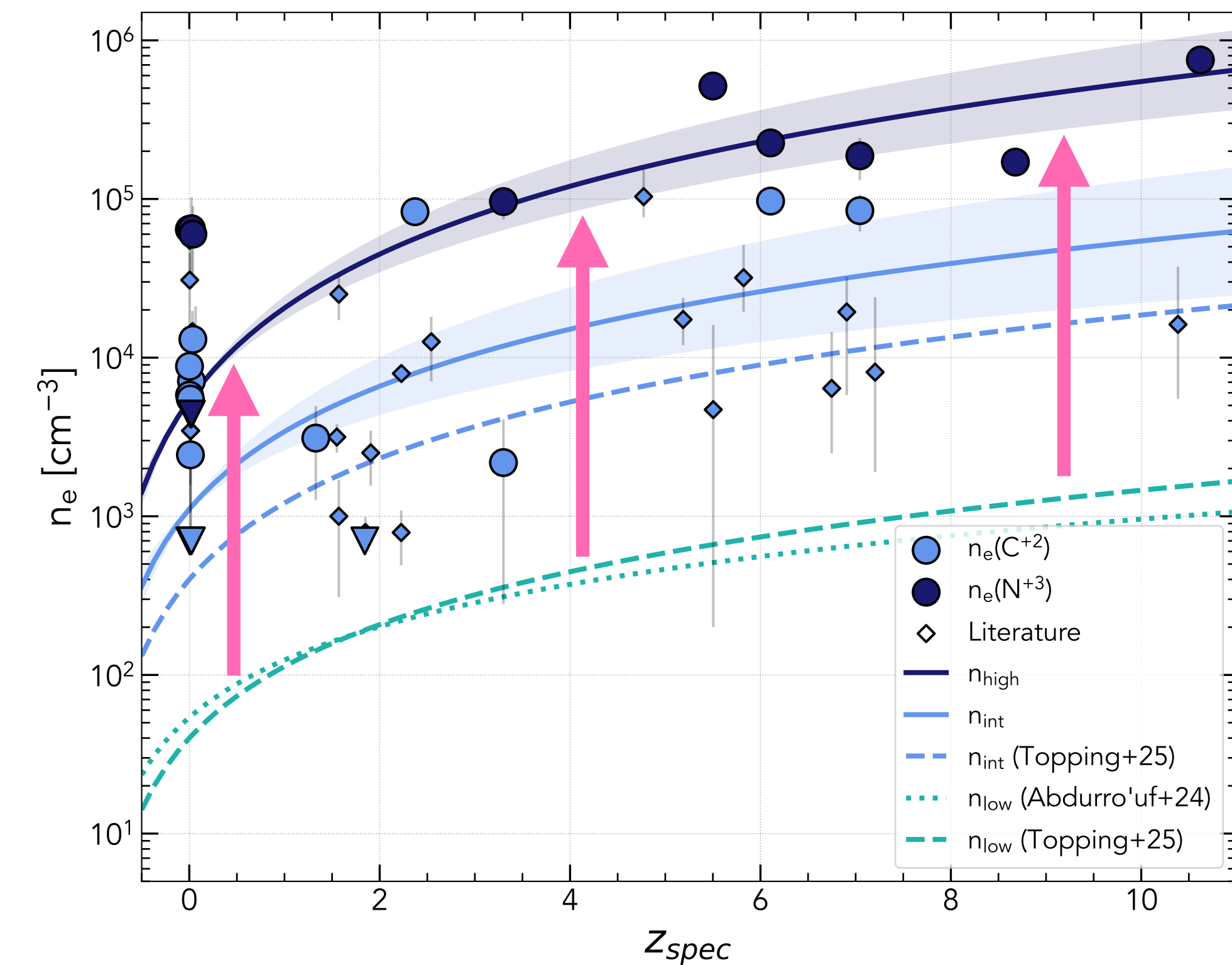
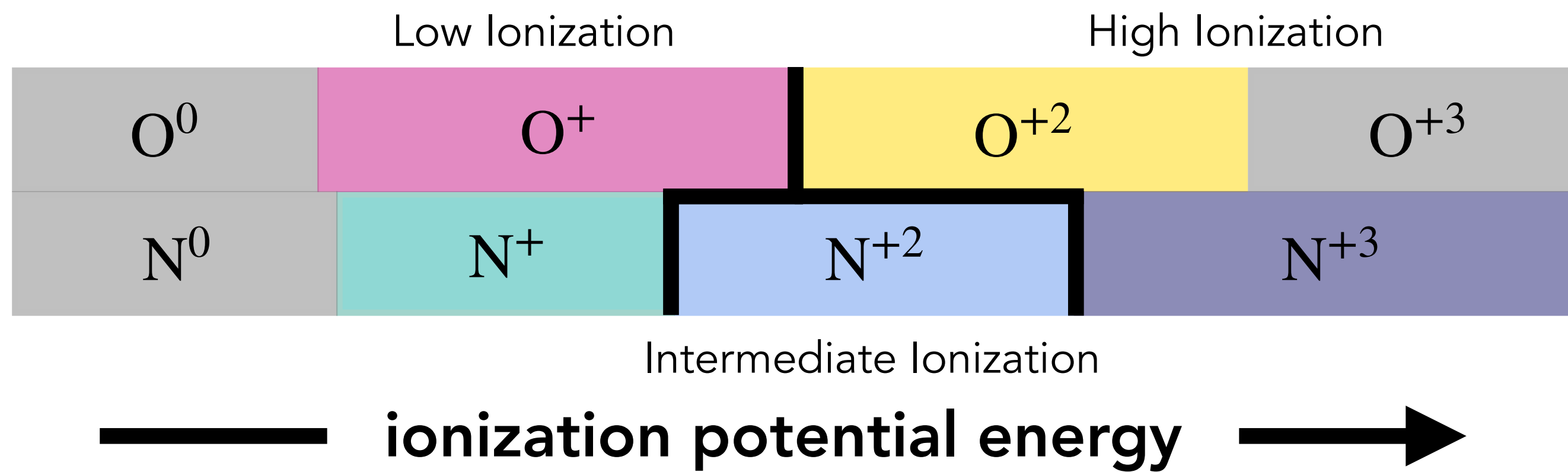


High-z sample produces the **anomalously high-N/O** seen in recent literature

Low-z sample is **consistent** with low-ionization scatter

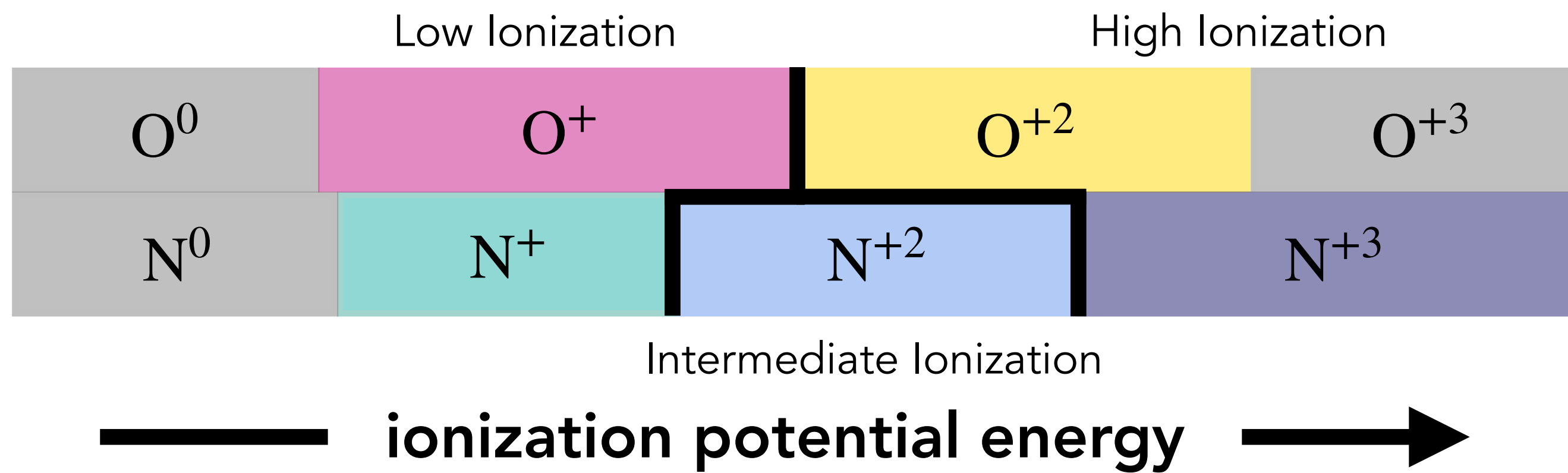


3-Zone Ionization Model



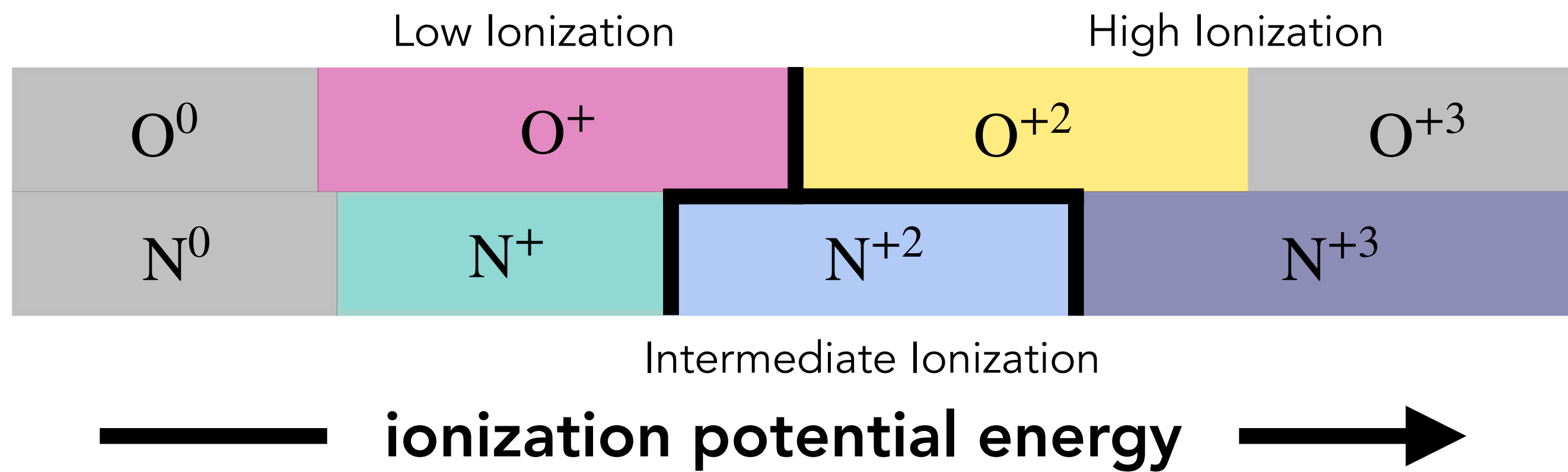
Density sensitive lines from **higher ionization** zones produce **higher densities**

3-Zone Ionization Model

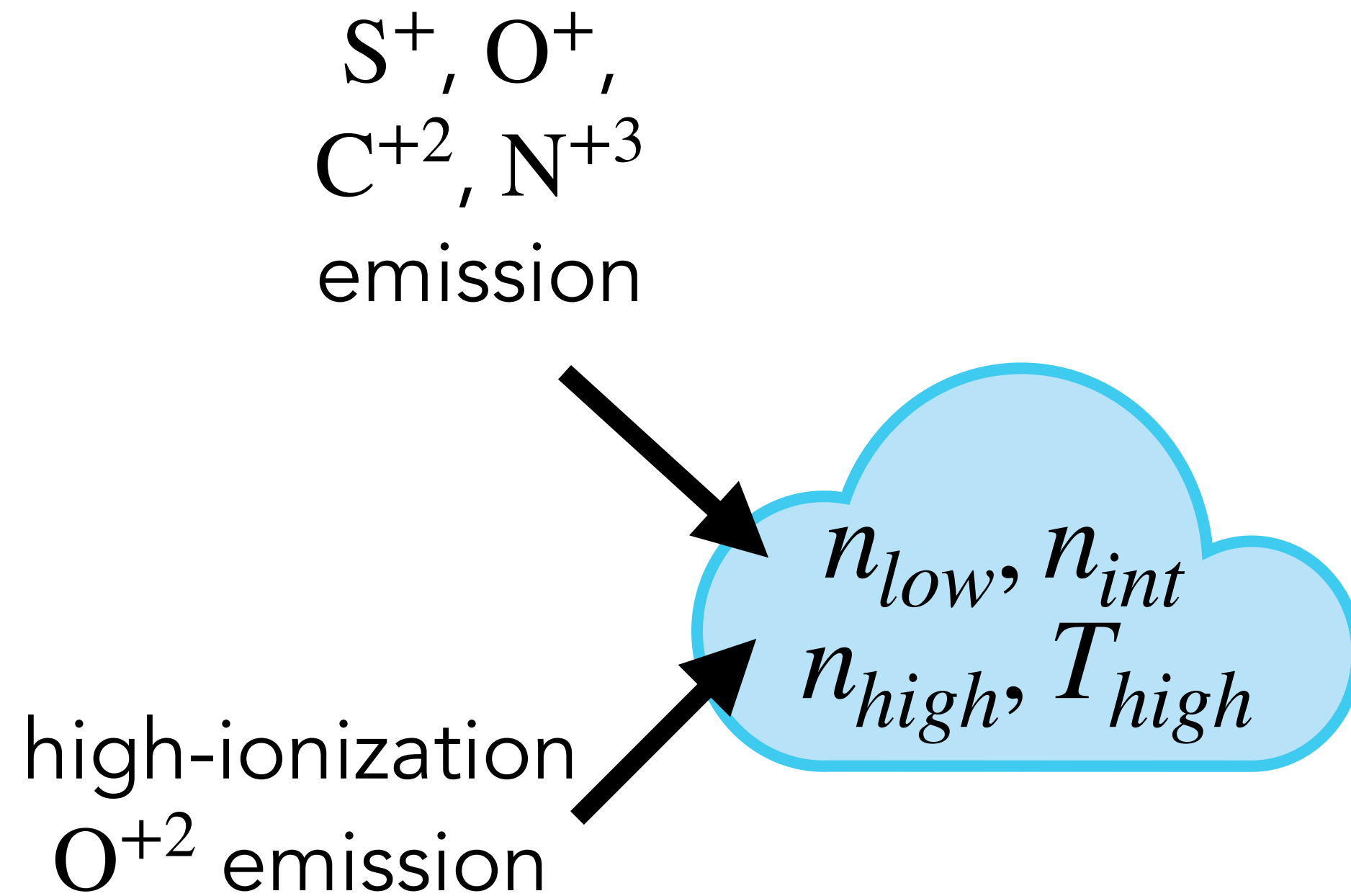


This model is more complicated, but also more accurate to the ionization structure

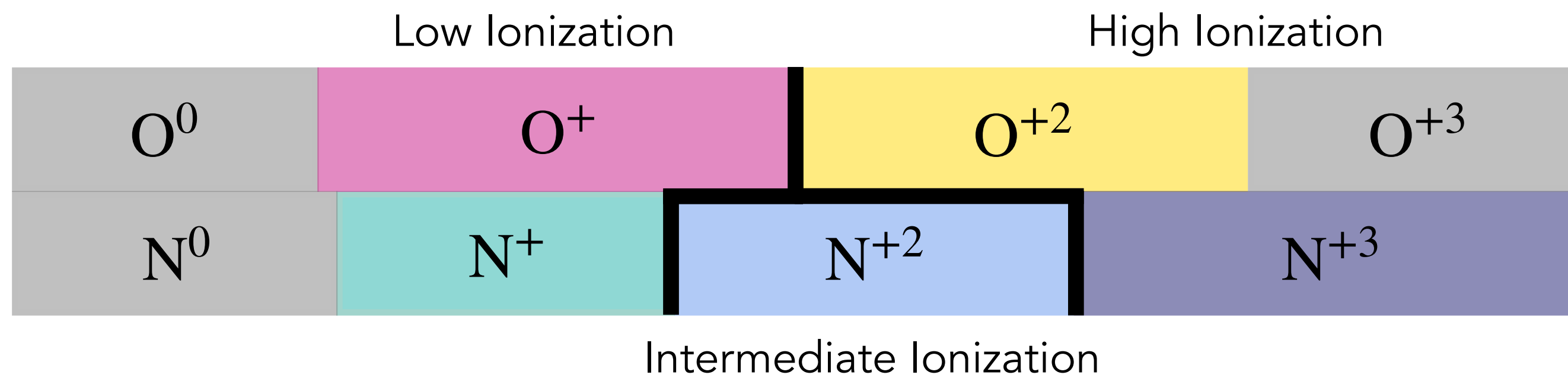
3-Zone Ionization Model



This model is more complicated, but also more accurate to the ionization structure



3-Zone Ionization Model

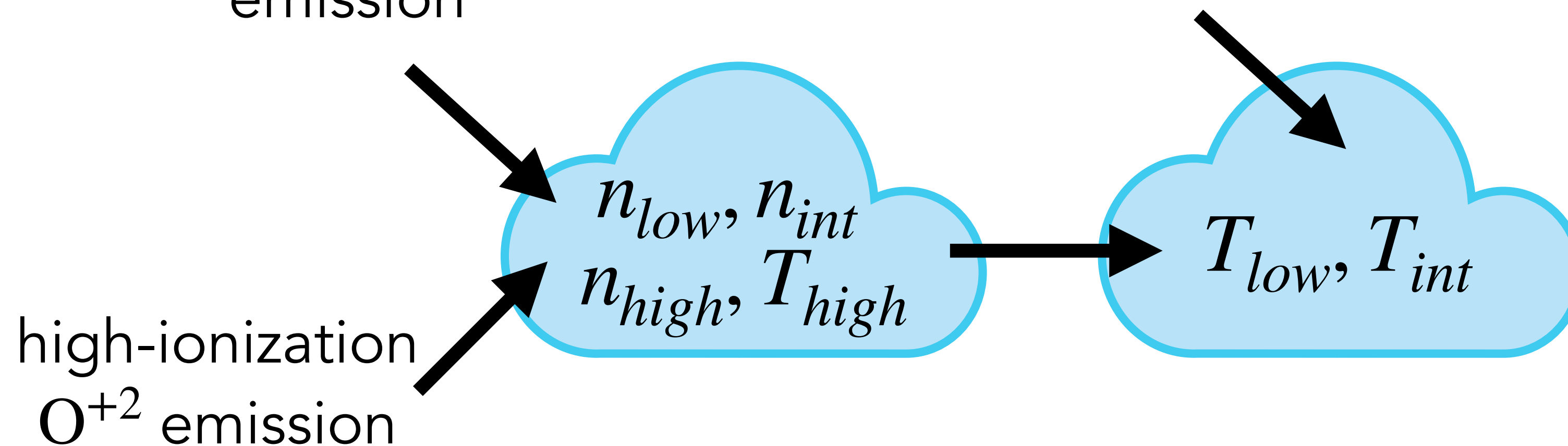


ionization potential energy

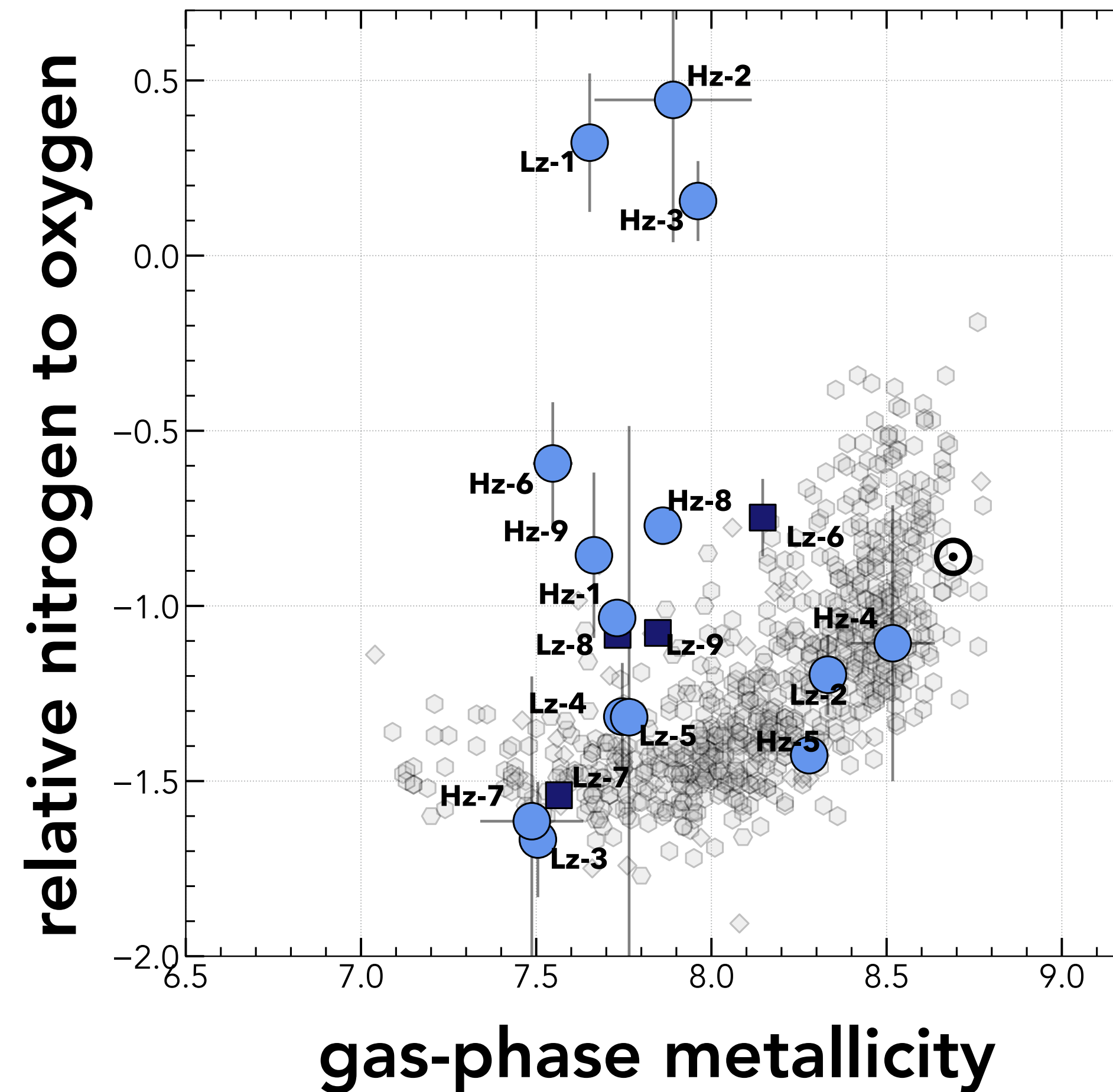
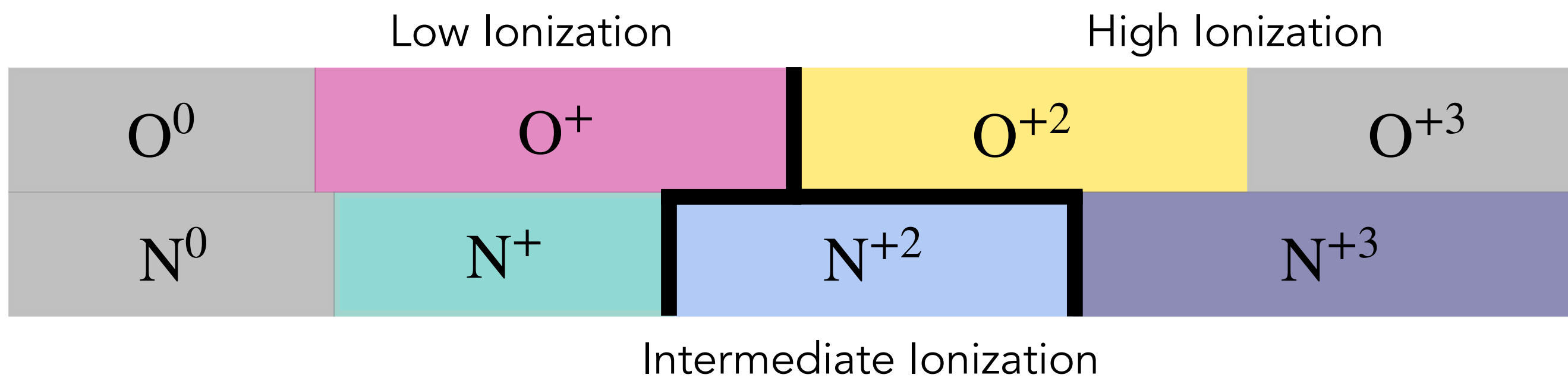
This model is more complicated, but also more accurate to the ionization structure

S^+ , O^+ ,
 C^{+2} , N^{+3}
emission

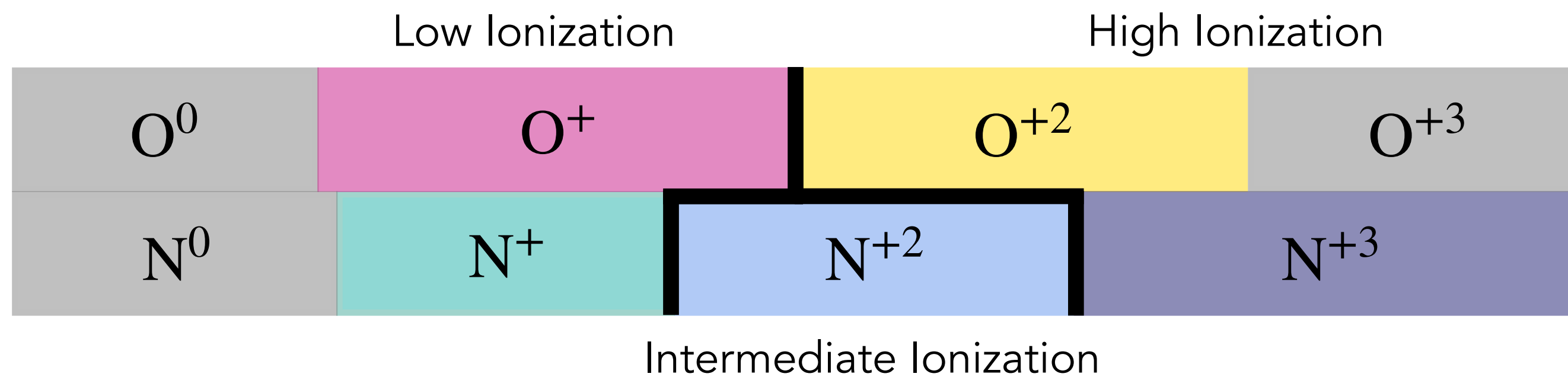
Garnett+92
 $T_e - T_e$ relations



3-Zone
Ionization
Model

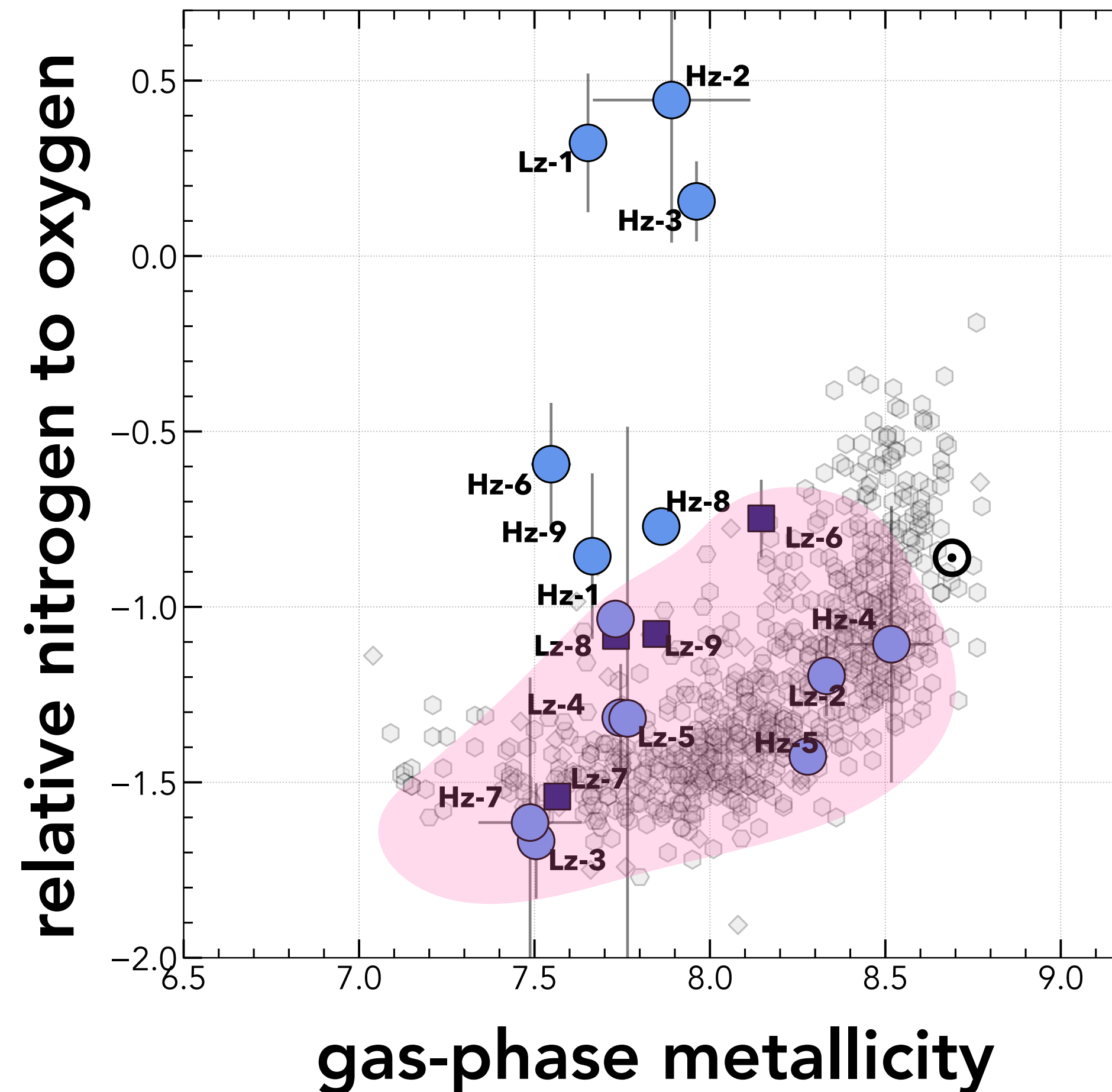


3-Zone
Ionization
Model

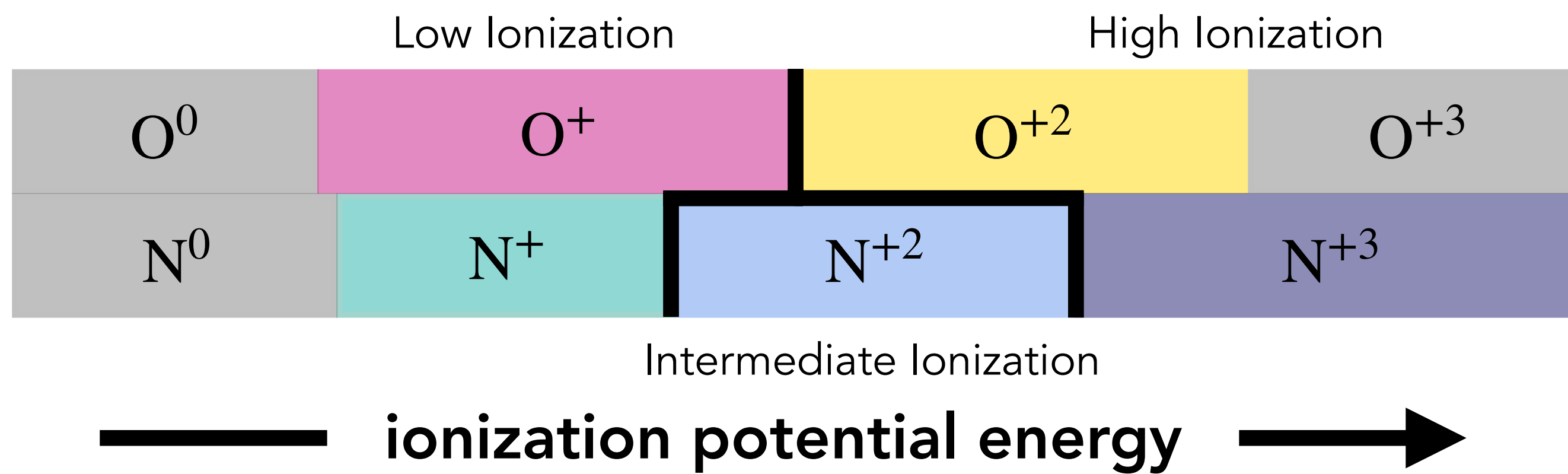


ionization potential energy →

Low-z sample remains
consistent with low-
ionization scatter

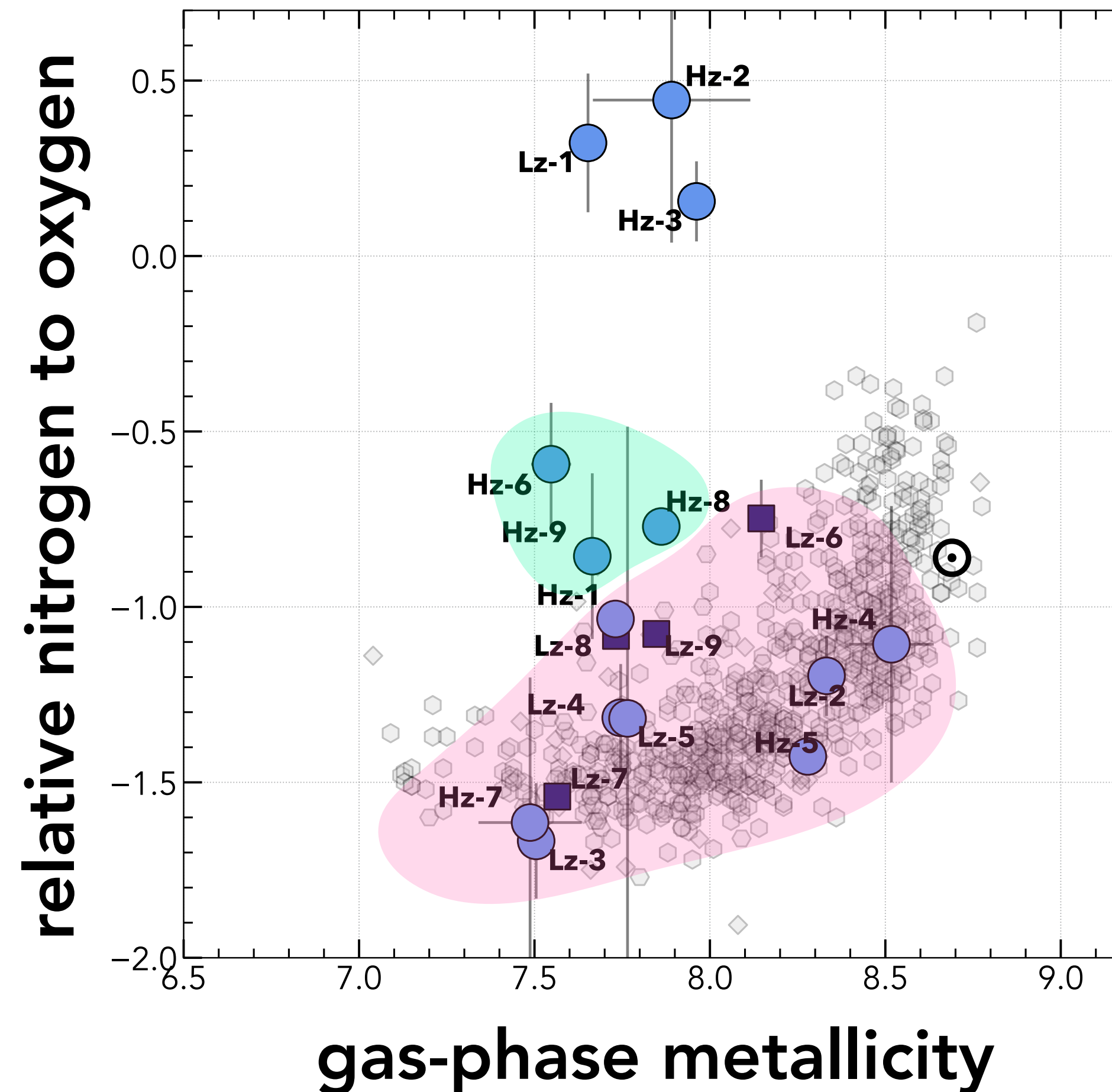


3-Zone
Ionization
Model

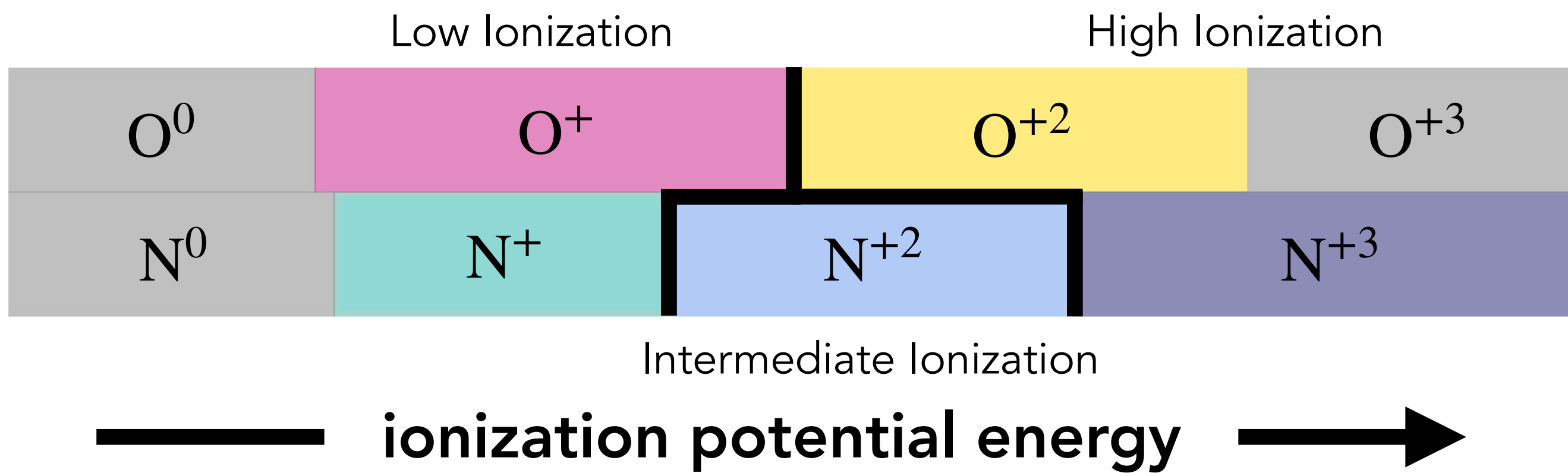


Some of the high-z sample moves **towards** the low-ionization scatter

Low-z sample remains **consistent** with low-ionization scatter



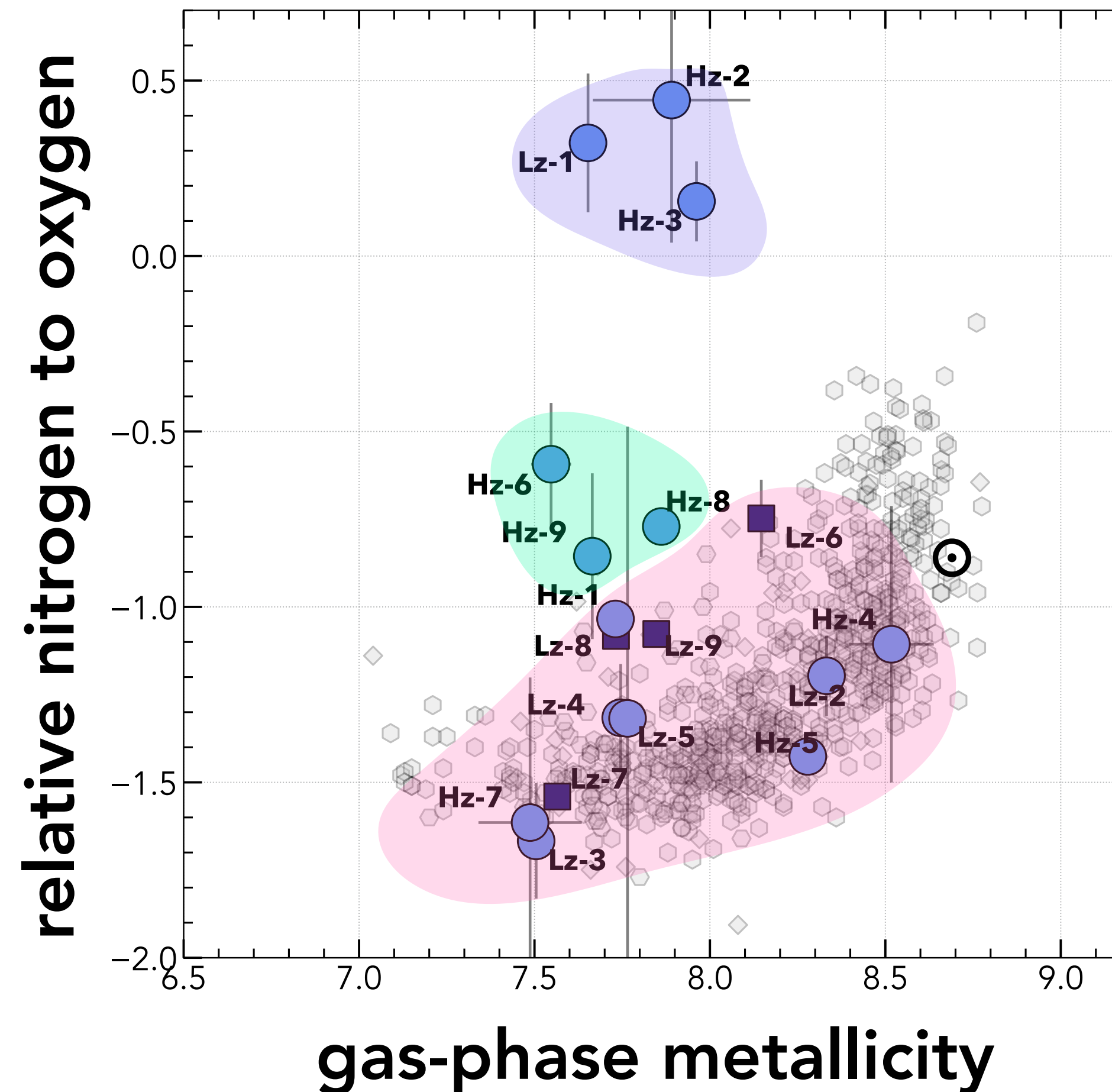
3-Zone Ionization Model



Some persist to be **anomalously high-N/O**

Some of the high-z sample moves **towards** the low-ionization scatter

Low-z sample remains **consistent** with low-ionization scatter



3-Zone Ionization Model

My photoionization modeling allowed me to determine that ...

- the ionization parameter, $\log U$, will be overestimated by 1.13 dex,
- temperatures derived using $[\text{OIII}] \lambda 4363 / \lambda 5007$ will be overestimated by 2200 K,
- and $12 + \log(\text{O}/\text{H})$ will be underestimated by 0.38 dex

... when the low-density limit is assumed for diagnostics that came from high-density gas. Next, I will continue the analyzing my models using a stratified nebula interpretation.

I applied my models to a sample of UV N emitters and find that...

- ICFs cannot, alone, resolve the discrepancy between low-ionization and high-ionization N/O, but are necessary for accurate abundance determinations.
- While a 2-Zone Model may be able produce N/O consistent with the local low-ionization scatter, at high- z this approach is insufficient to carefully constrain the N/O with high-ionization emission.
- The 3-Zone Model is the most appropriate method of determining N/O abundances given the extreme physical conditions of these objects.